Office of Energy Projects

July 2026

Columbia Gas Transmission, LLC **Docket No. CP26-19-000**

NKY Gate Enhancement Project

Environmental Assessment

Washington, DC 20426

NEPA Unique ID: EAXX-019-20-000-1770220686

FEDERAL ENERGY REGULATORY COMMISSION
WASHINGTON, D.C. 20426

OFFICE OF ENERGY PROJECTS

In Reply Refer To:
OEP/DG2E/Gas Branch 5
Columbia Gas Transmission, LLC
NKY Gate Enhancement Project
Docket No. CP26-19-000

TO THE INTERESTED PARTY:

The staff of the Federal Energy Regulatory Commission (FERC or Commission) has prepared an environmental assessment (EA) for the NKY Gate Enhancement Project, proposed by Columbia Gas Transmission, LLC (Columbia) in the above-referenced docket.[1] Columbia requests authorization to construct about 30.2 miles of new natural gas pipeline, abandon 48.5 miles of existing natural gas pipeline, and construct or abandon associated auxiliary and appurtenant facilities in Kentucky and Ohio. The project purpose is to enhance the safety, integrity, and reliability of Columbia's existing natural gas transmission system by replacing vintage pipeline infrastructure originally installed in the 1950s with new, modern pipeline facilities.

Any person wishing to comment on the EA may do so. To ensure consideration of your comments on the proposal prior to making a decision on the project, it is important that the Commission receive your comments **on or before 5:00pm Eastern Time on August 31, 2026.** Instructions for filing comments are provided on page 4.

FERC is the lead federal agency for authorizing interstate natural gas transmission facilities under the Natural Gas Act of 1938 (NGA) and the lead federal agency for preparation of the EA. The EA assesses the potential environmental effects of the NKY Gate Enhancement Project in accordance with the requirements of the National Environmental Policy Act (NEPA)[2] and the Commission's implementing regulations.[3] The principal purposes of the EA are to: identify and assess the potential effects on the natural and human environment; describe and evaluate reasonable alternatives; identify and recommend mitigation measures; and facilitate public involvement in the environmental review process. The EA concludes that approval of the proposed project would not constitute a major federal action significantly affecting the quality of the human environment.

[1] For tracking purposes under the National Environmental Policy Act, the unique identification number for documents relating to this environmental review is EAXX-019-20-000-1770220686.
[2] National Environmental Policy Act of 1969, as amended (Public Law [Pub. L.] 91-190. 42 U.S.C. 4321–4347, as amended by Pub. L. 94-52, July 3, 1975; Pub. L. 94-83, August 9, 1975; Pub. L. 97-258, §4(b), September 13, 1982; Pub. L. 118-5, June 3, 2023; Pub. L. 119-21, July 4, 2025).
[3] 18 Code of Federal Regulations (CFR) 380.

The EA addresses the potential environmental effects of the construction/installation, operation, and abandonment of the following project facilities:

- installing about 24.62 miles of new 36-inch-diameter natural gas transmission pipeline in northeastern Kentucky (Line AM36), primarily parallelling Columbia's existing Line AM04 and Line AM09 pipelines, crossing the Ohio River and terminating south of Interstate 275 in Ohio;

- installing about 0.10 mile of new 8-inch-diameter natural gas pipeline lateral, beginning at Line AM36 milepost (MP) 2.22, and terminating at the proposed Black River Meter Station in Bracken County, Kentucky;

- installing about 0.36 mile of new 24-inch diameter natural gas pipeline lateral, beginnings at Line AM36 MP 21.50 and terminating at the proposed Cold Spring Meter Station in Campbell County, Kentucky;

- installing about 0.06 mile of new 24-inch-diameter natural gas pipeline lateral, beginning at Line AM36 MP 24.60, and terminating a customer tie-in in Hamilton County, Ohio;

- installing about 5.02 miles of new 30-inch-diameter natural gas pipeline, beginning at Columbia's existing Moorefield Site, and primarily paralleling the existing Line EKY pipeline, and terminating at the existing Carlisle Meter Station in Nicholas County, Kentucky;

- abandoning in-place and by removal a total of about 22.48 miles of existing 24-inch-diameter, interstate, natural gas transmission pipeline (Line AM04), beginning at Columbia's existing Foster Launcher Receiver in Bracken County, KY and terminating before the Ohio River crossing east of Mary Ingles Highway (KY State Highway 445);

- abandoning in-place and by removal a total of about 20.50 miles of existing 26-inch-diamter, interstate, natural gas transmission pipeline (Line AM09),beginning at Columbia's existing Foster Regulator Station in Bracken County, Kentucky and terminating at Columbia's existing Cold Spring Meter Station;

- abandoning in-place and by removal a total of about 4.99 miles of existing 14-inch-diameter, interstate, natural gas transmission pipeline;

- abandoning by removal a total of about 0.57 mile of existing 20-inch-diameter interstate, natural gas transmission pipeline;

- constructing four new meter stations (Bracken Meter Station, Gubsur Mill Meter Station, Parkview Meter Station, UL47 Meter Station);

- modifying six existing stations (Foster Regulator Station; Black River Meter Station, Pendleton Meter Station, Alexandria Meter Station, Cold Spring Meter Station, Carlisle Meter Station); and

- installing tap facilities, launchers/receiver sites, mainline valves, and other associated pipeline facilities.

The Commission mailed a copy of the *Notice of Availability* of the EA to federal, state, and local government representatives and agencies; elected officials; Native American tribes; environmental and public interest groups; potentially affected landowners and other interested individuals and groups; and libraries and media outlets in the project area. The EA is only available in electronic format. It may be viewed and downloaded from the FERC's website (www.ferc.gov), on the natural gas environmental documents page (https://www.ferc.gov/industries-data/natural-gas/environment/environmental-documents). In addition, the EA may be accessed by using the eLibrary link on the FERC's website. Click on the eLibrary link (https://elibrary.ferc.gov/eLibrary/search), select "General Search" and enter the docket number in the "Docket Number" field, excluding the last three digits (i.e., CP26-19). Be sure you have selected an appropriate date range. For assistance, please contact FERC Online Support at FercOnlineSupport@ferc.gov or toll free at (866) 208-3676, or for TTY, contact (202) 502-8659.

The EA is not a decision document. It presents Commission staff's independent analysis of the environmental issues for the Commission to consider when addressing the merits of all issues in this proceeding. Under section 7(c) of the NGA, the Commission determines whether interstate natural gas transportation facilities are in the public convenience and necessity and, if so, grants a Certificate of Public Convenience and Necessity to construct and operate them. Section 7(b) of the NGA specifies that no natural gas company shall abandon any portion of its facilities subject to the Commission's jurisdiction without the Commission first finding that the abandonment will not negatively affect the present or future public convenience and necessity. The Commission bases its decisions on both economic issues, including need, and environmental effects.

Your comments should focus on the EA's disclosure and discussion of potential environmental effects, reasonable alternatives, and measures to avoid or lessen environmental effects. The more specific your comments, the more useful they will be. For your convenience, there are three methods you can use to file your comments to the

Commission. The Commission encourages electronic filing of comments and has staff available to assist you at (866) 208-3676 or FercOnlineSupport@ferc.gov. Please carefully follow these instructions so that your comments are properly recorded.

(1) You can file your comments electronically using the eComment feature on the Commission's website (www.ferc.gov) under the link to FERC Online. This is an easy method for submitting brief, text-only comments on a project;

(2) You can also file your comments electronically using the eFiling feature on the Commission's website (www.ferc.gov) under the link to FERC Online. With eFiling, you can provide comments in a variety of formats by attaching them as a file with your submission. New eFiling users must first create an account by clicking on "eRegister." You must select the type of filing you are making. If you are filing a comment on a particular project, please select "Comment on a Filing"; or

(3) You can file a paper copy of your comments by mailing them to the Commission. Be sure to reference the project docket number (CP26-19-000) on your letter. Submissions sent via the U.S. Postal Service must be addressed to: Debbie-Anne A. Reese, Secretary, Federal Energy Regulatory Commission, 888 First Street NE, Room 1A, Washington, DC 20426. Submissions sent via any other carrier must be addressed to: Debbie-Anne A. Reese, Secretary, Federal Energy Regulatory Commission, 12225 Wilkins Avenue, Rockville, Maryland 20852.

Filing environmental comments will not give you intervenor status, but you do not need intervenor status to have your comments considered. Only intervenors have the right to seek rehearing or judicial review of the Commission's decision. At this point in this proceeding, the timeframe for filing timely intervention requests has expired. Any person seeking to become a party to the proceeding must file a motion to intervene out-of-time pursuant to Rule 214(b)(3) and (d) of the Commission's Rules of Practice and Procedures (18 CFR 385.214(b)(3) and (d)) and show good cause why the time limitation should be waived. Motions to intervene are more fully described at https://www.ferc.gov/how-intervene.

For public inquiries and assistance with making filings such as interventions, comments, or requests for rehearing, contact the Office of Public Participation at (202) 502-6595 or OPP@ferc.gov. Additional information about the project is available from the FERC website (www.ferc.gov) using the eLibrary link. The eLibrary link also provides access to the texts of all formal documents issued by the Commission, such as orders, notices, and rulemakings.

In addition, the Commission offers a free service called eSubscription which allows you to keep track of all formal issuances and submittals in specific dockets. This can reduce the amount of time you spend researching proceedings by automatically providing you with notification of these filings, document summaries, and direct links to the documents. Go to https://www.ferc.gov/ferc-online/overview to register for eSubscription.

Commission Staff Page Limit and Deadline Certifications

I certify that Commission staff has considered the factors mandated by the National Environmental Policy Act and that this environmental document represents a good-faith effort to disclose the most important considerations required by the National Environmental Policy Act within the statutory page limit (42 U.S.C. § 4336a(e)) and the statutory deadline (42 U.S.C. § 4336a(g)). This certification reflects staff's expert judgment that the analysis contained within this environmental document is an accurate representation of the potential environmental effects of the proposed action and the analysis is substantially complete. In staff's judgement, any considerations addressed briefly or left unaddressed would not meaningfully inform the assessment of environmental effects.

Gertrude Fernandez Johnson, Director
Division of Gas -Environment &
Engineering

NKY GATE ENHANCEMENT PROJECT
ENVIRONMENTAL ASSESSMENT

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TECHNICAL ABBREVIATIONS AND ACRONYMS

APE	area of potential effect
ATWS	additional temporary workspace
BCC	Birds of Conservation Concern
BMP	best management practices
CFR	Code of Federal Regulations
CH_4	methane
CO	carbon monoxide
CO_2	carbon dioxide
CO_2e	carbon dioxide equivalents
Columbia	Columbia Gas Transmission, LLC
Commission	Federal Energy Regulatory Commission
dBA	A-weighted decibel
EA	environmental assessment
FERC	Federal Energy Regulatory Commission
FEMA	Federal Emergency Management Agency
FWS	U.S. Fish and Wildlife Service
GHG	greenhouse gas
HAP	hazardous air pollutant
HDD	horizontal directional drill
HUC	Hydrologic Unit Code
IPaC	Information for Planning and Consultation
KDFWR	Kentucky Department of Fish and Wildlife Resources
KEEC	Kentucky Energy and Environment Cabinet
KGS	Kentucky Geological Survey
KYSHPO	Kentucky Heritage Council
L_{dn}	day-night sound level
L_{eq}	24-hour equivalent sound level
MLV	mainline valve
MP	milepost
MSHCP	Multi-Species Habitat Conservation Plan
N_2O	nitrous oxide
NAAQS	National Ambient Air Quality Standards
NOAA	National Oceanic and Atmospheric Administration
Notice of Scoping	*Notice of Scoping Period Requesting Comments on Environmental Issues for the NKY Gate Enhancement Project*
NOx	nitrogen oxides
NSA	noise sensitive area
NRCS	Natural Resources Conservation Service

NRHP	National Register of Historic Places
ODNR	Ohio Department of Natural Resources
OSHPO	Ohio History Connection
PCB	polychlorinated biphenyls
PEM	palustrine emergent
PFO	palustrine forested
Plan	FERC's *Upland Erosion Control, Revegetation, and Maintenance Plan*
$PM_{2.5}$	particulate matter with an aerodynamic diameter less than or equal to 2.5 microns
$PM_{2.5}$	particulate matter with an aerodynamic diameter less than or equal to 2.5 microns
PM_{10}	particulate matter with an aerodynamic diameter less than or equal to 10 microns
Procedures	FERC's *Wetland and Waterbody Construction and Mitigation Procedures*
Project	NKY Gate Enhancement Project
PSD	Prevention of Significant Deterioration
PSS	palustrine scrub-shrub
ROW	right-of-way
SHPO	State Historic Preservation Office
SO_2	sulfur dioxide
SPCC	Spill Prevention, Control, and Countermeasure
SWPA	source water protection area
USACE	U.S. Army Corps of Engineers
U.S.C.	U.S. Code
USGCRP	U.S. Global Change Research Program
USGS	U.S. Geological Survey
USDA	U.S. Department of Agriculture
USDOT	U.S. Department of Transportation
USEPA	U.S. Environmental Protection Agency
VOC	volatile organic compound
WMA	Wildlife Management Area

A – PROPOSED ACTION

A.1 INTRODUCTION

The Federal Energy Regulatory Commission (Commission or FERC) staff has prepared this environmental assessment (EA) to assess the environmental effects of constructing/installing, operating, and abandoning the NKY Gate Enhancement Project (Project) facilities, proposed by Columbia Gas Transmission, LLC (Columbia). On November 10, 2025, Columbia filed an application in FERC Docket No. CP26-19-000 pursuant to Sections 7(b) and 7(c) of the Natural Gas Act, as amended, and Part 157 of the Commission's regulations. Columbia is seeking authorization to construct/install, operate, and abandon in-place and by removal interstate natural gas pipeline transmission facilities in Mason, Nicholas, Bracken, Pendleton, and Campbell Counties, Kentucky, and Hamilton County, Ohio.

A.2 PURPOSE AND NEED

Columbia states the purpose of the Project is to replace "vintage" pipeline infrastructure originally installed in the 1950s with modern pipeline facilities. According to Columbia, the Project is needed to address aging infrastructure, reduce potential safety and reliability risks, and ensure continued compliance with federal and state regulations.

A.3 SCOPE OF THE ENVIRONMENTAL ASSESSMENT

Commission staff reviewed the application, supplemental filings, and comments received to establish the scope of this EA. As presented in table 1 below, we[4] first considered whether each resource is present in the area of the proposed action and alternatives. If the resource is present, we evaluated whether further detailed analysis is warranted due to potential effects and if so, that resource is carried forward for discussion in the EA. If the resource is present but would not be affected to a degree that detailed analysis is required, that resource area is not carried forward and a rationale is provided. If the resource area is not present or would not be affected by the proposed action, it is not carried forward for further discussion.

The topics identified in table 1 as carried forward for further discussion are addressed in section B of this EA. Section B describes the affected environment as it currently exists, discusses the environmental consequences (effects) of the Project, and identifies measures proposed by Columbia to reduce these effects. In Section C, the EA assesses alternatives to the Project and in section D, we summarize our conclusions and present additional measures that we recommend the Commission adopt as mandatory environmental conditions of any Certificate of Public Convenience and Necessity (Certificate) it may issue to Columbia.

[4] "We," "us," and "our" refer to the Commission's environmental and engineering staff of the Office of Energy Projects.

Table 1 Summary of Resource Areas Affected		
Resource Area	**Presence of Resource**	**Determination**
Mineral Resources	☐ Present ☒ Not Present	☐ Carried Forward ☒ Not Carried Forward ***Rational if not carried forward:*** The nearest oil and gas well or other mineral extraction enterprise is located over 500 feet from any Project workspace
Geologic Hazards	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Soils	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Groundwater	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Surface Water	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Wetlands	☒ Present ☐ Not Present	☐ Carried Forward ☒ Not Carried Forward ***Rational if not carried forward:*** Of the 565.7 acres of land affected by the Project, less than one acre of wetland would be temporarily affected by construction of the Project, and less than one-tenth of an acre would be permanently affected by operation of the Project.
Fisheries	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Vegetation	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Wildlife	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Land Use	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Recreation	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Visual Resources	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Cultural Resources	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***

Table 1 Summary of Resource Areas Affected		
Resource Area	**Presence of Resource**	**Determination**
Socioeconomics	☒ Present ☐ Not Present	☐ Carried Forward ☒ Not Carried Forward ***Rational if not carried forward:*** Based on the scope of project activities and the small workforce size compared to the population of the project area, effects on socioeconomic conditions are expected to be temporary and minor.
Air Quality – Construction	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Air Quality – Operation	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward **Rational if not carried forward:**
Noise – Construction	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Noise- Operation	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***
Reliability and Safety	☒ Present ☐ Not Present	☒ Carried Forward ☐ Not Carried Forward ***Rational if not carried forward:***

The Commission is a federal regulatory agency and conducts a complete independent review of project proposals, including an environmental review of the proposed facilities. As the lead federal agency for the Project, FERC is required to comply with Section 7 of the Endangered Species Act, as amended, and Section 106 of the National Historic Preservation Act. We have considered these statutes in the preparation of this EA. FERC will use this document to consider the likely environmental effects that could result if it authorizes the Project. In addition to FERC, other federal, state, and local agencies may use this EA in approving or issuing any permits necessary for all or part of the proposed Project. Permits, approvals, and regulatory consultations for the Project are discussed in section A.10.

A.4 PUBLIC REVIEW AND COMMENT

On January 5, 2026, the Commission issued a *Notice of Scoping Period Requesting Comments on Environmental Issues for the NKY Gate Enhancement Project* (Notice of Scoping). The Notice of Scoping was sent to affected landowners; federal, state, and local government agencies; elected officials; environmental and public interest groups; Native American tribes; other interested parties; and local libraries and newspapers. Publication of the Notice of Scoping established a 30-day public comment period for submission of comments, concerns, and issues related to the environmental aspects of the Project. The public comment period closed on February 4, 2026. All comments received by the Commission are part of the public record for the Project and are available for viewing on the FERC website (www.FERC.gov), using the eLibrary link.

The Commission received comments from the Kentucky Department of Fish and Wildlife Resources (KDFWR) providing recommendations regarding the Project's scope of issues to address in the environmental review. Additionally, comments of support for the Project were received from numerous elected officials, political organizations, unions, local governments, individuals, and others. The Commission also received comments from potentially affected landowners, concerned citizens, and others regarding the purpose and need of the Project, general environmental and safety concerns, future land use and value, cultural resources, alternatives (including route deviations and removal of abandoned pipe), and loss of existing gas service. Comments within the scope of this EA are addressed in section B of this EA. Regarding comments received about loss of existing gas service, Columbia indicated that it would be disconnecting farm taps owned and operated by Duke Energy of Kentucky, Inc. and as a result, some individuals would lose existing gas service.[5, 6] This loss of service is outside the scope of the EA and is not addressed further. However, this comment may be addressed in any Order the Commission may issue.

A.5 PROPOSED FACILITIES

The Project involves the following activities/facilities:

- installing about 24.62 miles of new 36-inch-diameter natural gas transmission pipeline in northeastern Kentucky (Line AM36), primarily parallelling Columbia's existing Line AM04 and Line AM09 pipelines, crossing the Ohio River and terminating south of Interstate 275 in Ohio;

- installing about 0.10 mile of new 8-inch-diameter natural gas pipeline lateral (Black River Lateral), beginning at Line AM36 milepost (MP) 2.22, and terminating at the proposed Black River Meter Station in Bracken County, Kentucky;

- installing about 0.36 mile of new 24-inch diameter natural gas pipeline lateral (Cold Spring Lateral/Line AM24), beginning at Line AM36 MP 21.50, and terminating at the proposed Cold Spring Meter Station in Campbell County, Kentucky;

- installing about 0.06 mile of new 24-inch-diameter natural gas pipeline lateral (Kellogg Station Lateral), beginning at Line AM36 MP 24.60, and terminating a customer tie-in in Hamilton County, Ohio;

- installing about 5.02 miles of new 30-inch-diameter natural gas pipeline (Line EM-7), beginning at Columbia's existing Moorefield Site, and primarily paralleling the existing Line EKY pipeline, and terminating at the existing Carlisle Meter Station in Nicholas County, Kentucky;

[5] Columbia's March 11, 2026, response to FERC's March 5, 2026, Data Request. FERC eLibrary Accession No. 20260311-5145.
[6] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request, attachment 1-12. FERC eLibrary Accession No. 20260429-5302.

- abandoning in-place and by removal a total of about 22.48 miles of existing 24-inch-diameter, interstate, natural gas transmission pipeline (Line AM04), beginning at Columbia's existing Foster Launcher Receiver in Bracken County, KY and terminating before the Ohio River crossing east of Mary Ingles Highway (KY State Highway 445);

- abandoning in-place and by removal a total of about 20.50 miles of existing 26-inch-diameter, interstate, natural gas transmission pipeline (Line AM09), beginning at Columbia's existing Foster Regulator Station in Bracken County, Kentucky and terminating at Columbia's existing Cold Spring Meter Station;

- abandoning in-place and by removal a total of about 4.99 miles of existing 14-inch-diameter, interstate, natural gas transmission pipeline (Line EKY);

- abandoning by removal a total of about 0.57 mile of existing 20-inch-diameter interstate, natural gas transmission pipeline (Line E-Loop);

- constructing four new meter stations (Bracken Meter Station, Gubsur Mill Meter Station, Parkview Meter Station, UL47 Meter Station);

- modifying six existing stations (Foster Regulator Station; Black River Meter Station, Pendleton Meter Station, Alexandria Meter Station, Cold Spring Meter Station, Carlisle Meter Station); and

- installing tap facilities, launchers/receiver sites, mainline valves, and other associated pipeline facilities.

Figures 1-3 depict the locations of the proposed facilities.[7] Additional mapping can be found in Columbia's application and supplemental filings, available for viewing in the Commission's administrative record.[8]

[7] *Id*. at attachment 1-1.
[8] Columbia's November 10, 2025, Application, Resource Report 1, Appendix 1A. FERC eLibrary Accession No. 20251110-5151.

Figure 1 – Abandonment (Lines AM04, AM09, EKY, & E-Loop) & Construction (Lines AM36, AM24/Cold Spring Lateral, Black River Lateral, & Kellogg Station Lateral)

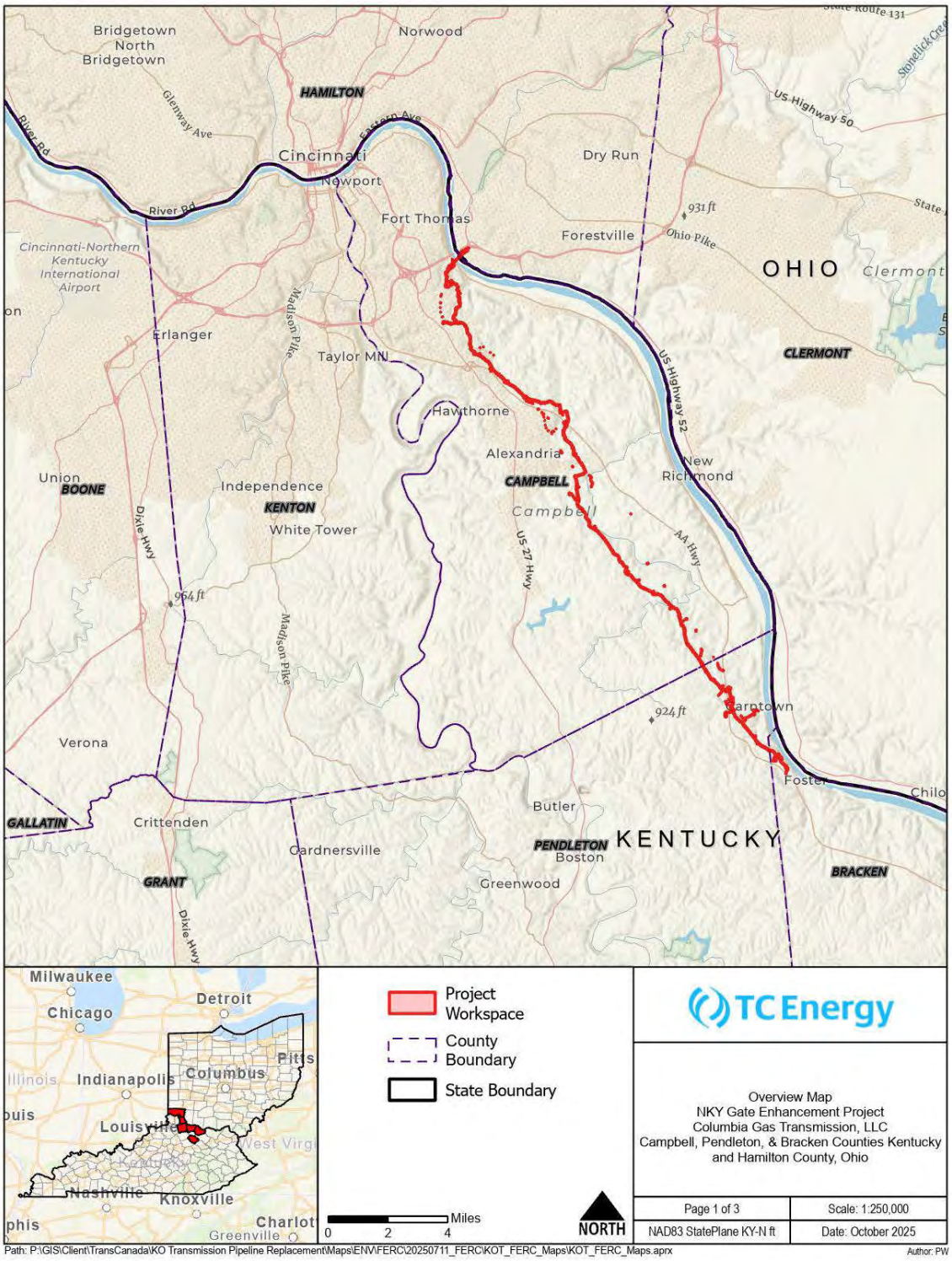


Figure 2 – Staging Area (SA-AM36-018)

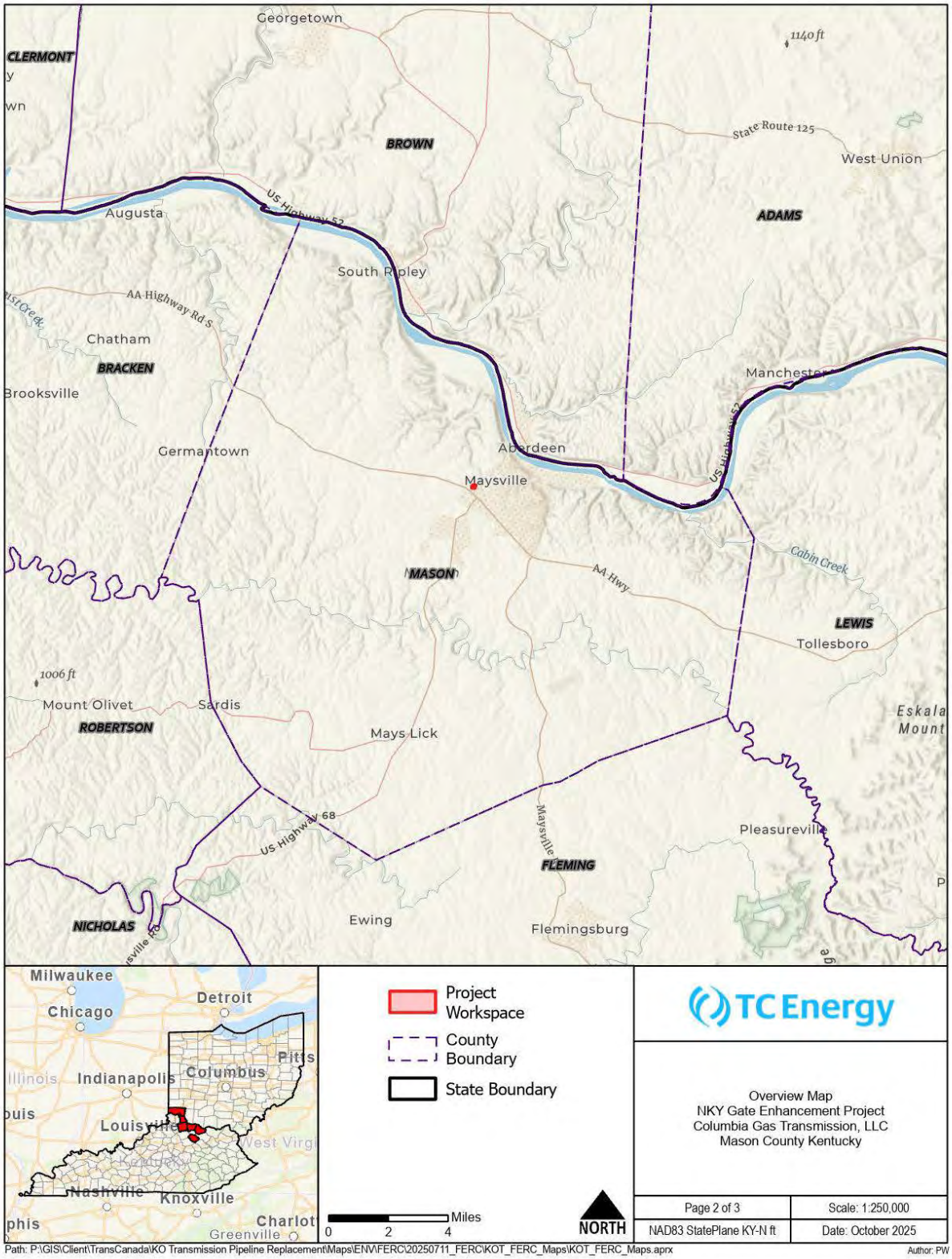
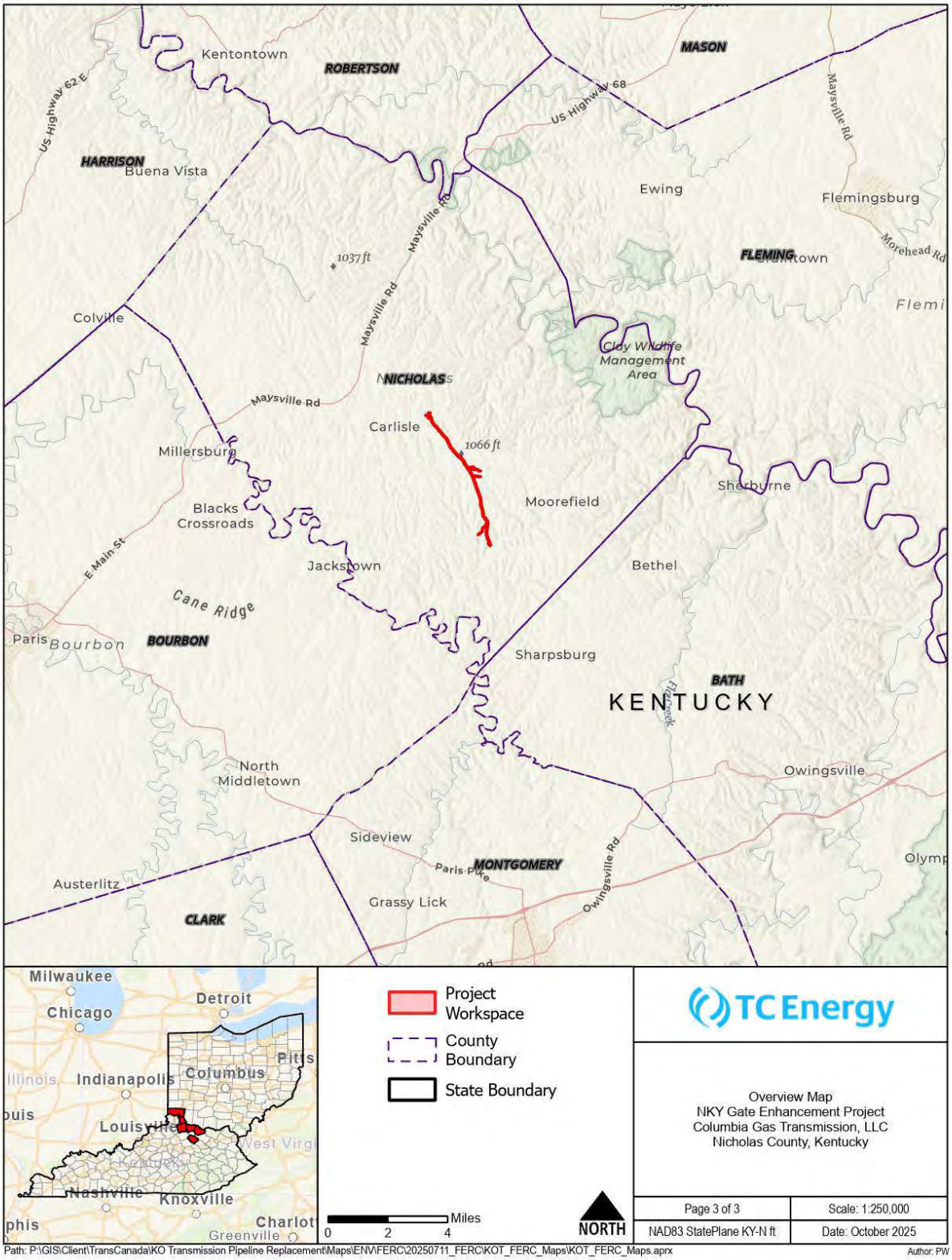


Figure 3 – Line EM-7



A.6 NON-JURISDICTIONAL FACILITIES

As part of the decision regarding whether to approve facilities under its jurisdiction, the Commission is required to consider all factors bearing on the public convenience and necessity. Occasionally, proposed projects have associated facilities that do not come under the jurisdiction of the FERC. These non-jurisdictional facilities may be integral to the need for the proposed facilities, or they may be minor, non-integral components of the proposed facilities.

The Project would include 21 non-jurisdictional electric utility upgrades and customer facilities at several of the Project above-ground facilities. Non-jurisdictional facilities for the Project include new overhead powerlines, belowground lines, power poles, transformers, meters, and other ancillary facilities. These facilities are addressed further in Section B.1 Environmental Trends and Planned Activities.

A.7 LAND REQUIREMENTS

Columbia would require the use of about 565.7 acres of land to complete the Project. These lands would encompass all temporary and permanent workspaces, access roads, and contractor yards. Of the 565.7 acres of land required for the Project, Columbia would require the permanent use of about 125.7 acres. These lands would be used to operate the Project including all new pipeline easements, permanent access roads, and new and expanded aboveground facilities.[9] Lands not permanently required for operation of the Project would be restored, released back to the landowner, and would be permitted to resume prior use.

For the proposed Line AM36 and the Cold Spring Lateral, Columbia would generally use a 125-foot-wide construction right-of-way (ROW). Columbia's proposed Line EM-7 is also a larger diameter pipeline; however, a 100-foot-wide construction ROW can be accommodated due to slightly less terrain-driven constraints. For Columbia's proposed Black River Lateral and Kellogg Station Lateral, Columbia would generally use a 75-foot-wide construction ROW. Construction ROWs would be expanded where required and may be reduced when adjacent to or crossing sensitive resources. Additional temporary workspaces (ATWS) would be used at road, wetland, waterbody, pipeline, and utility crossings, pipe bends, certain areas where topsoil segregation is required, at horizontal directional drill (HDD) entry and exit points, and when needed to stage and store equipment and ensure safe working conditions. Following construction, Columbia would retain a 50-foot-wide permanent easement generally centered on the pipeline to ensure access and safe operation.

A.8 PROJECT SCHEDULE AND WORKFORCE

Columbia anticipates that construction/installation of the proposed facilities could commence as early as January 2027, with the Project facilities being placed in to service by October 2027. To maintain existing natural gas service, abandonment activities would be completed after the new pipelines and facilities are placed into service. Columbia anticipates completing the Project (construction/installation, operation, and abandonment) by February 2028.

[9] *Id*. at table 1.2-1.

Generally, Columbia anticipates that construction activities would occur Monday through Saturday, between the hours of 7:00 AM and 7:00 PM. In order to address the potential for delays associated with weather, site conditions, specialized construction techniques, emergencies, or other atypical circumstances, Columbia may need to conduct construction activities on Sundays, federal holidays and/or between the hours of 7:00 PM and 7:00 AM. Nighttime construction is discussed further in section B.8 of this EA.

Columbia estimates a peak workforce of up to 400 staff and contractors would be needed to construct the Project with the construction workforce averaging approximately 75 workers during non-peak times. Abandonment activities would require the services of about 30 staff and contractors.[10] No permanent positions would be required to operate the Project.

A.9 CONSTRUCTION, OPERATION, ABANDONMENT, AND MAINTENANCE PROCEDURES

The proposed facilities would be designed, constructed, operated, abandoned, and maintained in accordance with the U.S. Department of Transportation (DOT) regulations in Title 42 CFR Part 192, *Transportation of Natural and Other Gas by Pipeline: Minimum Federal Safety Standards,* and other applicable federal and state regulations. To avoid, reduce, and mitigate effects on the environment, Columbia would implement FERC's *Upland Erosion Control, Revegetation and Maintenance Plan* (Plan),[11] and *Wetland and Waterbody Construction and Mitigation Procedures* (Procedures)[12] with Project-specific modifications. As summarized in Appendix A, we have reviewed Columbia's proposed workspaces, ROWs, and other modifications and find them to be acceptable. In addition, several other project-specific plans would be implemented to reduce environmental effects including Columbia's *Blasting Plan; Fugitive Dust Control Plan; Residential Construction Implementation Plan; Revegetation Plan; Spill Prevention, Control, and Countermeasure (*SPCC*) Plan; Plan for the Unanticipated Discovery of Contaminated Environmental Media; Plan for the Unanticipated Discovery of Historic Properties or Human Remains During Construction; Ohio River HDD Plan,*[13] *Karst Mitigation Plan, Winter Construction Plan, and Traffic Management Plan.*[14] We have reviewed these plans and find them to be acceptable.

With the exception of the special construction techniques described below, Columbia would employ conventional, industry, overland construction techniques to install the proposed pipelines. As proposed, pipeline construction would consist of a construction spread comprised of crews working in a sequential process of surveying and marking approved workspaces, clearing vegetation, grading work surfaces to ensure safety and stability, trenching, pipe stringing and bending, welding, lowering-in of the pipe and backfilling of spoil, hydrostatic

[10] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request. FERC eLibrary Accession No. 20260429-5302.
[11] A copy of FERC's Plan is available at https://www.ferc.gov/sites/default/files/2020-04/upland-erosion- control-revegetation-maintenance-plan.pdf.
[12] A copy of FERC's Procedures is available at https://www.ferc.gov/sites/default/files/2020-04/wetland-waterbody-construction-mitigation-procedures.pdf.
[13] Columbia's November 10, 2025, Application, Resource Report 1, appendix 1B. FERC eLibrary Accession No. 20251110-5151.
[14] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request, attachments 1-4, 5-27, and 6-29. FERC eLibrary Accession No. 20260429-5302.

testing, site cleanup, and restoration. Concurrently with these activities, Columbia work crews would implement erosion control devices including silt fences and water bars, and implement other measures to avoid, reduce, and mitigate construction effects on the environment. Depending on a variety of factors including available contractors, terrain, weather, and materials deliveries, these actions may occur relatively close in time or may be spread out over several weeks or longer. Following construction, Columbia would rehabilitate disturbed areas. These areas would be stabilized and revegetated. Permanent erosion and sediment controls would also be installed, and revegetation measures would be implemented according to the FERC Plan and Procedures, landowner requests, and Project-specific plans.

Special Construction Techniques

Horizontal Directional Drilling

Columbia would cross the Ohio River using a horizontal directional drill (HDD). An HDD allow for trenchless installation of pipe under a surface feature by use of a subsurface drill. A small-diameter pilot hole would be drilled under the feature to be crossed; the hole would then be successively enlarged through progressively sized reamings until a prefabricated segment of pipe can be inserted into the hole. The position of the drill would be electronically monitored, and directional corrections would be made to maintain its alignment between the desired entry and exit locations. Pipe sections would be welded on the exit side of the crossing and then pulled through the borehole.

During drilling, a slurry of drilling mud would lubricate the drill bit, remove drill cuttings, and stabilize the borehole. Drilling mud is under pressure during the HDD process; therefore, the potential exists for the mud to be released by migrating to the surface through a fracture in the underlying rock or substrate. Drilling mud consists of non-toxic bentonite clay and water, which can result in increased turbidity and sedimentation if it reaches a wetland or waterbody. Columbia proposes to utilize certain additives in the drill mud. Any additives used would be compliant with the National Sanitation Foundation/American National Standards Institute Standard 60.

The *Ohio River HDD Plan* includes measures to reduce the potential for, address, and mitigate any effects resulting from an inadvertent release of drilling muds or fluids. These measures include erosion controls at the HDD entrance/exit, inadvertent release containment kits, temporary halting of drilling operations (unless adequate containment and mitigation measures are already in place), release containment, and consulting regulatory agencies for final cleanup. Excess drill mud would be disposed of at an approved disposal facility in accordance with applicable federal, state, and local regulations.

Waterbody Crossings

Columbia would cross waterbodies conventional open-cut methods, conventional bores, or HDD (see above). Waterbodies less than 100 feet wide would typically be crossed via open-cut, wet ditch, crossing methods in accordance with our Procedures. Columbia would use dry-ditch methods, including flume crossings, and dam-and-pump crossings where open-cut methods are not feasible due to flows, or as prescribed by the U.S. Fish and Wildlife Service (FWS).

Generally, Columbia would abandon existing pipeline in place at stream crossings. However, pipeline would be removed via open cut methods at four locations where the existing line is exposed in the streambed and one location where it would be replaced with the proposed pipeline.

Agricultural and Residential Land Crossings

Columbia would cross agricultural and residential areas via modified open-cut methods. In agricultural areas, topsoil would be segregated and stockpiled, disturbed areas regraded to restore topography and drainage, soils tested for compaction and decompaction as needed, and topsoil redistributed. Excess rock would be removed to match adjacent undisturbed land, and agricultural areas would not be mulched, though an annual cover crop may be seeded where compatible. Columbia would coordinate with landowners to identify, stake, repair, and restore drain tiles, and ensure planned pipe depth does not obstruct future drain tile installation. In residential areas, construction would be completed as quickly and safely as reasonably practicable, access maintained where possible, safety fencing installed where needed, homeowners notified of utility interruptions, topsoil segregated unless waived or imported, and areas restored with debris removed and landscaping requests addressed where practicable.

Road and Railroad Crossings

Major roads, highways, and utility crossings[15] would be crossed using trenchless methods such as conventional jack-and-bore. With jack-and-bore, the pipeline would pass beneath the roadway or utility with little or no disturbance to traffic or ongoing operations. Lightly traveled, paved, and unimproved rural dirt or gravel roads[16] would typically be crossed via open cut crossing methods, with excavation by backhoe or similar equipment, compaction of backfill, and resurfacing. If open-cut construction requires extensive time to complete, detours and/or other traffic measures would be implemented in accordance with Columbia's Traffic Management Plan.[17]

Rugged and Karst Terrain

Where the Project crosses rugged topography (steep slopes), additional grading may be required to accommodate construction and safe working conditions. Additionally, temporary cut-and-fill may be used to establish a safe, level work terrace where the route crosses side slopes, with additional temporary workspace used as needed. Temporary erosion control measures would be more closely spaced and maintained until permanent measures are established. If large voids or unstable ground conditions associated with karst are identified, Columbia would implement mitigation measures and BMPs in accordance with its *Karst Mitigation Plan*.[18] Additionally, a qualified geotechnical professional would assess conditions

[15] Columbia's November 10, 2025, Application, Resource Report 8, appendix 8E. FERC eLibrary Accession No. 20251110-5151.
[16] *Id*.
[17] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request, attachment 5-27. FERC eLibrary Accession No. 20260429-5302.
[18] *Id*. at attachment 6-29.

and provide recommendations. Further information regarding potential karst terrain in the Project area is provided in section B of this EA.

Blasting

Blasting may be required where bedrock is encountered at depths that interfere with trenching. If necessary, a licensed subcontractor would conduct seismically monitored blasting in compliance with applicable regulations and implement mitigation measures such as blasting mats as appropriate, consistent with Columbia's Blasting Plan. The final blasting plan and schedule would be submitted for notification if blasting occurs within designated coldwater fisheries or within waterbodies identified as habitat for federally listed threatened or endangered species.

Winter Construction

Construction of the Project would occur during the 2026-2027 and 2027-2028 winter seasons. Therefore, Columbia has developed a Winter Construction Plan,[19] which includes specialized construction methods and procedures addressing snow handling and removal. In the event of heavy snow accumulation, snow could be stored off the right-of-way on an additional 25-foot-wide area adjacent to the construction workspace; however, Columbia would ensure that personnel and equipment would stay within the construction corridor to move snow, and care would be taken to minimize soil and vegetation removal from the construction corridor. Columbia would also establish gaps in topsoil piles to facilitate drainage of melting snow across the right-of-way. If inclement weather prohibits replacement of topsoil immediately following construction, Columbia would stabilize topsoil piles (e.g., mulching and erosion controls) until weather conditions improve. In the event that final cleanup is affected by winter weather, Columbia would implement measures to temporarily stabilize the right-of-way and reduce erosion potential until spring thaw conditions.

Pipeline Abandonment and Removal Procedures

Abandonment procedures may vary by site; however, the majority of the existing pipelines would be abandoned in place by cutting and capping the pipe with weld caps or steel plates and filled with grout beneath permitted road crossings. Pipeline abandonment by removal would be very similar to pipeline construction/installation and would generally include excavating the trench over the existing pipeline, cutting the existing pipe into segments, lifting the pipe out of the trench, and transporting the pipe to an authorized facility in accordance with applicable federal and state regulations. Prior to abandoning in place or by removal, the existing pipelines would be cleaned with a pig tool[20] to remove any residual materials. Containment would be placed under the pig receiver to capture the residual material, and any collected materials would be disposed of at an approved facility. Following pipe removal, pre-construction contours would be rehabilitated, and disturbed areas, except annual cropland, would be seeded with a perennial seed mix. Should additional backfill material be required to achieve pre-construction contours, Columbia would obtain suitable clean fill material from offsite

[19] *Id*. at attachment 1-4.

[20] Pipeline pigging allows the removal of debris, sediment, or other buildups from the pipeline's interior walls using a specialized device called a "pig."

sources and would temporarily store this material within the proposed contractor/staging/pipe yards, and/or other proposed temporary workspaces. In areas where topsoil segregation is required, the native topsoil would be spread over top of the imported subsoil.

Based on the age of the existing pipelines, Columbia believes there is potential to encounter asbestos-containing materials in the existing pipe coating materials. Columbia would treat all coal-tar and asphalt-coated pipe as asbestos-containing material and would require the contractor to implement control measures, in accordance with its Presumed Asbestos Containing Material Pipe Handling Plan, that ensures the pipeline is cut and removed in a manner that prevents waste material from entering the environment. Disposal or recycling of the removed pipe would be managed by a specialized waste team, and all asbestos-containing materials would be handled and disposed of in accordance with the U.S. Environmental Protection Agency Toxic Substance Control Act and all other applicable regulations.[21]

All removed piping and other materials would be stored, recycled, or disposed of by the contractor at authorized facilities in accordance with all federal, state, and local regulations.

We received numerous comments requesting Columbia to install portions of the new pipeline within existing easements after removal of the existing pipe. Columbia stated that the existing Line AM04 must remain in operation until the new Line AM36 is placed into service. Therefore, Columbia cannot install the new pipeline within the existing easement.[22] We also received comments concerning the final disposition of easements associated with pipeline to be abandoned. Columbia anticipates retaining the permanent easements associated with Lines AM04, AM09, EKY, and the E-Loop. However, Columbia is continuing landowner negotiations related to the lines, including discussions regarding the final disposition of permanent easements.[23]

A.10 PERMITS, APPROVALS, AND REGULATORY CONSULTATIONS

Federal, state, and local permits, authorizations, or clearances, as applicable, for the construction of the Project are identified in table 2. Columbia is responsible for obtaining all permits and approvals required to construct and operate its Project, regardless of whether they appear in the table.

[21] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request, at 8-9. FERC eLibrary Accession No. 20260429-5302.
[22] *Id*. at attachment 1-12.
[23] *Id*. at 6.

Table 2 Applicable Permits, Licenses, Authorizations, and Clearances for the Project			
Agency or Organization	**Permit/Approval**	**Submittal Date (anticipated)**	**Receipt Date (anticipated)**
Federal			
Federal Energy Regulatory Commission	Certificate of Public Convenience and Necessity	November 10, 2025	*Pending*
U.S. Army Corps of Engineers – Louisville District	*Rivers and Harbors Act* Section 10/*Clean Water Act* Section 404 Nationwide Permit 12	November 10, 2025	(November 10, 2026)
	Rivers and Harbors Act of 1899 Section 14 "Section 408" Permission"	November 10, 2025	(November 10, 2026)
U.S. Army Corps of Engineers – Huntington District	*Clean Water Act*, Section 404 Nationwide Permit 12	November 10, 2025	(November 10, 2026)
U.S. Fish and Wildlife Service – Kentucky and Ohio Ecological Services Field Office	*Endangered Species Act*, Section 7 Consultation; *Migratory Bird Treaty Act* Consultation	November 10, 2025	(September 10, 2026) (Kentucky Ecological Services Field Office) November 20, 2025 (Ohio Ecological Services Field Office)
Commonwealth of Kentucky			
Kentucky Energy and Environment Cabinet – Department of Environmental Protection	Section 401 Water Quality Certification	November 10, 2025	(September 10, 2026)
	Permit to Construct Across or Along a Stream	November 10, 2025	December 19, 2025
	Kentucky Pollution Discharge Elimination System (KPDES) Stormwater Construction General Permit	(October 1, 2026)	(October 31, 2026)
	Temporary Authorization for Water Withdrawal Permit	(1st Quarter 2027)	(1st Quarter 2027)
	KPDES General Permit for Discharge of Hydrostatic Test Water	(1st Quarter 2027)	(1st Quarter 2027)
Kentucky Department of Fish and Wildlife Resources	State Threatened and Endangered Species Consultation	November 10, 2025	November 21, 2025
Kentucky Heritage Council	*National Historic Preservation Act*, Section 106 Consultation	November 10, 2025 (Initial submittal) January 16, 2026 (Revised Submittal) March 23, 2026 (Addendum I to the Architectural Report)	March 24, 2026

Table 2 Applicable Permits, Licenses, Authorizations, and Clearances for the Project			
Agency or Organization	**Permit/Approval**	**Submittal Date (anticipated)**	**Receipt Date (anticipated)**
State of Ohio			
Ohio Environmental Protection Agency	Clean Water Act Section 401 General Water Quality Certification	Automatic with Nationwide Permit 12	N/A
	National Pollutant Discharge Elimination System General Permit for Discharges of Hydrostatic Test Water	(1st Quarter 2027)	(1st Quarter 2027)
Ohio Department of Natural Resources	Ohio Threatened and Endangered Species Consultation/Clearance	May 29, 2025	June 26, 2025
Ohio State Historic Preservation Office	Section 106 National Historic Preservation Act Consultation and Clearance	November 10, 2025	November 13, 2025
Source: Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request FERC eLibrary Accession No. 20260429-5302.			

B – ENVIRONMENTAL ANALYSIS

This section of the EA describes the affected environment as it currently exists and discusses the environmental consequences of the proposed Project. The environmental consequences of construction and operating the proposed pipelines and abandoning the existing pipelines would vary in duration and would include temporary, short-term, long-term, and permanent effects. Temporary effects generally occur during construction with the resource returning to a similar condition to that prior to construction, almost immediately following construction activities. Short-term effects could continue for up to 3 years following construction. Effects would be considered long-term if the resource required more than 3 years to recover. A permanent effect could occur because of any activity that modifies a resource to the extent that it would be affected for the life of the Project. When determining the significance of an effect, we consider the duration of the effect as well as the geographic, biological, and/or social context in which the effects would occur, and the intensity (e.g., severity) of the effect(s).

The analysis contained in this EA is based upon Columbia's application and our experience with the construction and operation of natural gas infrastructure. However, if the Project is approved and proceeds to the construction and abandonment phase, modifications (e.g., minor changes in workspace configurations) are not uncommon. These changes are often identified by a company once on-the-ground implementation work is initiated. Any Project modification would be subject to review and approval by FERC and any other permitting/authorizing agencies with jurisdiction.

B.1 ENVIRONMENTAL TRENDS AND PLANNED ACTIVITIES

To contextualize the Project's effects, we describe the existing environment in the project area (e.g., the environmental baseline), how the environment has changed over time, and other reasonably foreseeable projects in the area. Mason, Nicholas, Bracken, Pendleton, and Campbell Counties, Kentucky, and Hamilton County, Ohio share a humid, temperate climate typical of the south-central Appalachian and Interior Plateau region. Seasonal patterns include warm summers, cool winters, and year-round precipitation. The region is characterized by rolling hills, fertile valleys, and karst features, with underlying bedrock dominated by Ordovician and Silurian-aged sedimentary deposits that include limestone, shale, and dolomite that were subsequently overlain by Pleistocene and Holocene glacial deposits. These geologic features contribute to the area's rich soils, groundwater resources, and diverse habitats.[24]

The landscape is predominantly rural, with a mix of forests, agricultural lands, and small communities in the southern portion of the Project transitioning to denser suburban areas near the Ohio River. Population growth in Kentucky has been modest, with Mason County estimated at approximately 17,100 residents, Nicholas County at 7,500, Bracken County at 8,400, Pendleton County at 14,600, and Campbell County at 93,100 as of 2025. Population in Hamilton County, Ohio is estimated at approximately 830,600.[25] Steady increases in population reflect early agricultural settlement, later infrastructure development, and modern suburban and rural residential growth.

[24] https://websoilsurvey.sc.egov.usda.gov/App/HomePage.htm; https://www.uky.edu/KGS.
[25] https://data.census.gov/.

The project area is part of several major watersheds, including the Licking and Ohio Rivers. These watersheds experience nitrogen and sediment inputs from both agricultural and urban sources, which can affect water quality. The Kentucky Division of Water monitors these resources and has identified some streams in the area as impaired under Section 303(d) of the Clean Water Act, primarily due to nonpoint source pollution.[26]

According to the Kentucky Division for Air Quality and the U.S. Environmental Protection Agency (EPA), Hamilton County, Ohio is designated as maintenance for the 8-hour ozone (2015) standard and Campbell County, Kentucky is designated as maintenance for the 8-hour ozone (2015) standard and the 1-hour SO_2 (2010) standard. Mason, Nicholas, Bracken, and Pendleton, Counties, Kentucky are in attainment for all National Ambient Air Quality Standards (NAAQS), indicating relatively low baseline pollution. However, episodic events such as transboundary smoke from Canadian wildfires can temporarily degrade air quality.[27]

The predominant land cover changes in the project area over the last 20 years include a decrease in pasture/hay fields and deciduous forest, and an increase in cultivated croplands and medium-intensity development.[28] Planned activities in the vicinity of the Project that could affect the environmental baseline include construction of electric power facilities, commercial and residential development projects, industrial facilities, roadway improvements, and non-jurisdictional electrical facilities associated with this Project (see section A.6). Local planning documents and state infrastructure reports indicate ongoing and future investments in transportation and utility infrastructure.[29]

Construction of the 21 non-jurisdictional electrical facilities identified in section A.6 (e.g., powerlines and transformers/meters) would be concurrent with construction of the Project. These non-jurisdictional facilities may result in minor soil disturbances, and construction equipment emissions, but given the minor scope of activities, would not appreciably affect baseline conditions.

We identified twenty-two other planned activities/projects in the Project area. These projects include natural gas projects, commercial and residential development projects, industrial projects, and roadway projects. A FERC jurisdictional pipeline project, the Maysville Project under Docket No. CP26-25-000, involves installation of a 30-inch-diameter natural gas pipeline that would cross Mason County, approximately 0.62 miles from the Project's contractor/staging/pipe yard. Construction timeframes for both projects may overlap based on the anticipated construction schedules and therefore nearby residents may notice an increase in the presence of construction personnel and vehicle traffic in this vicinity; however, given the distance between the projects, we do not expect the Maysville Project to have any substantial effects on baseline conditions at the Project contractor yard.

[26] https://eec.ky.gov/Environmental-Protection/Water/Monitor/Pages/IntegratedReportDownload.aspx.
[27] https://www.epa.gov/outdoor-air-quality-data; https://eec.ky.gov/Environmental-Protection/Air/Pages/default.aspx.
[28] https://www.usgs.gov/centers/eros/science/national-land-cover-database.
[29] https://transportation.ky.gov/; local county planning offices.

Residential, industrial, and commercial projects include the 28-acre, Cold Spring Town Center development; the 71-acre, Coney Island multi-use commercial development, the 23-acre, Covington Central Riverfront mixed-use development; the 57-acre, Overlook at Sunrock residential development; the Riegler Blacktop Plant, a portable crushing and screening plant; grain loading dock; and several local utility pipeline and powerline replacement and improvement projects. These and other projects vary in size from less than one acre to more than 70 acres. These projects would generally be located between 1 and 13 miles from project workspaces with a majority of these projects located in Campbell County, Kentucky, which is the more developed/suburban, northern portion of the Project area. Additionally, the Kentucky Transportation Cabinet has authorized and will be undertaking roadway improvement projects; KY Highway 536 expansion, Brent Spence Bridge Corridor Project, I-71/75 and I-275 Interchange upgrade, and KY 8 Licking River Bridge Project. These projects would straighten and expand existing highways, install roundabouts, repair bridges, and upgrade, and expand an interstate highway interchange. These projects would range in length from 1,000 feet up to 5 miles. The KY 536 expansion would intersect with proposed pipeline work with the remaining projects located between four and nine miles from the Project. During construction, these projects would affect resources within the same sub-watershed as the Project, including groundwater resources through excavations and potential groundwater withdrawals, biological resources through vegetation clearing and habitat disturbance, and surface water and wetland disturbances through crossings, import of fill, or erosion events. However, given the footprints of these projects, their proximity to other developments, and their respective distances to Project workspaces, we expect their effects would be temporary and localized.

One commenter asserts that the Project's environmental effects should be considered in the context of effects from existing pipeline infrastructure in the area, especially as it relates to sinkholes.[30] Pursuant to NEPA, this environmental analysis focuses on the reasonably foreseeable environmental effects of the Project, i.e. the effects of constructing and operating the proposed pipelines and abandoning the existing pipeline. Previous activities that have occurred in the area are considered as part of our environmental baseline and our evaluation of resources occurs against this baseline.

B.2 GEOLOGY

The Project would be located within the Lexington Plain section of the Interior Low Plateaus physiographic province (U.S. Geological Survey [USGS], 2023). Topography across the Lexington Plain section ranges from low to moderate relief to very low relief. Elevations across the Project range from approximately 476 to 1,036 feet above mean sea level.

Pipeline construction and abandonment generally do not affect underlying geological resources; however, as described previously, Columbia proposes to install a segment of pipe that crosses the Ohio River via HDD. In exploratory borings probing sediments at depths where the HDD would be expected to penetrate, the most prevalent sediments encountered were sand and/or clay. The HDD site is underlain by limestone and shale, and bedrock was encountered at 4.5 feet below ground surface to 98 feet below ground surface.[31] Based on geotechnical data

[30] Joseph and Tracy Butts January 21, 2026, Comments at 4. FERC eLibrary Accession No. 20260121-5006.
[31] Columbia's November 10, 2025, Application, Resource Report 6, appendix 6D. FERC eLibrary Accession No. 20251110-5151.

collected to date, we do not anticipate that the Project, as designed, would be restricted by geological limitations.

During the HDD process, if pumped mud pressure exceeds the strength of the surrounding geologic materials or if the borehole intersects a preferential flow pathway, such as near-surface layers of fat clay,[32] drilling mud could leave the bore. The loss of mud would register with the HDD operator as a drop in mud pressure. While HDD operators can inject thickeners and binding agents to stop the loss, if the loss of mud persists it may emerge at the land surface (as a small, muddy spring) or waterbody (as a plume of sediment). This emergence is commonly referred to as an inadvertent return. In the event of an inadvertent return, Columbia would implement measures described in its Ohio River HDD Plan to limit effects on the Ohio River and associated sensitive resources. These measures include temporary halting drilling operations (unless adequate containment and mitigation measures are already in place), constructing containment, and consulting regulatory agencies for final cleanup. Any disposal or beneficial reuse of drilling fluids would be in conducted in accordance with applicable regulations.

Blasting

Bedrock of the Project area consists of Ordovician and Silurian-aged sedimentary deposits that include limestone, shale, and dolomite that were subsequently overlain by Pleistocene and Holocene glacial deposits. Because shallow bedrock is expected to be encountered in several areas throughout the Project, blasting is anticipated to occur. Blasting is discussed further in section B.3.

Paleontological Resources

The Project is located on Ordovician and Silurian-aged bedrock formations that are described as being abundant in fossils (KGS, 2025b; ODNR, 2025a). Cenozoic fossils are also abundant in Kentucky and Ohio and mainly occur in caves, sinkholes, and boggy deposits. To reduce the potential effects on paleontological resources, if discovered, Columbia would temporarily cease excavation in the area and notify the KGS or ODNR Division of Geological Survey, as well as FERC, so that all finds may be properly documented. Given that Columbia has committed to following these mitigation procedures to preserve finds, we conclude that the Project is unlikely to have significant effects on paleontological resources.

Geologic Hazards

Geologic hazards are natural, physical conditions that can, when active, result in damage to land or structures, or injuries to people. These include hazards that are seismic-related (earthquakes), landslides, and soil liquefaction. Other geological hazards include flooding and karst terrain. The Project would be located in an area of low seismic risk; and therefore, these and related hazards are not discussed further. Landslides, karst, and flooding are discussed below.

[32] Fat clay is clay that swells when moist, and when swollen with water offers little shear strength, forming the base for land slips in the event that strata are not level.

Landslides

A landslide is the movement of a mass of rock, debris, or earth down a slope and encompasses geologic processes such as debris or mud flows, rotational slides (slumps), translational slides, earth flows, rockfalls, or debris slides (Cruden, 1991; Cruden and Varnes, 1996). A site-specific geohazard assessment conducted for the Project identified previous landslide locations in the Project area and slopes with moderate to high landslide susceptibility, however, the analysis found a majority of the Project is within an area of low landslide susceptibility.[33] The geohazard assessment identified seven landslide locations associated with Line AM36, with three landslide locations identified with a moderate potential risk to affect pipeline construction and/or operation. Additionally, the geohazard assessment identified a landslide location associated with Line AM36 that was recommended to be avoided. Columbia would install Line AM36 via bore beneath the existing landslide to avoid the potential hazard. The geohazard assessment identified one moderate risk slope associated with Line EM-7.

To mitigate the risk of landslides during construction in areas of the Project with higher landslide susceptibility, Columbia would implement strategies including, but not limited to, reducing or distributing spoil placement to avoid concentrated loading, minimizing or temporarily retaining large slope cuts (e.g., cuts >5 feet high), avoiding or limiting blasting where practicable, managing groundwater and surface water where encountered, implementing subsurface and/or surface drainage improvements to facilitate efficient soil drainage where necessary, and monitoring conditions. Given Columbia's mitigation measures, we do not expect that landslides would significantly affect the Project.

Karst

Karst describes physical landscape features that are formed from the dissolution of soluble rocks. Areas with karst terrain may be more susceptible to subsidence events. Findings from the geohazard assessment conducted for the Project indicate that the potential for near-surface karst across the Project footprint is low. Some geologic formations underlying the Project do contain rock types that are soluble, such as limestone; however, no evidence of karst was identified within or adjacent to the Project during the geomorphic review of the Project. In addition, no documented or geomorphic evidence of underground mine areas was identified within the Project area (Kentucky Division of Mine Safety, 2023).[34] Should a hazardous karst feature (e.g., open voids including sinkholes) be encountered during construction, the feature would be assessed by a geotechnical professional to provide guidance to estimate the threat potential and Columbia would mitigate the threat as necessary in accordance with its Karst Mitigation Plan.[35] Therefore, we do not expect that karst hazards would significantly affect the Project, if discovered.

[33] Columbia's November 10, 2025, Application, Resource Report 6, section 6.5.2. FERC eLibrary Accession No. 20251110-5151..
[34] *Id*. at appendix 6C.
[35] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request, attachment 6-29. FERC eLibrary Accession No. 20260429-5302.

Flooding

According to the Federal Emergency Management Agency (FEMA), portions of Line AM36 and Line EM-7 cross the regulatory floodway and Zones A and AE of a 100-year floodplain (FEMA, 2025). Additionally, the Kellogg Station Launcher/Receiver and associated permanent access road are entirely within Zone AE. The remainder of the Project is outside the 500-year flood zone and therefore has a very low risk of flooding.[36] Based on the Project location and Columbia's limited permanent aboveground effects within floodplains, we expect that flooding would not significantly affect the Project.

B.3 SOILS

Soils present in Project workspaces include highly erodible soils, soils with steep slopes, soils with poor revegetation potential, soils with high rutting potential. Prime farmland, and shallow bedrock (bedrock within 60 inches of the ground surface) also occur in the Project area.

Approximately 97.0 acres of Project area soils are highly erodible due to water. None of the Project area soils are highly erodible by wind. Approximately 552.8 acres of Project area soils have a high potential for soil rutting (hazard). During construction, soils would be traversed by heavy equipment and disturbed to facilitate the installation of the proposed pipeline. As a result, soils could be compacted and rutted, and effects could be exacerbated where soils are saturated. Topsoil and subsoils could also be mixed, which could result in short- to long-term adverse effects on soil productivity (and revegetation). Soils could also erode during exposure caused by the removal of vegetative cover and soil handling associated with creating a workspace. If any of the above were to occur, effects could continue for one or more years. In addition, the presence of certain soil conditions (e.g., slope or low fertility) could result in poor revegetation of disturbed areas.

To avoid, reduce, and mitigate effects on soils, Columbia would implement numerous measures outlined in its Plan and Procedures. These measures include installing, monitoring, and maintaining temporary and permanent erosion control devices (such as slope breakers, sediment barriers, and trench plugs), managing runoff during and after ground disturbance, and restoring and revegetating disturbed areas as soon as practicable following final grading. In areas susceptible to saturation and compaction, Columbia would use timber mats or low-ground-pressure equipment as conditions warrant, and disturbed upland areas would be reseeded in accordance with its Project-specific Revegetation Plan.

While the Project would cross and disturb prime farmland; only 4.3 acres of prime farmland would be permanently affected by Project. Prime farmland soils that are currently used for agricultural purposes would continue to be used for agricultural. Therefore, we conclude that Project effects on the availability of prime farmland and farmland of statewide importance would be temporary and not significant. Agricultural activities are addressed further in section B.7 .

[36] Columbia's November 10, 2025, Application, Resource Report 6, section 6.5.4. FERC eLibrary Accession No. 20251110-5151.

Areas mapped as having shallow bedrock encompass approximately 483.2 acres of the Project area, of which 360.0 acres of these soils are also classified as rocky.[37] In areas where stony/rocky soils or shallow bedrock interfere with conventional excavation or rock trenching methods, blasting may be required to excavate the trench. Columbia stated it would use the minimum explosive charge necessary to fracture bedrock and minimize shot rock from leaving the construction corridor. As necessary, excess rock would be removed from the construction corridor or deposited on site, if a beneficial reuse is identified and agreed to by the landowner, and in accordance with applicable permit conditions. Columbia has developed a Project-specific Blasting Plan and would adhere to all local, state, and federal regulations for all blasting activities.

Additionally, if contaminated media is discovered during construction, Columbia would implement its Plan for the Unanticipated Discovery of Contaminated Environmental Media. We have reviewed this plan and found it acceptable.

Based on Columbia's implementation of avoidance, reduction, and mitigation measures, we conclude that while constructing and operating the Project could result in adverse effects on soils, these effects would not be significant.

B.4 WATER RESOURCES

Groundwater

The Project area overlies minor aquifers categorized as "other rocks," which generally consist of karst or limited availability/localized fractures and minor karst (Kentucky Energy and Environment Cabinet [KEEC], 2025a). These minor aquifers form in regions where rocks, like limestone and dolostone are chemically dissolved by water forming large passageways and caves. The Great Miami Buried Valley Aquifer, a U.S Environmental Protection Agency (USEPA) designated sole source aquifer, is crossed by the Project components in Ohio (USEPA, 2025a).[38] The Project would not affect wellhead protection areas (KGS, 2025c; Ohio Environmental Protection Agency, 2025).

Twenty-three private water supply wells were identified within 150 feet of the Project workspace.[39] KGS Well Record Information and Ohio Water Well Locater records indicate that seven of these wells (KGS, 2025c; ODNR, 2025b) are within the Project workspace. Columbia would offer pre- and post-construction testing for water supply wells within 150 feet of the construction workspace as negotiated with the landowner.[40]

Columbia would not withdraw groundwater for construction or operational water needs, except as necessary for trench dewatering. However, ground disturbing activities including surface grading and trenching could affect surface drainage and groundwater recharge patterns,

[37] Columbia's November 10, 2025, Application, Resource Report 7, Appendix 7D. FERC eLibrary Accession No. 20251110-5151.
[38] Columbia's November 10, 2025 Application, Resource Report 2, section 2.2.3. FERC eLibrary Accession No. 20251110-5151.
[39] *Id*. at table 2.2-1.
[40] *Id*. at section 2.2.7.

potentially causing minor fluctuations in groundwater levels and/or increased turbidity in shallow groundwater. While shallow groundwater may be intercepted by pipeline construction, effects on deeper other rocks aquifers are not anticipated. If trench dewatering becomes necessary, any resulting changes in water levels and/or turbidity in shallow groundwater would be localized and temporary because water levels quickly re-establish equilibrium and turbidity levels rapidly subside.

Should an inadvertent release of HDD drilling fluids intersect with groundwater, localized increases in groundwater turbidity, observable in groundwater withdrawals (including private and municipal wells), may occur. In the event of an inadvertent return, Columbia would implement the contingency plan specified in the Ohio River HDD Plan, which includes measures for monitoring drilling fluid circulation and returns, and responding to meaningful losses of drilling fluid circulation. With implementation of these measures, we expect that any volume of drilling fluid to be lost would be minor when compared to the volume of the aquifer, and because dissolved solids would be removed from the groundwater through natural filtration processes, we conclude that any resulting effects would be temporary and minor.

Groundwater Contamination

Three leaking underground storage tanks were identified within 0.5 mile from the Project (KEEC, 2025b; Ohio Department of Commerce, 2025; USEPA, 2025b). According to state records, these releases have been remediated. The storage tank leaks occurred within 0.12, 0.24, and 0.30 mile of Line AM36 MP 20.09, Line AM09 MP 20.39 and Line AM04 MP 20.30, respectively. The distance between these sites and the Project indicates that contamination would not likely be encountered. If localized pre-existing contamination is encountered during construction of the Project, Columbia would adhere to its Plan for the Unanticipated Discovery of Contaminated Environmental Media. In addition, the measures outlined in the SPCC Plan, developed by Columbia, are designed to avoid spills and expedite cleanup of a spill should one occur. SPCC Plan measures limit fueling and vehicle storage exclusion zones near wells and the imposition of secondary containment at storage facilities.

Based on the groundwater resources known to occur in the Project area and the Project's potential effects to these resources, we conclude that effects to groundwater during construction and operation of the Project would not be significant.

Surface Waters

Constructing and abandoning the proposed facilities would require the crossing of 252 waterbodies including 13 perennial streams, 41 intermittent streams, 197 ephemeral streams, and 1 pond. Of the 252 waterbodies that would be crossed, 229 would be considered minor (less than 10 feet wide), 22 would be considered intermediate (greater than 10 feet wide, but less than 100 feet wide), and 1 (the Ohio River) would be considered a major waterbody (greater than 100 feet wide). Appendix B summarizes the waterbody crossings for each Project facility.

Columbia would install the proposed pipelines across streams using open cut trenching methods at 99 locations. One crossing would be completed via HDD and two would be completed using bores. Columbia would abandon pipeline by removal (open cut) at five stream

crossings and abandon pipeline in place at five stream crossings. Existing culverts and/or timber mats would be used for permanent or temporary access at 139 stream crossings, and one new culvert would be installed along one permanent access road.

Open cut crossings using wet ditch methods would temporarily disturb streambeds and subsequently increase erosion, turbidity, and sedimentation which in turn would temporarily, but adversely affect water quality. Open cut crossings using dry ditch methods would have similar, but less intense effects on waterbodies.

Columbia identified 59 additional temporary workspaces that would be sited within 50 feet of a surface water feature and has provided justification for deviations from section V.B.2 of our Procedures. Steep terrain, pipeline points of inflection, proximately to residential areas and space limitations are among the site-specific factors cited.

During the scoping period, several commenters noted that removal of vegetation, ground disturbance, and multiple stream crossings would lead to accelerated erosion and water quality issues in the streams. To reduce and mitigate effects on waterbodies, Columbia would implement numerous measures as described in its Procedures as well as other applicable construction plans. All open cut crossings would be completed in accordance with Columbia's Procedures, and last no longer than 24 to 48 hours. Mitigation measures include the installation of erosion control devices, the monitoring of crossing activities, and the inspection/maintenance of crossing equipment and machinery.

As described previously, Columbia would cross the Ohio River via HDD. The use of HDD would avoid streambed disturbance, flow interruptions, and associated effects on the river. An inadvertent release of drilling fluids into the Ohio River would likely dissipate quickly resulting in only minor and temporary effects on the river.

Given that Columbia would complete waterbody crossings in accordance with its Procedures, implement measures to avoid and reduce effects to larger streams, and implement its *Ohio River HDD Plan,* we conclude that the effects to surface waters would not be significant.

Sensitive Surface Waters

The Ohio River is the only impaired waterbody that would be crossed by the Project. It is listed as impaired for aquatic life habitat, water supply, recreation by *Escherichia coli,* dioxins, and PCBs. The Ohio River also supports species protected under Section 7 of the Endangered Species Act (ESA), commercial fisheries, public water supplies, and is listed by the U.S. Army Corps of Engineers as a Section 10 of the Rivers and Harbors Act Navigable Water. As described previously, effects on the Ohio River would be substantially reduced due to the use of an HDD.

Source Water Protection Areas

Source water protection areas (SWPAs) are defined as designated surface and subsurface zones surrounding public water supply intakes to prevent contaminants from entering the surface water drainage area and compromising the quality of public drinking water. Three SWPAs are located within 1 mile of the Project area, of which one, the Northern KY Water District –

Newport Plant SWPA, is crossed by Line AM36, Line AM04 and Line AM09 workspaces. In addition, two surface water intakes in the Ohio River, one associated with the Northern KY Water District – Newport Plant SWPA and the other associated with the Northern KY Water District, are located 1 mile and 2 miles (respectively) downstream of the Project's Ohio River crossing (KEEC, 2025c).

To avoid and reduce potential adverse effects to the Northern KY Water District – Newport Plant SWPA, Columbia would implement its Blasting Plan, SPCC, and Plan for the Unanticipated Discovery of Contaminated Environmental Media, and maintain compliance with applicable federal, state, and local permits and regulations specific to waterbody crossings. If blasting is required, Columbia would consult the local water authorities that may be affected prior to blasting. The HDD of the Ohio River would substantially reduce potential effects on the two downstream water intakes. Additionally, and as described previously, Columbia would implement measures to avoid, reduce, and mitigate effects associated with an inadvertent return of drilling fluids with the Ohio River. Therefore, based on Columbia's implementation measures, and distance from the Project to surface water intakes, we conclude the Project would not result in significant effects on SWPAs or surface water intakes for public supply.

Water Use

Columbia would require about eight million gallons of water for hydrostatic testing of the new pipeline and aboveground facilities. Water would be sourced from clean municipal water sources, held in tanks, and reused until all tests are complete. After depressurizing a test section, used test water would be discharged through a filtration/energy-dissipation structure[41] into a well-vegetated upland area. Columbia would discharge water according to its Plan and Procedures, as well as applicable federal and state regulations and permit conditions.

Columbia anticipates using a total of 5,000 gallons per day during construction of the Project to control fugitive dust emissions and an estimated 727,000 gallons of water for HDD drilling mud.[42] All water utilized for dust control would be acquired from municipal water sources. Columbia would only apply water for dust control, when necessary, thereby reducing its demand for water. Therefore, we conclude that the Project would not have significant effects on water use and water quality would not be significantly affected by these releases. Fugitive dust control is discussed further in section B.8.

B.5 FISHERIES, VEGETATION, AND WILDLIFE

Fisheries

The Project would cross waterbodies supporting commonly occurring warm water fish populations and aquatic invertebrates. Waterbodies and fisheries classifications are summarized in Appendix B. Additionally, the Ohio River supports commercial fisheries. Columbia would largely avoid effects to commercial fisheries by using an HDD to cross the river.

[41] Columbia's November 10, 2025, Application, Resource Report 1, appendix 1B. FERC eLibrary Accession No. 20251110-5151.
[42] Columbia's November 10, 2025, Application, Resource Report 2, section 2.3.5. FERC eLibrary Accession No. 20251110-5151.

The Project would temporarily affect fish inhabiting crossed waterbodies. Fish and other aquatic species may experience increased rates of stress, injury, and mortality due to flow disturbances and temporary increases in sedimentation and turbidity. The removal of vegetative cover could increase water temperatures; however, as fish are highly mobile, this effect should be minor. As described previously, Columbia would complete stream crossings within 24 to 48 hours and would implement measures and devices to reduce erosion, turbidity, and sedimentation in waterbodies. Following the completion of a waterbody crossing, Columbia would rehabilitate and stabilize streambeds and banks. Based on the characteristics of the fish populations known to occur in the waterbodies to be crossed and the measures Columbia would implement to reduce effects on these populations, we conclude that effects to fisheries resources and aquatic habitats would not be significant.

Vegetation

The Project area consists of maintained grasses and shrubs (utility rights-of-way and developed lands); unmanaged grasses, wildflowers, and shrubs (open lands/natural areas); deciduous woodlands (forested); cultivated crops and pasture (agricultural vegetation), and wetland vegetation. Of the vegetation affected by the Project, about 45 percent would be deciduous woodlands and 40 percent would be maintained grasses and shrubs. The remaining 15 percent of vegetation affected consists of unmanaged grasses, wildflowers, and shrubs (open lands/natural areas); cultivated crops and pasture (agricultural), and wetland vegetation. A summary of vegetation/habitat types affected by each facility is provided in Appendix C.

Constructing, operating, and abandoning the proposed facilities would temporarily and permanently affect vegetation. Vegetation would be temporarily cleared resulting in increased rates of erosion and loss of wildlife habitat. Unless otherwise requested by a landowner (for areas located outside of the existing aboveground facilities and ROWs), temporary workspace would be revegetated in accordance with the Columbia's Plan and Procedures using a seed mix derived from consultations with the local NRCS Field Service Centers. About 70.1 acres of deciduous woodlands within the permanent operational easements would be permanently converted to maintained grasses and shrubs (utility rights-of-way and developed lands). All other deciduous woodlands would be allowed to revegetate following completion of the Project. Because woodlands can take up to 30 years to recover, the resulting effects can be considered long-term. Additionally, the temporary clearing, long-term regrowth, and permanent vegetation conversions can result in the introduction and spread of pest/nuisance, exotic, and invasive plant species.

To reduce and mitigate effects on vegetation, Columbia would implement several measures as identified in its Plan and Procedures and other Project-specific plans. Specifically, Columbia would implement erosion control devices, segregate soils in agricultural areas, manage non-native or invasive plant species at Project aboveground facilities, monitor revegetation success, and address any revegetation related issues.[43] Given the steps Columbia would take to reduce and mitigate effects to vegetation, we conclude effects to vegetation would not be significant.

[43] Columbia's November 10, 2025, Application, Resource Report 3, section 3.3.4. FERC eLibrary Accession No. 20251110-5151.

Wildlife

A variety of commonly occurring mammals, birds, reptiles, and amphibians occupy the Project area. Species include white-tailed deer, eastern cottontail, fox squirrel, gray squirrel, gray fox, red fox, mink, muskrat, raccoon, striped skunk, opossum, white-tailed deer, white-footed mouse, deer mouse, least shrew, prairie vole, mourning dove, brown-headed cowbird, dickcissel, eastern kingbird, loggerhead shrike, northern mockingbird, black racer, black rat snake, common garter snake, common snapping turtle, red-eared slider, American bullfrog, American toad, green frog, and Jefferson salamander (USDA, 1996).

Equipment activity and noise, human presence, ground disturbance, and other Project-related activities would result in the temporary and permanent loss of wildlife habitat and likely result in wildlife avoiding the Project area. Proposed nighttime construction would also affect bats and other nocturnal wildlife in the surrounding areas, disrupting their foraging/feeding and other normal behaviors. In general, available wildlife habitat would be temporarily and permanently lost, and wildlife would experience increased rates of stress, injury, and mortality. Less mobile wildlife would especially experience increased mortality. Additionally, wildlife avoiding project activities would likely relocate to other similar nearby habitats which could increase predation and crowding. However, these effects would be temporary as wildlife would likely return to affected lands following completion of the Project.

To reduce effects on wildlife resulting from their entrapment in open pipeline trenches, Columbia would conduct daily inspections of open trenches and arrange for the safe relocation of wildlife if encountered in the trench or workspaces. Additionally, Columbia would reduce nighttime effects by using handheld equipment/tools and pipe lifting equipment; limiting night-time lighting by directing lighting fixtures only towards active workspace, and restricting equipment deliveries to daytime operations.

Several comments received during scoping expressed concerns that the removal of mature trees and vegetation would have adverse effects on wildlife habitat available to threatened and endangered species, migratory birds, and wild turkeys that use the trees for roosting. Further, commenters noted that tree removal could lead to accelerated growth of invasive Amur bush honeysuckle which inhibits the growth of trees. As discussed, wildlife could experience adverse effects resulting from the proposed activities. We note that Columbia would collocate a majority of the new pipeline with existing rights-of-ways and utility corridors where wildlife is generally accustomed to human disturbance; thereby reducing effects on wildlife.

No protected or sensitive wildlife areas were identified within 0.25 mile of the Project alignment. However, the Project is in the vicinity of a state Wildlife Management Area (WMA), the Clay WMA, located 0.55 mile northeast of Line EM-7 (Carlisle Meter Station) (KDFWR, 2025). Project construction activities that produce noise and light may temporarily disturb wildlife in the Clay WMA but are not anticipated to continue once construction concludes. Given the distance of the Project from the Clay WMA, we conclude any indirect effects on wildlife in the Clay WMA that may result from construction of the Project would be short-term and not significant.

Given the developed and disturbed nature of the Project area, wildlife's habituation to the existing pipelines and associated maintenance activities, and the effects to wildlife habitats and wildlife resulting from construction and abandonment activities as described above, we conclude that effects on wildlife would be minor and insignificant.

Migratory Birds

Migratory birds are protected under the Migratory Bird Treaty Act (16 U.S. Code [U.S.C.] 703-711); bald and golden eagles are additionally protected under the Bald and Golden Eagle Protection Act (16 U.S.C. 668-668d). Executive Order 13186 (66 Federal Register 3853) directs federal agencies to identify where unintentional take is likely to have a measurable negative effect on migratory bird populations and to avoid or minimize adverse effects on migratory birds through enhanced collaboration with the FWS. On March 30, 2011, the FWS and the Commission entered a Memorandum of Understanding that focuses on avoiding, minimizing, or mitigating adverse effects on migratory birds and strengthening migratory bird conservation through enhanced collaboration between the two agencies.

The FWS's Information for Planning and Consultation (IPaC) database identified 16 Birds of Conservation Concern (BCC) that may potentially occur in the Project area. The Project's effect on migratory birds would be similar to that of general wildlife and may result in the temporary or permanent displacement of individuals. Loss of wetland and wooded habitat and nighttime lighting could adversely affect migratory birds. Vegetation clearing activities during bird nesting season could result in nest abandonment from increased human presence and noise or direct mortality of nesting birds or nestlings and fledglings. Loss of habitat would force migratory birds to seek refuge elsewhere – increasing competition for mates, nest site selection, and food resources. Nighttime lighting would contribute to light pollution that could alter the migratory behavior of birds by drawing them towards developed landscapes and away from high-quality stopover habitat. Excess time spent searching for stopover sites and selection of low-quality stopover habitat can reduce fitness by limiting food resources, expending extra energy, and increasing the risk of anthropogenic-related mortality (McLaren, et al. 2018).

Columbia has indicated that any clearing required for the Project would be done in accordance with the Migratory Bird Treaty Act and consultation with FWS. In addition, revegetation of all temporarily disturbed areas would promote re-establishment of lost migratory bird nesting and foraging habitat. Given this, and Columbia's nighttime minimization measures outlined above, we conclude that the Project would not have population-level effects on migratory birds or significant measurable negative effects on any migratory bird species or their habitat.

Eagles

Based on the KDFWR's and ODNR's occurrence data, no nesting bald eagles have been identified within the Project area. Additionally, no bald eagles or bald eagle nests were observed during Columbia's field surveys conducted from May 2024 through June 2024, in September 2024, and in April 2025. If a bald eagle nest is found near the Project prior to or during

construction, Columbia would coordinate with KDFWR, ODNR, and the FWS and adhere to the conservation measures in FWS's 2007 *National Bald Eagle Management Guidelines.*[44]

Federally Listed Threatened and Endangered Species

Federal agencies are required under Section 7 of the Endangered Species Act to ensure that any actions authorized, funded, or carried out by the agency would not jeopardize the continued existence of a federally listed endangered or threatened species or result in the destruction or adverse modification of the designated critical habitat of a federally listed species. As the lead federal agency authorizing the Project, FERC is required to consult with the FWS to determine whether federally listed endangered or threatened species or designated critical habitat are found in the vicinity of the Project, and to evaluate the proposed action's potential effects on those species and/or critical habitats.

Using the FWS's IPaC database, Columbia identified eight federally listed species occurring or potentially occurring in the Project area. These species are the: gray bat (*Myotis grisescens*, endangered), Indiana bat (*Myotis sodalis*, endangered), northern long-eared bat (*Myotis septentrionalis*, endangered), fanshell (*Cyprogenia stegaria*, endangered), pink mucket (*Lampsilis abrupta*, endangered), sheepnose mussel (*Plethobasus cyphyus*, endangered), snuffbox mussel (*Epioblasma triquetra*, endangered), and Short's goldenrod (*Solidago shortii*, endangered). Four other species proposed for listing: tricolored bat (*Perimyotis subflavus*, proposed endangered), eastern hellbender (*Cryptobranchus alleganiensis alleganiensis*, proposed endangered), salamander mussel (*Simpsonaias ambigua*, proposed endangered), and monarch butterfly (*Danaus plexippus*, proposed threatened) could also potentially occur within the Project area.

Columbia's Multi-Species Habitat Conservation Plan (MSHCP) which was developed through coordination with the FWS for evaluation of federally listed species in Columbia's service areas also the orange pimpleback (*Lethobasus cooperianus*), ring pink (*Obovaria retusa*), and rough pigtoe (*Pleurobema plenum*) as occurring or potentially occurring in the Project area.

Gray Bat, Indiana Bat, Northern Long-eared Bat, and Tricolored Bat

The gray bat inhabits caves year-round and therefore does not have suitable habitat in the Project area, as there are no hibernacula on record or caves in the Project area. Therefore, we have determined the Project will have *no effect* on the species. The Project area has suitable habitat for the Indiana bat, the northern long-eared bat, and the tricolored bat due to the presence of summer roosting habitat. Tree removal is planned when most bats are hibernating (between November 16 and March 31), and there are no suitable winter hibernacula within the Project area. Should any clearing activities occur outside of this period, effects would be mitigated through an increased contribution to the Imperiled Bat Conservation Fund for forested habitat clearing in Kentucky in accordance with the Intra-Service Biological Order. Due to the presence of adjacent suitable habitat and timing of tree removal, as currently planned, we have determined that the Project *may affect, but is not likely to adversely affect*, the Indiana bat and the northern long-eared bat, and *would not likely jeopardize the continued existence* of the tricolored bat.

[44] FWS's 2007 National Bald Eagle Management Guidelines can be found at https://www.fws.gov/media/national-bald-eagle-management-guidelines.

Fanshell, Orange Pimpleback, Pink Mucket, Ring Pink, Rough Toe, Sheepnose Mussel, Snuffbox Mussel, and Salamander Mussel

The orange pimpleback, the ring pink, and the rough toe were only identified by the MSHCP and not the IPaC for the portion of the Project area located in Bracken and Pendleton counties, Kentucky. Suitable habitat for these species is not present within the Project area in Bracken and Pendleton counties, Kentucky. Therefore, we have determined that the Project would result in *no effect* on these species. Suitable habitat within the Project area may exist for the fanshell, the pink mucket, the sheepnose mussel, the snuffbox mussel, and the salamander mussel in the Ohio River and Twelvemile Creek. The Ohio River would be crossed via HDD, thereby substantially reducing potential effects to mussel species that may be present in that waterbody. Columbia would conduct additional mussel surveys in May/June 2026 to determine presence of federally listed mussel species in Twelvemile Creek. If mussel species are found during the surveys, Columbia would consult with the FWS to identify any recommended measures to avoid and minimize adverse effects to the species. Given Columbia's commitment to work with the FWS, we have determined that the Project *may affect, but is not likely to adversely affect* the fanshell, the pink mucket, sheepnose mussel, snuffbox mussel, and *would not likely jeopardize the continued existence* of the salamander mussel if present in Twelvemile Creek.

Short's Goldenrod

Suitable habitat for the Short's goldenrod within the Project area may exist; however, surveys conducted during the flowering season for this plant species did not identify any individuals within the Project area. Therefore, we have determined the Project *may affect, but is not likely to adversely affect,* the Short's goldenrod.

Eastern Hellbender

Suitable habitat for the eastern hellbender is present within perennial waters in the Project area. This species only appears in the IPaC review for the portion of the Project area located in Mason County, Kentucky. The Project facilities in Mason County, Kentucky include only a contractor/staging/pipe yard with no suitable habitat for the eastern hellbender. Therefore, we conclude the Project is *not likely to jeopardize the continued existence* of this species.

Monarch Butterfly

The monarch butterfly has the potential to be present within the Project area. Given that surveys conducted during the flowering season for this plant species did not identify milkweed, a plant necessary to the monarch's life cycle, and given an abundance of similar habitat is located adjacent to the Project area, we have concluded the Project is *not likely to jeopardize the continued existence* of this species.

Columbia has stated that it continues to consult with the FWS and given that we have determined that they Project *may affect* federally listed threatened and endangered species, concurrence with our determinations from the FWS is required. Therefore, **we recommend that any Commission Order approving the Project include a mandatory condition requiring that Columbia not begin Project construction until ESA consultation is complete** (see section D, recommended condition no. 12).

State Listed Species

Columbia identified twenty-three Kentucky and Ohio state listed species as potentially occurring within the counties affected by the Project. These species include alligator gar (*Atractosteus spatula*), barn owl (*Tyto alba*), buffalo clover (*Trifolium reflexum*), closed gentian (*Gentiana andrewsii* var. *andrewsii*), fanshell, lake sturgeon (*Acipenser fulvescens*), little brown bat (*Myotis lucifugus*), longsolid (*Fusconaia subrotunda*), northern long-eared bat, northern metalmark (*Calephelis borealis*), peregrine falcon (*Falco peregrinus*), pink mucket, pocketbook (*Lampsilis ovata*), running buffalo clover (*Trifolium stoloniferum*), sheepnose mussel, Short's goldenrod, six-banded longhorn beetle (*Dryobius sexnotatus*), spotted sandpiper (*Actitis macularius*), stemless evening-primrose (*Oenothera triloba*), timid sedge (*Carex timida*), tricolored bat, vesper sparrow (*Pooecetes gramineus*), and white walnut (*Juglans cinerea*). The fanshell, the northern long-eared bat, the pink mucket, the sheepnose mussel, the Short's goldenrod, and the tricolored bat were discussed above. The remaining state assessed species are discussed below. Columbia is expected to complete consultation with KDFWR and provide updates prior to commencing construction.

Alligator Gar and Lake Sturgeon

The alligator gar and the lake sturgeon were only identified by the ODNR for the portion of the Project area located in Hamilton County, Ohio. Suitable habitat for these species is not present within the Project area; therefore, effects on these species are unlikely.

Barn Owl, Peregrine Falcon, Spotted Sandpiper, and Vesper Sparrow

The barn owl was only identified by the ODNR in Hamilton County, Ohio. Suitable habitat for this species is not present within the Project area; therefore, effects on the barn owl are unlikely. Suitable foraging and nesting habitat is present in the Project area in the vicinity of the Ohio River for the peregrine falcon and the spotted sandpiper, as well as migratory habitat for the spotted sandpiper. Given that the Ohio River would be crossed via HDD, adverse effects to the peregrine falcon and the spotted sandpiper are unlikely. Suitable habitat is present in the Project area for the vesper sparrow due to its generalist habitat preferences. Tree removal would be conducted outside of the primary bird nesting season (between April 15 and August 1), or Columbia would conduct nest surveys no more than two weeks prior to clearing activities in areas where clearing must occur within the season. Due to the presence of adjacent suitable habitat and timing of tree removal, adverse effects to the vesper sparrow are unlikely.

Buffalo Clover, Closed Gentian, Running Buffalo Clover, Stemless Evening-primrose, Timid Sedge, and White Walnut

The buffalo clover was only identified by the ODNR for the portion of the Project area located in Hamilton County, Ohio. Suitable habitat for this species is not present within the Project area; therefore, effects on the buffalo clover are unlikely. Suitable habitat is present in the Project area along delineated forested wetlands abutting the Ohio River for the closed gentian and in forested areas along the Ohio River for the white walnut. Given that the Ohio River would be crossed via HDD, effects on the closed gentian and the white walnut are unlikely. Suitable habitat in the Project area for running buffalo clover, stemless evening-primrose and timid sedge may exist; however, surveys conducted during the flowering season for these plant species did not identify any individuals within the Project area. Therefore, adverse effects on the running buffalo clover, the stemless evening-primrose, and the timid sedge are unlikely.

Little Brown Bat

The Project area has suitable habitat for the little brown bat due to the presence of summer roosting habitat. Tree removal is planned to occur when most bats are hibernating (between November 16 and March 31) and there are no suitable winter hibernacula within the Project area. Due to presence of adjacent suitable habitat and planned timing of tree removal, adverse effects to the little brown bat are unlikely.

Longsolid and Pocketbook

Suitable habitat within the Project area may exist for the longsolid and the pocketbook in the Ohio River and Twelvemile Creek. Given that the Ohio River would be crossed via HDD and additional Phase 1 surveys would be conducted based on recommendations from the FWS to determine presence of federally listed mussel species in Twelvemile Creek, adverse effects to the longsolid and the pocketbook are unlikely.

Northern Metalmark and Six-banded Longhorn Beetle

The northern metalmark has the potential to be present within the Project area. Given that surveys conducted during the flowering season for this plant species identified one lone roundleaf ragwort, a plant necessary to the metalmark's life cycle, and given an abundance of similar habitat is located adjacent to the Project area, adverse effects on the northern metalmark are unlikely. The six-banded longhorn beetle was only identified for the portion of the Project area located in Hamilton County, Ohio. Suitable habitat for this species is not present within the Project area; therefore, effects on the six-banded longhorn beetle are unlikely.

B.6 CULTURAL RESOURCES

In addition to accounting for effects on cultural resources under NEPA, Section 106 of the National Historic Preservation Act (NHPA), as amended, requires FERC to consider the effects of its undertakings on historic properties listed, or eligible for listing on the National Register of Historic Places (NRHP), and to afford the Advisory Council on Historic Preservation an opportunity to comment. Columbia, as a non-federal party, is assisting us in meeting our obligations under Section 106 and the implementing regulations at 36 CFR 800. The Section

106 process is coordinated at the state level by the State Historic Preservation Office. In Kentucky, the Kentucky Heritage Council serves as the State Historic Preservation Office (KYSHPO). In Ohio, the Ohio Historic Preservation Office (OSHPO) serves that role.

The Area of Potential Effects

The area of potential effects (APE) is the "geographic area or areas within which an undertaking may directly or indirectly cause alterations in the character or use of historic properties, if any such properties exist" (36 CFR 800.16(d)). The direct APE for archaeological sites includes all areas of potential effects where ground-disturbing activities are possible, while the indirect APE is considered to be the geographic areas from which any permanent infrastructure has the potential to affect, diminish or alter the visual, auditory, vibratory, or atmospheric setting of a NRHP-listed or NRHP-eligible property. The APE for direct effects totals 572.8 acres, and comprises the pipeline corridors, the total footprint for above-ground facilities, access road corridors, temporary workspace, and contractor/staging/pipe-yards. The APE for indirect effects is considered to be the geographic areas from which any permanent infrastructure has the potential to visually diminish or alter the setting of an NRHP listed or NRHP-eligible property (36 CFR § 800.16[d]). The Project would involve modifications to existing above-ground facilities, or construction of new facilities with a low visible range across the landscape. As such, an assessment of indirect effects was deemed not warranted. The APE is sufficient to account for all potential effects to historic properties by the proposed Project.

Cultural Resources Investigations

In its effort to identify historic properties in the Project APE and to account for any effects to those properties by the proposed Project, Columbia conducted cultural resources investigations prior to fieldwork, which included background research examining databases maintained by the SHPO listing recorded archaeological resources, previously conducted archaeological surveys, and historic structures. The background review revealed that a total of four previously recorded archeological sites and eight architectural resources structures are mapped within the environmental survey area. The field investigations identified a total of 15 archeological sites and 35 architectural resources within the APE. Columbia has determined these resources do not possess sufficient integrity, nor the potential to yield information important in prehistory or history. Therefore, Columbia has recommended them as not eligible for listing in the NRHP.

In a letter dated November 13, 2025, the OSHPO concurred that there would be no historic properties affected within the portions of Project APE in Ohio. We agree.

We received comments from Phillip W. McCartney and Elizabeth McMillin-McCartney identifying the presence of a historic rock wall on their property and within the Kentucky APE, as well as a possible burial mound, attributed to the Adena culture. In an email dated December 5, 2025, the KYSHPO requested revisions to Columbia's Phase I survey report and notified Columbia of a possible undocumented resource, a portion of a rock fence. Columbia conducted additional intensive investigations to document these resources. The cultural resources investigation included a review of existing records, consultation with appropriate agencies, and a field survey of an additional 0.71 acre of the project area. No archaeological sites, historic

structures, or other cultural resources eligible for listing in the National Register of Historic Places were identified within the APE. Therefore, the proposed project is not anticipated to have any adverse effects on cultural resources. Should any previously unknown cultural materials be discovered during project activities, work would be halted in the vicinity of the find and the appropriate authorities would be notified in accordance with applicable laws and regulations.

On March 2026, Columbia submitted an addendum report to the KYSHPO. In a letter dated March 24, 2026, the Kentucky SHPO concurred with Columbia's finding that *no adverse effect* to cultural resources would result from the Project. We agree.

Tribal Outreach

Columbia contacted the following Native American tribes, providing a Project description and mapping: Cherokee Nation, Delaware Nation, Eastern-Band of Cherokee Indians, Miami Tribe of Oklahoma, Osage Nation, Santee Sioux-Nation Nebraska, Seneca-Cayuga Nation, Absentee-Shawnee Tribe of Indians of Oklahoma, Eastern Shawnee Tribe of Oklahoma, and Shawnee Tribe. In an email dated November 17, 2025, the Delaware Nation indicated there are no known cultural resources or historic properties associated with the tribe within the proposed Project area, and therefore, will not actively participate in consultation. No additional comments have been received to date.

Unanticipated Discovery Plan

Columbia developed a Project-specific plan titled *Plan for the Unanticipated Discovery of Cultural Resources and Human Remains* (Unanticipated Discovery Plan), which outlines the procedures to follow, in accordance with state and federal laws, in the event that unanticipated cultural resources or human remains are discovered during construction of the Project, including consultation with FERC, the SHPO, and tribes regarding discoveries. The Unanticipated Discovery Plan was submitted to FERC, the KYSHPO, and the OSHPO. We find the Unanticipated Discovery Plan acceptable.

Compliance with the National Historic Preservation Act

FERC has completed its compliance requirements with Section 106 for the Project.

B.7 LAND USE, RECREATION, AND VISUAL RESOURCES

Land Use

As described previously, constructing and abandoning the proposed facilities would require the use of about 565.7 acres of land, including the permanent use of about 125.7 acres of land to operate the Project, which consists of the new permanent pipeline easements, permanent access roads, and new and expanded aboveground facilities.[45] Construction and abandonment activities would temporarily preclude existing land uses. Following construction, affected lands,

[45] Columbia's November 10, 2025, Application, Resource Report 8, section 8.2, FERC eLibrary Accession No. 20251110-5151.

except for those permanently converted to industrial land (9.1 acres),[46] would be able to resume pre-construction uses. The land within the Project area can be generally characterized as deciduous woodland (forested lands), open land, agricultural land, industrial land, wetlands, open water, and residential land.

Forested lands are comprised of deciduous trees. Agricultural production in the area consists primarily of row crops and some small areas of improved pasture utilized for livestock grazing. Industrial lands encompass most developed lands that are not characterized as residential and include existing roads, utilities, railways, commercial business, industrial, and other highly developed areas. Residential lands are developed lands that include both single and multiple family dwellings and may be in developed subdivisions as well as rural areas. Wetlands are comprised of palustrine emergent, scrub shrub, and forested wetlands. Open water includes waterbodies with crossings greater than 100 feet in width and ponds. See Appendix D for a summary of land uses within the Project area.

The Commission received numerous comments concerning the Project's effects on land value and future use. Specifically, commenters expressed concerns that permanent pipeline easements would lower home property values and adversely affect future development. The effect that a pipeline easement may have on property value and future use/development is a damage-related issue that would be negotiated between the parties during the easement acquisition process. The easement acquisition process is designed to provide fair compensation to the landowner for the right to use the property for pipeline construction and operation. Appraisal methods used to value land are typically based on the characteristics of the property and any improvements. The effect a pipeline could have on a property's value would depend on many factors including the size of the tract, the values of adjacent properties, the presence of other utilities, the current value of the land, and the current land use. Therefore, due to the many factors involved in the value assessment of a property and the presence of and the proximity to existing pipeline infrastructure, we conclude that significant negative effects on property values are not anticipated.

Forest

A total of 253.4 acres[47] of forested land would be affected by the Project. We received comments regarding canopy loss and clearing. Maintenance of the new operational right-of-way for the proposed Project pipelines (70.1 acres) would preclude the reestablishment of trees and shrubs following construction, thereby permanently converting forest to open land. In addition, a total of 2.4 acres of forest would be converted to industrial land to allow permanent operation of the new and expanded aboveground facilities and for permanent access roads.[48] Temporary workspace areas that would be cleared for construction would result in long-term effects due to the time required for trees to become reestablished. Columbia would facilitate this reestablishment via guidelines outlined in the FERC Plan.

[46] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request, Appendix E, FERC eLibrary Accession No. 20260429-5302.
[47] Columbia's November 10, 2025, Application, Resource Report 8, section 8.2. FERC eLibrary Accession No. 20251110-5151.
[48] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request, Appendix E, FERC eLibrary Accession No. 20260429-5302.

Open Land

A total of 225.9 acres of open land would be affected by the Project. Operation of the proposed Project pipelines would not result in a change of land use from open land, as the open land located within the new operational right-of-way would be restored and allowed to return to pre-construction uses. A total of 3.45 acres of open land would be permanently converted to industrial land for the operation of the new and expanded aboveground facilities, as well as for permanent access roads.[49] Following the completion of construction activities, open land within all other temporary workspaces would be reseeded in accordance with the FERC Plan, as well as guidance specific to the Project area from the NRCS Field Service Center and Columbia's Project-specific Revegetation Plan.[50]

Agricultural Land

A total of 36.6 acres[51] of agricultural land would be affected by the Project. All agricultural land affected by the Project would be restored to its original use, including the permanent pipeline rights-of-way. Columbia would implement the FERC Plan to minimize adverse effects on agricultural land. Columbia would work with landowners in these areas to ensure that proper restoration of any affected agricultural area occurs, including replacement of segregated topsoil, stone removal, and compliance with reseeding specifications. Within agricultural land affected by the Project, Columbia would negotiate with and compensate landowners for any damages and/or temporary or permanent loss of production as a result of the Project. Additionally, Project construction could disrupt and/or could damage existing drain tiles and irrigation systems. In its application, Columbia stated that drain tiles were identified within the construction workspace. If irrigation systems or drain tiles are damaged by Project-related activities, Columbia would work with landowners to repair or replace those damaged sections.[52]

An important element of right-of-way restoration in active agricultural areas is timely replanting of crops or other cover vegetation. Resumption of agricultural operations following Project construction and/or planting of a cover crop would aid in the restoration of soil structure and productivity that could take several years to achieve success, depending on site-specific conditions and land use practices. FERC staff would monitor restoration efforts and require corrective action where necessary to ensure compliance with the Commission's orders. Per the FERC Plan, revegetation of agricultural areas would be considered successful when crop growth and vigor, based on visual inspection, are similar to adjacent undisturbed portions of the same field, unless the easement agreement specifies otherwise. FERC also requires the size, density, and distribution of rock remaining on the construction area following restoration should be the same as adjacent areas not disturbed by construction.

[49] Columbia's November 10, 2025, Application, Resource Report 8, section 8.2. FERC eLibrary Accession No. 20251110-5151.
[50] Columbia's November 10, 2025 Application, Resource Report 1, appendix 1D. FERC eLibrary Accession No. 20251110-5151.
[51] Columbia's November 10, 2025, Application, Resource Report 8, section 8.2. FERC eLibrary Accession No. 20251110-5151.
[52] Note the Commission does not adjudicate disputes regarding compensation for damages.

Industrial

Industrial land accounts for 26.7 acres of the Project area. Of that amount, a total of 2.7 acres of industrial land would be permanently affected for operation of new and expanded aboveground facilities and permanent access roads; about 2.0 acres of industrial land would be permanently maintained as operational right-of-way for new pipeline facilities.[53]

Wetlands

Wetlands account for 1.1 acres of the Project area. Of that amount, 0.50 acre of wetlands would be maintained as new operational right-of-way, but only a 10-foot corridor centered on the pipeline(s) would be maintained in an herbaceous state in accordance with the FERC Procedures.[54] Trees located within 15 feet of the pipeline that have roots that could threaten the structural integrity of the pipelines would be selectively removed. In areas crossed by HDD, the permanent right-of-way would not be maintained in order to avoid and minimize effects on wetlands.

Residential Land

Residential land accounts for approximately 20.2 acres of the total Project area. Of that amount, 2.3 acres would be maintained as new operational right-of-way, and 3.3 acres of permanent effects would be associated with new and existing aboveground facilities and permanent access roads.[55] We received comments regarding residential easements and alternatives to reduce residential disturbance. Columbia committed to reducing the overall footprint on the landowners' properties, including the amount of tree clearing that would be required.[56] Also, as described in Columbia's application, construction activities in residential areas would be completed as quickly and safely as practicable to minimize disturbances to residents. Columbia would make all reasonable efforts to maintain access to the residences during construction; however, if access is temporarily impeded, Columbia would coordinate with landowners to minimize the disturbance. Additional mitigation measures to reduce residential land effects are discussed below.

Open Water

Open water accounts for 1.8 acres[57], or less than 1 percent, of the Project area. That amount is entirely associated with the permanent right-of-way for Line AM36 at the Ohio River crossing (MP 24.21)[58]. The Ohio River would be avoided via HDD during Project construction and operation and would not result in a change of land use designation from open water.

[53] Columbia's November 10, 2025, Application, Resource Report 8, section 8.2. FERC eLibrary Accession No. 20251110-5151.
[54] *Id.*
[55] *Id.*
[56] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request, Appendix E. FERC eLibrary Accession No. 20260429-5302.
[57] Columbia's November 10, 2025, Application, Resource Report 8, section 8.2. FERC eLibrary Accession No. 20251110-5151.
[58] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request, Appendix E. FERC eLibrary Accession No. 20260429-5302.

Columbia would implement BMPs and would adhere to the FERC Procedures and its Ohio River HDD Plan to minimize effects on open water resources.

Road and Utility Crossings

The Project would cross 55 roads, 2 railroads, and 167 utilities comprising overhead telephone, transmission, and/or power lines, buried water lines, buried gas pipelines, and buried sewage lines.[59] Roads include county roads, paved private roads, state highways, and maintained unpaved private roads. Paved roads would be crossed via open-cut, subsurface bores, or the HDD method.

We received comments regarding property access and maintenance effects on the road used for project equipment and traffic. Columbia committed to coordinating with the jurisdictional authorities regarding road use and traffic during construction and would obtain all necessary permits. In addition, Columbia's proposed mitigation measures to minimize effects on traffic and roads used for the Project as part of their Project Traffic Management Plan.[60] We also received comments regarding project interference with county sewer lines. Columbia would coordinate with local authorities during construction and obtain all necessary permits.[61] Columbia would maintain safe and accessible conditions at road crossings during construction activities. The surrounding highways would be used to transport Project materials to the site. Columbia would implement mitigation measures outlined in the Traffic Management Plan[62] to minimize project traffic effects. Columbia would install construction entrances at public paved roads that intersect with the construction corridor at access roads. Trenches in proximity to public road crossings would be fenced or covered with steel plates during all non-working hours to control livestock and limit public access. Temporary and permanent erosion control devices would be installed adjacent to paved roads, as needed, to prevent construction corridor sediment from being washed onto roads during rain events. Traffic control measures would be used as necessary to ensure the safety of local traffic. Minor, temporary traffic effects could occur during construction activities at roadway crossings; however, no permanent effects on existing roadways or utilities crossed would occur as a result of the Project.

Residential Areas

Columbia provided a list of all structures within 100 feet of the Project, along with the milepost location, structure type, and approximate distance from the Project. Of the 473 structures identified, 155 structures are located within 25 feet of Project workspaces, including residences, commercial buildings, sheds, barns, and other unoccupied structures.[63] Columbia provided site-specific Residential Construction Plans for the residences located within 25 feet of Project workspaces. These plans are provided in Appendix E. We have reviewed the site-specific residential plan drawings and have found them acceptable. We encourage potentially

[59] Columbia's November 10, 2025 Application, Resource Report 8, Appendix 8E & section 8.2.1. FERC eLibrary Accession No. 20251110-5151.
[60] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request, attachment 1-12. FERC eLibrary Accession No. 20260429-5302.
[61] *Id*.
[62] *Id*. at attachment 5-27.
[63] Columbia's November 10, 2025 Application, Resource Report 8, appendix 8G. FERC eLibrary Accession No. 20251110-5151.

affected residents located within 25 feet of Project workspaces to review and comment on these plans.

Project activities would result in short-term effects on residential areas, including removal of existing vegetation and landscaping from the construction workspace, increased construction related traffic on local roads, as well as dust and noise generated during construction. For residences located within 25 feet of the Project area, Columbia would implement mitigation measures outlined in the Residential Construction Implementation Plans[64] during construction to reduce effects on residences.

After construction, residential lands would be restored to pre-construction conditions to the extent practicable, or as agreed upon with the landowner during right-of-way easement negotiations. If any damage to residential property results from construction, Columbia would be responsible for repairs to the damaged property, in accordance with landowner easements. Most uses of the land would be able to continue in accordance with easement agreements. However, new buildings, sheds, wells, reservoirs, pools, large trees, obstructions, or structures would not be permitted within permanent pipeline easements. Construction of features such as lawns, roads or driveways, utility lines, and properly gated fences are generally permissible uses within the permanent easements.

Based on Columbia's implementation of minimization measures to avoid and reduce effects in residential areas, we have determined that while Project construction would temporarily disrupt residential activities, these effects would not be significant.

Public Land, Recreation, and Visual Resources

Public Land

No national or state parks or forests, monuments, preserves, historical parks, memorials, battlefields, military parks, cemeteries, recreation areas, seashores, lakeshores, rivers, parkways, conservation lands, federal Indian reservations, National Wildlife Refuges, National Wilderness Areas, or designated Natural Landmarks are within 0.25 mile of the Project.[65]

The Lewis and Clark National Historic Trail is a nearly 4,900-mile national historic trail that commemorates the Lewis and Clark Expedition of 1803 to 1806, extending from the Ohio River in Pittsburgh, Pennsylvania to the Pacific Ocean. The National Park Service administers the Lewis and Clark National Historic Trail as a component of the National Trails System through collaboration with state and local partners and communities but does not directly own any land associated with the trail. The Project directly crosses the Lewis and Clark National Historic, as the historic trail directly follows the Ohio River; this section of the trail, however, only follows the Ohio River without specific roads or paths specified for travelers to follow.[66] As the portion of the Project intersecting with the Ohio River would be crossed using a HDD, the

[64] Columbia's November 10, 2025, Application, Resource Report 1, Appendix 1B. FERC eLibrary Accession No. 20251110-5151.
[65] Columbia's November 10, 2025, Application, Resource Report 8, section 8.4.1. FERC eLibrary Accession No. 20251110-5151.
[66] *Id*.

Ohio River would not be affected by Project activities. Visual effects associated with construction and operation of the nearby Kellogg Station Launcher/Receiver are not anticipated to be significant, as discussed further below.

Recreation

No natural, recreational, or scenic areas including the Nationwide Rivers Inventory, National Wild and Scenic Rivers System, the National Trails System, wilderness areas designated under the Wilderness Act, or natural areas designated by state agencies are within 0.25 mile of the Project with the exception of the Lewis and Clark National Historic Trail, which is a component of the National Trails System, as discussed above.[67]

The Ohio River Scenic Byway is a 943-mile driving route designated under the National Scenic Byways Program that follows the Ohio River through Illinois, Indiana, and Ohio along various state and US routes. While the Project does not directly cross this byway, the construction of the Kellogg Station Launcher/Receiver and terminus of Line AM36 is proximate to Interstate 275 (0.05 mile and 0.11 mile, respectively)[68], which serves as a portion of the associated byway and would be utilized for access to the Project area during construction and operation. Effects are not anticipated as no road modifications or improvements would be required, and the majority of effects would be temporary and associated with the construction phase of the Project. Traffic may temporarily increase along Interstate 275 during construction. Therefore, we conclude that effects on this Scenic Byway would be minor and localized primarily to temporary traffic due to construction activities. Visual effects associated with construction and operation of the Kellogg Station Launcher/Receiver are not anticipated to be significant, as further discussed below.

Visual Resources

Temporary effects on visual and/or aesthetic resources associated with the Project would occur during construction as a result of the presence of construction equipment, ground disturbance, and pipeline replacement and installation activities. Given that the replacement activities involve both the abandonment of existing lines and modifications of existing facilities, the majority of effects on visual resources would be temporary in nature and comprised of the presence of construction equipment on previously disturbed land. Dust generated from Project construction activities would also affect the visual character of the landscape. See section B.8, Air Quality and Noise, below for further discussion.

As identified above, the proposed Line AM36 crosses the Lewis and Clark National Historic Trail. The Kellogg Station Launcher/Receiver is proposed on the Ohio side of the historic river trail, approximately 600 feet east. Additionally, the proposed Kellogg Station Launcher/Receiver and terminus of Line AM36 is proposed approximately 0.05 mile east of the river trail and would be installed between a paved/graveled road and parking lot in a predominately industrial area. Therefore, we do not anticipate visual effects on the Lewis and

[67] *Id.*

[68] Columbia's November 10, 2025, Application, Resource Report 8, section 8.4.2. FERC eLibrary Accession No. 20251110-5151.

Clark National Historic Trail resulting from construction and operation of the Project to be significant.

The Ohio River Scenic Byway (Interstate 275) would be utilized for access to the Project area during construction and operation. The proposed Kellogg Station Launcher/Receiver would be installed approximately 0.11 mile southwest of the Scenic Byway and would be installed between a paved/graveled road and parking lot in a predominately industrial area. Therefore, we conclude visual effects on the Ohio River Scenic Byway resulting from the Project would not be significant.

Long-term maintenance of the new permanent rights-of-way associated with the proposed Line AM36, Line EM-7, and laterals, as well as the operation of the proposed aboveground facilities would result in permanent effects on visual resources. During operation, permanent rights-of-way associated with Line AM36, Line EM-7, and the proposed laterals, would be consistent with the existing landscape of the previously maintained rights-of-ways of the pipelines that would be abandoned, as well as natural gas infrastructure in the surrounding area. Modification activities associated with existing facilities would have negligible effects on visual resources to nearby residences, as the affected land is within or in close proximity to existing industrial land. The proposed locations of new aboveground facilities are predominantly rural, forested, or highly congested areas of mixed-use development (i.e., industrial and residential). Therefore, the overall visual effects from construction and operation of the proposed aboveground facilities would be minimal due to the presence of existing vegetative or physical buffers between the proposed facility and nearby residences, or the location within or immediately adjacent to existing industrial and/or land disturbed by natural gas infrastructure from the existing lines to be abandoned. Overall, visual effects from construction of the aboveground facilities would be minimal, as the aboveground facilities are primarily located in rural or forested areas, or within areas of existing industrial facilities or development. Therefore, we conclude that constructing, operating, and abandoning the proposed facilities would not result in a significant effect on visual resources.

B.8 AIR QUALITY AND NOISE

Air Quality

The term “air quality” refers to relative concentrations of pollutants in the ambient air. Pollutants of concern are primarily ground-level ozone, carbon monoxide (CO), oxides of nitrogen (NO_x), sulfur dioxide (SO_2), and respirable and fine particulate matter (inhalable particulate matter with an aerodynamic diameter less than or equal to 10 microns [PM_{10}] and less than or equal to 2.5 microns [$PM_{2.5}$]). Ozone is not directly emitted into the atmosphere from an emissions source; rather, ozone develops because of a chemical reaction between NO_x and volatile organic compounds (VOC) in the presence of sunlight.

As well as being the reactant to form ground-level ozone, VOCs are a subset of organic compounds that are emitted during fossil-fuel combustion and can cause a variety of health effects, from irritation to more serious health effects. Fossil fuels would be used in construction equipment for the Project. Hazardous air pollutants (HAP) are also emitted during fossil-fuel

combustion and contain compounds that are known or suspected of causing cancer and other serious health effects.

Additionally, fugitive dust would be generated during Project construction from soil handling, wind-blown dust from stockpiles, and road dust. Most of the fugitive dust would be particulate matter larger than 10 microns, but a portion would be PM_{10} and $PM_{2.5}$.

Fugitive leaks from pipeline components as well as blowdowns – necessary to remove gas in piping for maintenance or startup - are sources of pollutant emissions including methane gas (CH_4) which is a greenhouse gas, as well as small amounts of VOC and HAP.

The term "greenhouse gas" (GHG) refers to the gases and aerosols that occur in the atmosphere both naturally and because of human activities, such as the burning of fossil fuels. GHGs produced by fossil-fuel combustion are primarily carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O). GHG's status as a pollutant is not related to toxicity, as they are non-hazardous to health at normal ambient concentrations, but are a pollutant due to the ability of GHGs to absorb infrared radiation (USEPA 2024), which concentrates heat in the atmosphere.

Emissions of GHGs are typically quantified and regulated in units of carbon dioxide equivalents (CO_2e). The CO_2e accounts for the global warming potential of each GHG. Global Warming Potential is the measure of a particular GHG's ability to absorb solar radiation as well as its residence time within the atmosphere.[69]

Existing Air Quality

The Project is located in Mason, Nicholas, Bracken, Pendleton, and Campbell Counties, Kentucky, and Hamilton County, Ohio, all of which have humid subtropical climate zones with a mean daily temperature ranging from 31.4 degrees Fahrenheit in January to 75.9 degrees Fahrenheit in July (National Oceanic and Atmospheric Administration [NOAA], 2025).

Ambient air quality is protected by the Clean Air Act of 1970, as amended in 1977 and 1990. The USEPA oversees the implementation of the Clean Air Act and establishes National Ambient Air Quality Standards (NAAQS) to protect human health and welfare. These standards have been developed for seven "criteria air pollutants" including NO_2, CO, ground-level ozone, SO_2, $PM_{2.5}$, PM_{10}, and lead, and include levels for short-term (acute) and long-term (chronic) exposures. Areas in compliance with the NAAQS are designated as attainment, areas not in compliance are designated as nonattainment, and areas previously designated as nonattainment that have since demonstrated compliance with the NAAQS are designated as maintenance for that pollutant. Areas for which insufficient data is available to determine attainment status are termed unclassified.

Hamilton County, Ohio is designated as maintenance for the 8-hour ozone (2015) standard and Campbell County, Kentucky is designated as maintenance for the 8-hour ozone (2015) standard and the 1-hour SO_2 (2010) standard. The remainder of the Project area is considered as attainment/unclassified with respect to all criteria pollutants. The Project does not

[69] Global warming potentials: of $CO_2 = 1$, $CH_4 = 28$, and $N_2O = 265$ (Table A-1 of 40 CFR 98).

fall within an ozone transport region. The USEPA has delegated the Clean Air Act Permitting Regulations to each State, as discussed below.

Permitting/Regulatory Requirements

The Clean Air Act is the basic federal statute governing air quality. Air emissions and equipment would be subject to the NAAQS as well as other federal and state air-quality regulations. We have reviewed the following federal requirements and determined that they are not applicable to the Project: New Source Review and Prevention of Significant Deterioration (PSD) (and Federal Class I Areas), Title V, National Emissions Standards for Hazardous Air Pollutants, New Source Performance Standards, and the Greenhouse Gas Reporting Rule. These regulations are not applicable because Project facilities are not major above-ground stationary sources. Additionally, a General Conformity Determination is not required as the Project emissions are below the 100 tpy VOC and NO_x thresholds for ozone in Hamilton County, Ohio and Campbell County, Kentucky and the 100 tpy threshold for SO_2 in Campbell County Kentucky as seen in Table 3.[70] It is important to note that emissions listed in Table 3 account for multiple years of construction; we have conservatively assumed all construction emissions would occur in a single year when comparing against General Conformity Applicability thresholds.

Construction of the proposed Project in Kentucky would be authorized under 401 Kentucky Administrative Regulations (KAR) Chapter 52, which establishes air permitting exemptions and registration requirements for air contaminant sources. The proposed heaters and associated equipment would qualify for exemption from air permitting and registration because projected emissions at each facility are below the *de minimis* thresholds in 401 KAR 52:070 Section 1(1)(a). In addition, the proposed sources are not subject to the requirements under 40 C.F.R. Parts 60, 61, or 63, consistent with 401 KAR 52:070 Section 1(1)(b). In Ohio, the proposed Kellogg Station Launcher/Receiver meets the criteria for exemption under Ohio Administrative Code 3745-15-05 and would not require an air permit.

Columbia may utilize open burning during the construction of the Project as a means of disposing of cleared/cut forested vegetation. Kentucky state laws allow the burning of forested areas outside of Fire Hazard Seasons between February 15 to April 30 and October 1 to December 15, outside of emergency drought burn bans, and outside of ozone season restrictions between May 1 to September 30 for Campbell County.[71] In addition, Pendleton County laws require that Columbia notify Kentucky State Police of any open burning activities with the exception of small recreational fires,[72] and Campbell County requires that Columbia obtain a burn permit via Campbell Fire/Rescue before performing any open burning activities in the county.[73] Ohio state laws prohibit open burning for forest clearing in restricted areas, including

[70] General Conformity Determinations in maintenance areas are subject to General Conformity Applicability thresholds defined under 40 CFR 93.153(b) (2).
[71] Kentucky State rules on open burning are defined under 401 KAR 63:005 and KRS 149.400.
[72] Pendleton County open burning laws are defined under Pendleton County Fiscal Court Ordinance No. 341.1.
[73] Campbell County open burning permits are required by Campbell Fire/Rescue at https://www.cfrky.org/burn-permit.

Hamilton County, as defined in OAC 3745-19. Any open burning would be conducted by Columbia in accordance with all necessary permits and applicable local and state regulations.

Construction Emissions

The Project would result in effects on air quality resulting from construction including intermittent and temporary emissions from fossil-fueled construction equipment, pipeline venting, and fugitive dust (PM_{10} and $PM_{2.5}$) generated from soil disturbing activities. The amount of dust generated during construction would be a function of precipitation, vehicle numbers and types, vehicle speeds, and roadway characteristics. Dust emissions would be greater during dry periods and in areas of fine-textured soils.

Combustion-related emissions for Project construction would include NO_x, CO, VOCs, SO_2, PM, small amounts of HAPs, and GHGs. The USEPA requires manufacturers of on- and off-road engines to certify their products to engine emission standards based on the year of manufacture. For diesel engines, the emission standards have been phased in over the past two decades in four steps, referred to as Tier 1 to Tier 4. The engine must comply with the emission standards throughout its life. In 2010, the USEPA required the sulfur concentration in diesel fuels be lowered from historical concentration of 500 parts per million to 15 parts per million (ultra-low sulfur diesel fuel), which allows diesel engines to meet current Tier 4 emission requirements.

To reduce fugitive dust emissions, Columbia would adhere to its Fugitive Dust Control Plan, which prescribes applying water to disturbed surfaces, vehicle speed restrictions, using gravel tracking pads at egress points, and restoring disturbed areas.

Columbia may utilize open burning as a means of clearing forested land during construction of the Project. Open burning entails the controlled burning of waste, trees, or other combustible materials without emissions reduction measures. Open burning emissions estimates found in Table 3 conservatively assume that all forest land affected by Project construction is burned.

Table 3 summarizes the total estimated emissions of criteria pollutants, total HAPs, and GHGs from construction equipment and material deliveries. The GHG emissions associated with the Project construction are principally from CO_2. Construction activities would begin as early as January 2027. At any one location along the Project pipelines, construction emissions would occur for a limited period as the proposed and existing pipelines are progressively constructed or abandoned, respectively. Pipeline abandonments would have limited or no emissions, except where pipeline would be abandoned by removal and emissions would be similar to the typical pipeline construction. Where aboveground facilities are constructed and at HDD locations, construction activities would be static, and construction emissions would occur over a longer period.

Table 3 Construction-Related Emissions for the Project								
Construction Activity	**CO (tons)**	**NO_X (tons)**	**SO_2 (tons)**	**VOC (tons)**	**PM_{10} (tons)**	**$PM_{2.5}$ (tons)**	**HAP (tons)**	**CO_2e (tons)**
Off-Road Equipment	2.19	7.27	<0.01	0.49	0.32	0.30	0.36	2,525.98
On-Road Equipment	31.77	3.45	0.06	2.11	0.06	0.06	0.32	4,928.93
Construction Activity Fugitive Dust	--	--	--	--	97.66	10.29	--	--
Roadway Fugitive Dust	--	--	--	--	1.11	0.1	--	--
Venting/Blowdown/ Purging	--	--	--	7.02	--	--	0.19	20427.98
Open Burn	133.05	3.81	--	23.47	16.16	16.16	--	151.67
Project Total	**167.01**	**14.52**	**0.08**	**33.09**	**115.31**	**26.93**	**0.88**	**28,034.55**
Construction-Related Emissions by County								
Campbell County, Kentucky Subtotal	124.82	10.21	0.05	20.66	94.98	21.23	0.46	8,154.39
Hamilton County, Ohio Subtotal	8.00	1.33	0.01	1.97	0.68	0.32	0.12	4,615.63
Source: Columbia's November 10, 2025, Application, Resource Report 9, appendix 9A. FERC eLibrary Accession No. 20251110-5151. Note: The Emissions by County section only includes counties designated maintenance/nonattainment; therefore, county subtotals will not sum to the Project Total.								

Through the implementation of the work practices described above, compliance with state air regulations, the short duration and low magnitude of the construction emissions, and emissions below the General Conformity Applicability Thresholds, emissions during construction of the Project would be minor, and the effect of these emissions would be localized. Therefore, we conclude these emissions generated during construction would not have a significant effect on regional air quality.

Operational Emissions

Air quality effects would result from new operational emissions for natural gas fired equipment at the new and modified aboveground facilities and fugitive components being constructed as part of the Project. All aboveground facilities would have fugitive components and the Bracken, Pendleton, Alexandria, Parkview, Cold Spring, and UL47 Meter Stations would have new natural gas fired heaters. Fugitive emissions would also result from the proposed Project pipelines. Table 4 summarizes the estimated operational emissions from the Project.

Table 4 Operating Emissions for the Project								
Emission Source	**CO (tons)**	**NO_X (tons)**	**SO_2 (tons)**	**VOC (tons)**	**PM_{10} (tons)**	**$PM_{2.5}$ (tons)**	**HAP (tons)**	**CO_2e (tons)**
Component Fugitives	--	--	--	0.45	--	--	0.01	1,311.17
Venting	--	--	--	0.04	--	--	<0.01	112.92
Heaters (6)	9.03	10.74	0.06	0.59	0.82	0.82	0.20	12,832.82
Pipeline Fugitives	--	--	--	<0.01	--	--	<0.01	2.84
Project Total	**9.03**	**10.74**	**0.06**	**1.08**	**0.82**	**0.82**	**0.21**	**14,259.75**

Source: Populated from Columbia's November 10, 2025, Application, Resource Report 9, Appendix 9B. FERC eLibrary Accession number 20251110-5151

CO = carbon monoxide; NOx = nitrogen oxides; SO_2 = sulfur dioxide; VOC= volatile organic compound; PM_{10} = particulate matter with an aerodynamic diameter of 10 microns or less; $PM_{2.5}$ = particulate matter with an aerodynamic diameter of 2.5 microns or less; HAP = hazardous air pollutant; CO_2e = carbon dioxide equivalent.

The Project operational emissions would be limited to only fugitive emissions and minor emissions from new natural gas heaters at Meter Stations Bracken, Pendleton, Alexandria, Parkview, Cold Spring, and UL47 listed in table 4 and would not trigger requirements under PSD or require a Title V operating permit; therefore, we conclude that operation of the Project would not have significant effects on local or region air quality.

Climate Change

Climate change is the variation in the Earth's climate (including temperature, precipitation, humidity, wind, and other meteorological variables) over time. Climate change is driven by accumulation of GHGs in the atmosphere due to the increased consumption of fossil fuels (e.g., coal, petroleum, and natural gas) since the early beginnings of the industrial age and accelerating in the mid- to late-20th century.[74] The GHGs produced by fossil fuel combustion are CO2, methane, and nitrous oxide.

In 2017 and 2018, the U.S. Global Change Research Program (UCORP)[75] issued its Climate Science Special Report: Fourth National Climate Assessment, Volumes I and II.[76] This report and the recently released report by the Intergovernmental Panel on Climate Change, Climate Change 2021: The Physical Science Basis, state that climate change has resulted in a

[74] Intergovernmental Panel on Climate Change, United Nations, *Summary for Policymakers* of Climate Change 2021: The Physical Science Basis. (Valerie Masson-Delmotte et al., eds.) (2021), https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_SPM.pdf (IPCC Report) at SPM-5. Other forces contribute to climate change, such as agriculture, forest clearing, and other anthropogenically driven sources.

[75] The U.S. Global Change Research Program comprised representatives from 13 federal departments and agencies and issued reports every 4 years that described the state of the science relating to climate change and the effects of climate change on different regions of the United States and on various societal and environmental sectors, such as water resources, agriculture, energy use, and human health.

[76] U.S. Global Change Research Program, Climate Science Special Report, Fourth National Climate Assessment | Volume I (Donald J. Wuebbles et al. eds) (2017), https://repository.library.noaa.gov/view/noaa/19486 (USGCRP Report Volume I); U.S. Global Change Research Program, Fourth National Climate Assessment, Volume II Effects, Risks, And Adaptation In The United States (David Reidmiller et al. eds.) (2018), https://repository.library.noaa.gov/view/noaa/19487 (USGCRP Report Volume II).

wide range of effects across every region of the country and the globe. Those effects extend beyond atmospheric climate change alone and include changes to water resources, agriculture, ecosystems, human health, and ocean systems.[77] According to the Fourth Assessment Report, the United States and the world are warming; global sea level is rising and oceans are acidifying; and certain weather events are becoming more frequent and more severe.[78] These effects have accelerated throughout the end of the 20th and into the 21st century.[79]

GHG emissions do not result in proportional local and immediate effects; it is the combined concentration in the atmosphere that affects the global climate. These are fundamentally global effects that feed back to local and regional climate change effects. Thus, the geographic scope for analysis of GHG emissions is global rather than local or regional. For example, a project 1 mile away emitting 1 ton of GHGs would contribute to climate change in a similar manner as a project 2,000 miles distant also emitting 1 ton of GHGs.

Climate change is a global phenomenon; however, for this analysis, we will focus on the existing and potential climate change effects in the Project area. The USGCRP's Fourth Assessment Report notes that the following observations of environmental effects are attributed to climate change in the Midwest Region (which includes the Project area in Ohio):[80]

- increases in warm-season absolute humidity and precipitation have eroded soils, created favorable conditions for pests and pathogens, and degraded the quality of stored grain;
- threats from a changing climate are interacting with existing stressors such as invasive species and pests to increase tree mortality and reduce forest productivity;
- stormwater management systems, transportation networks, and other critical infrastructure are already experiencing effects from changing precipitation patterns and elevated flood risks; and
- at-risk communities in the Midwest are becoming more vulnerable to climate change effects such as flooding, drought, and increases in urban heat islands and tribal nations are especially vulnerable because of their reliance on threatened natural resources for their cultural, subsistence, and economic needs.

The USGCRP's Fourth Assessment Report notes that the following observations of environmental effects are attributed to climate change in the Southeast Region (which includes the Project area in Kentucky):[81]

- combined effects of changing extreme rainfall events and sea level rise are already increasing flood frequencies, which effects property values and infrastructure viability;
- changing winter temperature extremes, wildfire patterns, sea levels, hurricanes, floods, droughts, and warming ocean temperatures are expected to redistribute species and greatly modify ecosystems; and

[77] IPCC Report at SPM-5 to SPM-10.
[78] USGCRP Report Volume II at 73-75.
[79] USGCRP Report Volume II at 99 (describing accelerating flooding rates in Atlantic and Gulf Coast cities).
[80] USGCRP Report Volume I and II.
[81] *Id*.

- occurrence of more frequent extreme heat episodes and changing seasonal climates.

The USGCRP's Fourth Assessment Report notes the following projections of climate change effects in the Midwest region with a high or very high level of confidence:[82]

- projected changes in precipitation, coupled with rising extreme temperatures before mid-century, will reduce Midwest agricultural productivity to levels of the 1980s without major technological advances;
- effects will result in the loss of economically and culturally important tree species, such as paper birch and black ash, and are expected to lead to the conversion of some forests to other forest types or even to non-forested ecosystems by the end of the century;
- climate change is expected to worsen existing conditions and introduce new health threats by increasing the frequency and intensity of poor air quality days, extreme high temperature events, and heavy rainfall; extending pollen seasons; and modifying the distribution of disease-carrying pests and insects; and
- the annual cost of adapting urban stormwater systems to more frequent and severe storms is projected to exceed $500 million for the Midwest by the end of the century.

The USGCRP's Fourth Assessment Report notes the following projections of climate change effects in the Southeast with a high or very high level of confidence:[83]

- projected changes in precipitation and sea level will lead to daily high tide flooding by the end of the century;
- loss of forested and non-forested ecosystems puts ecological resources that people depend on for livelihood, protection, and well-being increasingly at risk; and
- extreme heat and changing seasonal climates are projected to increase exposure-linked health effects and economic vulnerabilities in the agricultural, timber, and manufacturing sectors.

It should be noted that while the effects described above taken individually may be manageable for certain communities, the effects of compound events (such as simultaneous heat and drought, wildfires associated with hot and dry conditions, or flooding associated with high precipitation on top of saturated soils) can be greater than the sum of the parts.[84]

The GHG emissions associated with construction and operation of the Project are identified and quantified in tables 3 and 4 and Appendix F. Emissions of GHGs are typically expressed in terms of CO_2e.[85] Construction of the Project may result in emissions of about

[82] USGCRP Report Volume II.
[83] *Id.*
[84] *Id.*
[85] GHG gases are converted to CO_2e by means of the GWP; the measure of a particular GHG's ability to absorb solar radiation; and its residence time within the atmosphere, consistent with the USEPA's established method for reporting GHG emissions for air permitting requirements that allows a consistent comparison with federal regulatory requirements.

28,035 tons (25,433 metric tons) of CO_2e.[86] Operation of proposed Project equipment would result in emissions of approximately 14,263 tons per year (12,939 metric tons per year) of CO_2e. Upstream and downstream GHG emissions are outside the scope of the environmental review for this project.[87]

Construction and operation of the Project would increase the atmospheric concentration of GHGs in combination with past, current, and future emissions from all other sources globally and contribute incrementally to future climate change effects. To assess effects on climate change associated with the Project, Commission staff considered whether it could identify discrete physical effects resulting from the Project's GHG emissions or compare the Project's GHG emissions to established targets designed to combat climate change.

To date, Commission staff have not identified a methodology to attribute discrete, quantifiable, physical effects on the environment resulting from the Project's incremental contribution to GHGs. Without the ability to determine discrete resource effects, Commission staff are unable to assess the Project's contribution to climate change through any objective analysis of physical affect attributable to the Project. Additionally, Commission staff have not been able to find an established threshold for determining the Project's significance when compared to established GHG reduction targets at the state or federal level. Ultimately, this EA is not characterizing the Project's GHG emissions as significant or insignificant.[88] However, as we have done in prior NEPA analyses, we disclose the Project's GHG emissions in comparison to national and state GHG emission inventories.

In order to provide context of the Project emissions on a national level, we compare the Project's GHG emissions to the total GHG emissions of the United States as a whole. At a national level, 5,489 million metric tons of CO_2e were emitted in 2022 (inclusive of CO_2e sources and sinks) (EPA, 2024). Construction emissions from the Project could potentially increase CO_2e emissions based on the national 2022 levels by 0.0005 percent; in subsequent years, the Project operations could potentially increase emissions nationally by 0.0002 percent.

In order to provide context of the Project emissions on a state level, we compare the Project's GHG emissions to the state GHG inventories. At the state level, Kentucky energy related CO_2 emissions in 2024 were 106.9 million metric tons and Ohio energy related CO_2 emissions in 2024 were 186.3 million metric tons (Energy Information Administration 2026). Project construction could potentially increase CO_2e emissions based on the Kentucky and Ohio 2024 levels by 0.0200 percent and 0.0022 percent respectively; in subsequent years, Project operations could potentially increase emissions by 0.0120 percent and 0.00003 percent respectively.

When states have GHG emissions reduction targets, we compare the project's operational GHG emissions to those state goals to provide additional context. Neither Kentucky nor Ohio

[86] See table B.8.1. Figures presented here are converted from U.S. tons to metric tons. A metric ton is about equal to 1.1 ton.

[87] *Seven Cnty. Infrastructure Coal. v. Eagle Cnty., Colo.*, 605 U.S. 168, 186-90 (2025).

[88] *e.g.*, *Driftwood Pipeline LLC*, 183 FERC ¶ 61,049, at P 63 (2023) ("…there currently are no accepted tools or methods for the Commission to use to determine significance, therefore the Commission is not herein characterizing these emissions as significant or insignificant.)

have formally adopted executive or legislative statewide goals for GHG emissions reduction targets.

Noise

Constructing the Project would affect noise levels in the Project area. Noise is generally defined as sound with intensity greater than the ambient or background sound pressure level. The FERC has adopted the USEPA noise criterion and uses it to evaluate the potential noise effects from the proposed Project at noise sensitive areas (NSAs). NSAs are defined as homes, schools, churches, or any location where people reside or gather. The USEPA has indicated that a day-night average sound level (L_{dn}) of 55 decibels on the A weighted scale (dBA) protects the public from indoor and outdoor activity interference. Due to the 10 dBA nighttime penalty added prior to the logarithmic calculation of the L_{dn}, for nighttime construction activities to meet the 55 dBA L_{dn} limit, it must be designed such that actual constant noise levels on a 24-hour basis do not exceed 48.6 dBA equivalent sound level (L_{eq}) at any NSA. Therefore, we require that construction activities that could or may occur during nighttime hours be performed with the goal that the activity contributes noise at a level at or below 55 dBA L_{dn} or 48.6 dBA L_{eq}. In general, a person's threshold of perception for a perceivable change in loudness on the A-weighted sound level is about 3 dBA, whereas a 6 dBA change is clearly noticeable, and a 9 dBA change is perceived as either twice or half as loud.

Construction Noise

Project effects on noise would be temporary. Columbia indicated that construction and abandonment activities would generally be conducted during the daytime (7 AM to 7 PM) Monday through Saturday, except for specialized construction techniques, emergencies, or other atypical circumstances such as potential delays associated with weather or site conditions. Columbia stated activities such as tie-ins at critical locations, HDD activities, stream crossings, hydrostatically testing piping, and recompression and service transferring activities could occur at nighttime.[89] Nighttime construction activities for the Project would be temporary in nature and are not expected to occur for the duration of construction. These effects would be minimized through the implementation of Columbia's proposed noise mitigation measures including the use of handheld equipment/tools and pipe lifting equipment and limiting deliveries to daytime hours to the extent practicable.

Construction activities would begin as early as January 2027. At any one location along the Project pipelines, construction noise would occur for a limited period as the proposed and existing pipelines are progressively constructed or abandoned, respectively. Pipeline abandonments would have limited or no noise, except where pipeline would be abandoned by removal and noise would be similar to the typical pipeline construction. Where aboveground facilities are constructed and at HDD locations, construction activities would be static, and construction noise would occur over a longer period.

Columbia proposes employing in-place compression or mobile recompression services during construction at the Foster Meter Station. During tie-in construction, an in-place (or

[89] Columbia's November 10, 2025, Application, Resource Report 1, section 1.3.1. FERC eLibrary Accession Number 20251110-5151.

mobile) compression unit is used to capture and transfer gas that would otherwise have to be vented during construction. Mobile recompression services could occur during both daytime and nighttime hours. Table 5 summarizes the predicted sound levels at the nearest NSAs to the Foster Meter Station during the proposed recompression activities.

Table 5
Noise Modeling Results at NSAs for the Foster Regulator Station (Recompression)

NSAs	Distance and Direction to Proposed Recompression Activities	Measured L_d (dBA)	Measured L_n (dBA)	Measured Ambient L_{dn} (dBA)	Modeled Sound Level of Recompression with Mitigation L_{dn} (dBA)	Sound Level of Ambient plus Modeled Noise L_{dn} (dBA)	Potential Increase Above Ambient (dB)
1	1,340 SE	48.4	43.3	50.9	54.6	56.1	5.2
2	1,395 SE	48.4	43.3	50.9	54.0	55.7	4.8
3	2,135 S	48.4	43.3	50.9	49.7	53.4	2.5
4	2,170 SW	48.4	43.3	50.9	36.3	51.0	0.1
5	2,150 SW	48.4	43.3	50.9	43.4	51.6	0.7

Source: Populated from Columbia's November 10, 2025, Application, Resource Report 9. FERC eLibrary Accession number 20251110-5151.

Columbia incorporated noise mitigation measures in the noise modeling of the Foster Regulator Station to decrease noise effects at the NSAs. The predictive noise modeling demonstrates that with the inclusion of two 12-foot-high STC 25 freestanding sound panels (total of 64 feet long), the estimated mitigated noise levels would be below 55 dBA L_{dn} at all NSAs. Columbia stated it would implement the sound panels incorporated in the noise modeling; therefore, with the implementation of Columbia's mitigation measures, we conclude that Project recompression activities at the Foster Meter Station would not have a significant effect on noise in the Project area.

Noise would occur during HDD activities at the AM36 HDD Ohio River Crossing which would last for about 117 days[90]. Based on the results of the unmitigated noise modeling of the proposed HDD construction activities at the Ohio River crossing HDD entry and exit sites, the noise levels were predicted to comply with our noise criterion at the nearest NSAs with the exception of NSA AM36a-01. Table 6 summarizes the predicted sound levels at the nearest NSAs to the Ohio River crossing entry and exit sites during HDD activities. The modeled sound level attributable to HDD activities at NSA AM36a-01 is 64.7 dBA Ldn. However, given the high ambient noise levels at this location (62.3 dBA L_{dn}), the potential increase above ambient would be 4.4 dB. This change in noise level would be perceptible but falls below a level of increase generally considered to be clearly noticeable (e.g., 6 dBA). Therefore, we conclude that the noise would not be significant.

[90] Columbia November 10, 2025, Application, HDD Feasibility Report. FERC eLibrary Accession Number 20251110-5151.

Table 6 Noise Modeling Results at NSAs for the Line AM36 HDD Entry/Exit Sites								
HDD Site	**NSAs**	**Distance and Direction from Proposed HDD Entry/Exit Borehole (feet)**	**Measured Ambient L_d (dBA)**	**Measured Ambient L_n (dBA)**	**Measured Ambient L_{dn} (dBA)[91]**	**Modeled Sound Level of HDD Activities L_{dn} (dBA)**	**Sound Level of Ambient plus Modeled Noise L_{dn} (dBA)**	**Potential Increase Above Ambient (dB)**
Line AM36 HDD Entry	AM36a-01	255 NW	59.6	54.7	62.3	64.7	66.7	4.4
	AM36a-02	1,535 NW	59.6	54.7	62.3	49.1	62.5	0.2
	AM36a-03	1,960 N	59.6	54.7	62.3	46.8	62.4	0.1
	AM36a-04	2,290 E	59.6	54.7	62.3	47.7	63.4	1.1
	AM36a-05	2,185 SW	60.4	56.5	63.7	47.9	63.8	0.1
	AM36a-06	2,055 SW	60.4	56.5	63.7	48.3	63.8	0.1
	AM36a-07	2,109 SW	60.4	56.5	63.7	47.5	63.8	0.1
	AM36a-08	2,430 SW	60.4	56.5	63.7	47.3	63.8	0.1
Line AM36 HDD Exit	AM36b-01	1,395 SW	60.4	56.5	63.7	52.0	64.0	0.3
	AM36b-02	1,245 W	60.4	56.5	63.7	53.1	64.1	0.4
	AM36b-03	1,265 W	60.4	56.5	63.7	52.9	64.0	0.3
	AM36b-04	1,185 W	60.4	56.5	63.7	53.3	64.1	0.4
	AM36b-05	1,165 W	60.4	56.5	63.7	53.4	64.1	0.4
	AM36b-06	1,125 W	60.4	56.5	63.7	53.6	64.1	0.4
	AM36b-07	1,200 NW	60.4	56.5	63.7	53.6	64.1	0.4
	AM36b-08	845 N	60.4	56.5	63.7	55.3	64.3	0.6
	AM36b-09	885 N	60.4	56.5	63.7	53.8	64.1	0.4
	AM36b-10	940 N	60.4	56.5	63.7	52.5	64.0	0.3
	AM36b-11	1,205 NE	60.4	56.5	63.7	45.7	63.8	0.1
	AM36b-12	1,235 NE	60.4	56.5	63.7	38.6	63.7	0.0
	AM36b-13	1,215 NE	60.4	56.5	63.7	40.1	63.7	0.0
	AM36b-14	1,135 NE	60.4	56.5	63.7	41.4	63.7	0.0
	AM36b-15	970 E	60.4	56.5	63.7	42.4	63.7	0.0
	AM36b-16	660 E	60.4	56.5	63.7	43.7	63.7	0.0
	AM36b-17	805 SE	60.4	56.5	63.7	45.6	63.8	0.1
	AM36b-18	1,030 SE	60.4	56.5	63.7	43.8	63.7	0.0
	AM36b-19	1,555 SE	60.4	56.5	63.7	44.6	63.8	0.1
	AM36b-20	1,855 SE	60.4	56.5	63.7	43.1	63.7	0.0
	AM36b-21	1,725 SW	60.4	56.5	63.7	48.9	63.8	0.1
Source: Populated from Columbia's November 10, 2025, Application, Resource Report 9. FERC eLibrary Accession number 20251110-5151. Note: Traffic noise from the nearby highway I-275 contributed heavily to measured ambient sound.								

[91] Ambient noise levels are relatively high primarily due to traffic noise. See Columbia November 10, 2025, Application, Resource Report 9, Appendix 9J. FERC eLibrary Accession Number 20251110-5151.

Campbell County noise regulations establish maximum permissible sound levels for continuous sound; these local noise regulations apply to the MLV-1/Gubsur Mill Tap Meter Station, Parkview Meter Station, UL47 Meter Station, Alexandria Meter Station, Cold Spring Meter Station, and Ohio River crossing HDD exit site.92 As discussed, Columbia indicated it would implement the recommended noise control measures at the Foster Regulator Station for the proposed recompression activities. Columbia stated it would coordinate with Campbell County regarding acceptable noise control measures for construction noise levels at the Ohio River crossing HDD exit site, Alexandria Meter Station, Cold Spring Meter Station, MLV-1/Gubsur Mill Meter Station, Parkview Meter Station, and UL47 Meter Station.Through the implementation of Columbia's measures, and the temporary and localized nature of construction activities, we conclude that Project construction and abandonment activities would not have a significant effect on noise in the Project area.

Commenters expressed concern about disturbance from vibrations during HDD construction. Vibrations from HDD construction attenuate with distance and would only have a minor effect at any NSAs (Reilly et al., 2020). Furthermore, HDD activities would require about 117 days and would not last the entire construction period.

Operational Noise

The Project would result in permanent (ongoing) noise effects associated with operation of the Alexandria, Cold Spring, Gubsur Mill, Parkview, Pendleton, and UL47 meter stations where new noise emitting equipment is proposed. The tables in Appendix G summarize the predicted sound levels at the nearest NSAs to the Project meter stations during Project operation where new noise emitting equipment is proposed and those that have NSAs within 0.50-mile of the meter station.

Based on the results of the noise modeling, noise levels were predicted to comply with our noise criterion at the nearest NSAs during Project operation at all of the Project meter stations where new noise emitting equipment is proposed. Operational noise levels were predicted to comply with the Campbell County noise regulations at all of the Project meter stations where new noise emitting equipment is proposed with the exception of the Alexandria Meter Station. Columbia would coordinate with Campbell County regarding acceptable noise control measures at the Alexandria Meter Station.

Given the proximity of the nearest NSAs to the Cold Spring and Pendleton Meter Stations (180 feet and 280 feet, respectively) and to ensure that noise levels are consistent with the modeling, **we recommend that any Commission Order approving the Project include a mandatory condition requiring Columbia to file noise surveys no later than 60 days after placing the Cold Spring Meter Station and Pendleton Meter Station in service.** (see section D, recommended condition no. 13).

Based on the proposed operational noise levels, and our recommendation, we conclude that the noise attributable to operation of the Project would not result in significant effects.

[92] Columbia November 10, 2025, Application, Resource Report 9, section 9.3.1. FERC eLibrary Accession Number 20251110-5151.

B.9 RELIABILITY AND SAFETY

The transportation of natural gas by pipeline involves some risk to the public in the event of an accident and subsequent release of gas. The greatest hazard is a fire or explosion following a major pipeline rupture. Methane, the primary component of natural gas, is colorless, odorless, and tasteless. It is not toxic, but is classified as a simple asphyxiate, possessing a slight inhalation hazard. If breathed in high concentration, oxygen deficiency can result in serious injury or death.

Commenters raise concerns about the proximity of high-pressure gas facilities to residences and barns. One commenter asserts that the Project is in tension with pipeline safety setback practices and questions whether the Project complies with PHMSA requirements. The pipeline and aboveground facilities associated with the project must be designed, constructed, operated, and maintained in accordance with the DOT Minimum Federal Safety Standards in 49 CFR Part 192. These regulations are intended to ensure adequate protection for the public and to prevent natural gas facility accidents and failures. We also note that the existing pipelines have operated reliably and safely in close proximity to residences, commercial buildings, and other structures for several decades. Additionally, much of the new pipeline would be located in the same location or adjacent to the existing pipelines. As discussed further below, the replacement of "vintage" pipe with new pipe would increase the safety and reliability of the system.

The DOT pipeline standards are published in Parts 190-199 of Title 49 of the CFR. For example, Part 192 of 49 CFR specifically addresses natural gas pipeline safety issues, prescribes the minimum standards for operating and maintaining pipeline facilities, and incorporates compressor station design, including emergency shutdowns and safety equipment. Part 192 also requires a pipeline operator to establish a written emergency plan that includes procedures to minimize the hazards in a natural gas pipeline emergency.

The operator must also establish a continuing education program to enable customers, the public, government officials, and those engaged in excavation activities to recognize a gas pipeline emergency and report it to appropriate public officials.

Facilities associated with Columbia's Project must be designed, constructed, operated, and maintained in accordance with DOT standards, including the provisions for written emergency plans and emergency shutdowns. Columbia would provide the appropriate training to local emergency service personnel before the facilities are placed in service.

The Project would replace older pipeline infrastructure originally installed in the 1950s with modern pipeline facilities to address aging infrastructure, reduce potential safety and reliability risks, and ensure continued compliance with federal and state regulations. Therefore, we conclude that Columbia's facility and pipeline construction and operation would represent a decrease in risk to the public and we are confident that with the options available in the detailed design of Columbia's facilities, that they would be constructed and operated safely.

B.10 POLYCHLORINATED BIPHENYLS

From approximately 1950 to the early 1970s, PCB-containing compounds were used by some interstate natural gas transmission companies as a lubricant, hydraulic fluid, or sealant for turbines and air compressors. As part of normal operation, PCBs could leak or blow by pressure seals and enter the transmission pipeline. PCBs may also be present in natural gas pipelines due to the historical practice of oil fogging, performed in the late 1940s through 1960s (USEPA, 2004). Older pipeline segments and associated facilities in operation at the time that PCBs were employed in the natural gas transmission industry may contain PCBs at levels requiring abandonment and disposal procedures consistent with USEPA's regulations found in 40 CFR 761.

Columbia indicated that the potential for PCB contamination along the existing Project pipelines and associated facilities exists.[93] During construction activities, Columbia would sample/dispose removed and abandoned pipelines in accordance with Toxic Substances Control Act regulations. Where applicable, Columbia would collect pipeline liquid samples prior to 72 hours of the last transmission of gas in accordance with 40 CFR 761.60. If PCBs are discovered, Columbia would comply with the requirements presented in 40 CFR 761.60(b)(5) and adhere to any applicable federal, state, and local regulations. In addition, Columbia would implement its *Plan for the Unanticipated Discovery of Contaminated Environmental Media* in the event PCB contamination is encountered.[94]

Based on Columbia's implementation of measures to minimize effects in the event PCBs are discovered during Project construction and abandonment activities, we conclude that PCB contamination resulting from the Project is not anticipated to affect the surrounding environment.

[93] Columbia November 10, 2025, Application, Resource Report 12, section 12.0. FERC eLibrary Accession Number 20251110-5151.
[94] Columbia's April 29, 2026, response to FERC's April 9, 2026, Environmental Information Request. FERC eLibrary Accession No. 20260429-5302.

C – ALTERNATIVES

In accordance with NEPA and Commission policy, we sought to identify and evaluate alternatives to the Project and determine whether they would be reasonable and preferable to the proposed action while meeting the Project's purpose. In this section we evaluate the No-Action Alternative and Pipeline Route Alternatives. As the proposed Project involves the replacement of existing pipelines, and the construction and modification of associated facilities, we did not consider System Alternatives which generally use existing, modified, or proposed pipeline systems or existing compression to meet the purpose of a Project.

Our evaluation of alternatives is based on Project-specific information provided by Columbia, publicly available information, and our expertise and experience regarding the siting, construction, and operation of natural gas projects and such projects' potential effects on the environment. It is important to recognize that not all conceivable alternatives can meet the Project's purpose or are technically and economically feasible and practical; therefore, an alternative that does not meet the Project's purpose and/or is not feasible and practical cannot be considered a viable alternative.

The evaluation criteria used for developing and reviewing alternatives were:

- ability to meet the Project's stated purpose;
- technical and economic feasibility and practicality; and
- offers a significant environmental advantage over the proposed action.

Through environmental comparison and application of our professional judgement, each alternative is considered to a point where it becomes clear that the alternative could or could not meet the three evaluation criteria. To ensure a consistent environmental comparison and to normalize the comparison factors, we generally use desktop sources of information (e.g., publicly available data, geographic information system data, aerial mileage) and assume the same general workspace requirements.

Our evaluation considers effects on both the natural and human environments. The natural environment includes vegetation, habitat, wildlife, waterbodies and wetlands, noise and air quality, and other common environmental resources. The human environment includes site availability, land use, residences, roads, utilities, and other nearby development. In recognition of the competing interests and the differing nature of effects resulting from an alternative that sometimes exists (i.e., effects on the natural environment versus effects on the human environment), we also consider other factors that are relevant to a particular alternative or discount or eliminate factors that are not relevant or may have less weight or significance.

Determining if an alternative provides a significant environmental advantage requires a comparison of the effects on each resource as well as an analysis of effects on resources that are not common to the alternatives being considered. The determination must then balance the overall effects and all other relevant considerations. In comparing the effect between resources (factors), we also considered the degree of effect anticipated on each resource. Ultimately, an

alternative that results in equal or minor advantages in terms of environmental effect would not compel us to recommend shifting the effects from the current landowner to a new landowner.

C.1 NO-ACTION ALTERNATIVE

NEPA requires the Commission to consider and evaluate the No-Action Alternative. In instances involving federal decisions on proposals for projects, selecting the no-action alternative would mean the proposed activity would not take place and the resulting environmental effects from taking no-action would be compared with the effects of permitting the activity. Further, the No-Action Alternative provides a benchmark for decisionmakers to compare the magnitude of environmental effects of the proposed activity and alternatives.

Under the no-action alternative, the environmental effects associated with the Project and analyzed in this EA would not occur. Columbia would not construct or abandon any component of the Project and, consequently, would be unable to meet the Project's stated purpose. If the purpose of the Project is not met, the "vintage" pipeline infrastructure originally installed in the 1950s would not be replaced and the issues associated with aging infrastructure, safety and reliability risks, and continued compliance with federal and state regulations would not be addressed. Therefore, we do not recommend the no-action alternative. However, the Commission will ultimately determine the Project need and could choose the no-action alternative.

C.2 PIPELINE ROUTE ALTERNATIVES

Typically, major route alternatives deviate from the proposed route for an extended distance, often miles away from the proposed route. The proposed Project involves the replacement of existing pipeline facilities; therefore, we did not evaluate any major route alternatives. The existing Line AM04 and Line AM09 would remain in operation until the new Line AM36 is placed into service; therefore, we did not consider alternatives involving replacement in the same location/trench. The proposed Line EM-7 would primarily be replaced in the same location/trench, and any significant deviations from the existing pipeline route would not take advantage of previously disturbed and maintained lands and existing permanent easements. Installation of a pipeline within or along an existing cleared right-of-way/easement is generally preferable to creating new right-of-way because effects associated with construction and operation would be limited to the existing corridor and immediately adjacent areas which have already been disturbed and are being maintained. Therefore, we did not evaluate any major route alternatives to the proposed Line EM-7.

Minor route alternatives are generally identified and evaluated to avoid or minimize effects on localized sensitive resources and are informed by field surveys, stakeholder feedback, and constructability considerations. Because Columbia is collocating 100 percent of Line EM-7 with existing rights-of-way, we did not identify or evaluate minor route alternatives. We received comments from Ms. Janalee Shaw[95] and Ms. April Webster[96] identifying specific minor route alternatives for Line AM36 that did not involve placement of the new pipe within or adjacent to the existing easements for Lines AM04 and AM09.

[95] Janalee Shaw December 15, 2025, Comments at 4-5. FERC eLibrary Accession No. 20251215-5209.
[96] April Webster December 15, 2025, Comments at 1. FERC eLibrary Accession No. 20260127-5144.

Ms. Shaw identified a minor route alternative that would avoid effects on one of two streams that would be crossed by the proposed AM36 pipeline. Our evaluation of this route alternative determined that both the proposed route and the route alternative would cross both streams, both of which are minor (less than 10 feet wide each), and one being intermittent. Project effects on these waterbodies as discussed above would be temporary and minor. Because implementing this alternative would not result in a significant environmental advantage when compared to the proposed route and we do not recommend Ms. Shaw's alternative.

Ms. Webster identified a minor route alternative that would cross her driveway rather than a ridgetop on her property. Our evaluation of this route alternative determined that the proposed pipeline would be placed within the existing trench after this portion of Line AM09 is removed. By placing the new pipeline in the existing easement, effects on the environment are substantially reduced. Therefore, implementing this alternative would not result in a significant environmental advantage when compared to the proposed route and we do not recommend Ms. Webster's alternative.

C.3 ALTERNATIVES CONCLUSION

Based on the preceding analysis, we conclude that the proposed Project is the preferred alternative to meet the Project's purpose.

SECTION D – CONCLUSIONS AND RECOMMENDATIONS

Based on the analysis in this EA, we have determined that if Columbia constructs, operates, and abandons the Project in accordance with its application and supplements and our recommended mitigation measures, approval of the Project would not constitute a major federal action significantly affecting the quality of the human environment. We recommend that the Commission's Order contain a finding of no significant impact and include the mitigation measures listed below as conditions in any Certificate the Commission may issue.

1. Columbia shall follow the construction and abandonment procedures and mitigation measures described in its application and supplements (including responses to staff data requests) and as identified in the EA, unless modified by the Order. Columbia must:

    a. request any modification to these procedures, measures, or conditions in a filing with the Secretary of the Commission (Secretary);
    b. justify each modification relative to site-specific conditions;
    c. explain how that modification provides an equal or greater level of environmental protection than the original measure; and
    d. receive approval in writing from the Director of the Office of Energy Projects (OEP), or the Director's designee, **before using that modification**.

2. The Director of OEP, or the Director's designee, has delegated authority to address any requests for approvals or authorizations necessary to carry out the conditions of the Order, and take whatever steps are necessary to ensure the protection of environmental resources during abandonment, construction, and operation of the Project. This authority shall allow:

    a. the modification of conditions of the Order;
    b. stop-work authority and authority to cease operation; and
    c. the imposition of any additional measures deemed necessary to ensure continued compliance with the intent of the conditions of the Order as well as the avoidance or mitigation of unforeseen adverse environmental impact resulting from Project abandonment activities, construction, and operation.

3. **Prior to any construction**, Columbia shall file an affirmative statement with the Secretary, certified by a senior company official, that all company personnel, EIs, and contractor personnel would be informed of the EI's authority and have been or will be trained on the implementation of the environmental mitigation measures appropriate to their jobs before becoming involved with construction and restoration activities.

4. The authorized facility locations shall be as shown in the EA, as supplemented by filed alignment sheets and maps. **As soon as they are available, and before the start of construction**, Columbia shall file with the Secretary any revised detailed survey alignment maps/sheets at a scale not smaller than 1:6,000 with station positions for all facilities approved by the Order. All requests for modifications of environmental conditions of the Order or site-specific clearances must be written and must reference

locations designated on these alignment maps/sheets.

Columbia's exercise of eminent domain authority granted under NGA section 7(h) in any condemnation proceedings related to the Order must be consistent with these authorized facilities and locations. Columbia's right of eminent domain granted under NGA section 7(h) does not authorize it to increase the size of its natural gas pipelines and facilities to accommodate future needs or to acquire a right-of-way for a pipeline to transport a commodity other than natural gas.

5. Columbia shall file with the Secretary detailed alignment maps/sheets and aerial photographs at a scale not smaller than 1:6,000 identifying all route realignments or facility relocations, and staging areas, pipe storage yards, new access roads, and other areas that would be used or disturbed and have not been previously identified in filings with the Secretary. Approval for each of these areas must be explicitly requested in writing. For each area, the request must include a description of the existing land use/cover type, documentation of landowner approval, whether any cultural resources or federally listed threatened or endangered species would be affected, and whether any other environmentally sensitive areas are within or abutting the area. All areas shall be clearly identified on the maps/sheets/aerial photographs. Each area must be approved in writing by the Director of OEP, or the Director's designee, **before construction in or near that area**.

   This requirement does not apply to extra workspace allowed by the FERC Plan and/or minor field realignments per landowner needs and requirements which do not affect other landowners or sensitive environmental areas such as wetlands.

   Examples of alterations requiring approval include all route realignments and facility location changes resulting from:

   a. implementation of cultural resources mitigation measures;
   b. implementation of endangered, threatened, or special concern species mitigation measures;
   c. recommendations by state regulatory authorities; and
   d. agreements with individual landowners that affect other landowners or could affect sensitive environmental areas.

6. **Within 60 days of the acceptance of the authorization and before construction begins**, Columbia shall file an Implementation Plan with the Secretary for review and written approval by the Director of OEP, or the Director's designee. Columbia must file revisions to the plan as schedules change. The plan shall identify:

   a. how Columbia would implement the construction procedures and mitigation measures described in its application and supplements (including responses to staff data requests), identified in the EA, and required by the Order;
   b. how Columbia would incorporate these requirements into the contract bid documents, construction contracts (especially penalty clauses and specifications),

and construction drawings so that the mitigation required at each site is clear to onsite construction and inspection personnel;

c. the number of EIs assigned, and how the company would ensure that sufficient personnel are available to implement the environmental mitigation;
d. company personnel, including EIs and contractors, who would receive copies of the appropriate material;
e. the location and dates of the environmental compliance training and instructions Columbia would give to all personnel involved with construction and restoration (initial and refresher training as the Project progresses and personnel change);
f. the company personnel (if known) and specific portion of Columbia' organization having responsibility for compliance;
g. the procedures (including use of contract penalties) Columbia would follow if noncompliance occurs; and
h. for each discrete facility, a Gantt or PERT chart (or similar project scheduling diagram), and dates for:
    (1) the completion of all required surveys and reports;
    (2) the environmental compliance training of onsite personnel;
    (3) the start of construction; and
    (4) the start and completion of restoration.

7. Columbia shall employ at least one EI for the Project. The EI shall be:

    a. responsible for monitoring and ensuring compliance with all mitigation measures required by the Order and other grants, permits, certificates, or other authorizing documents;
    b. responsible for evaluating the construction contractor's implementation of the environmental mitigation measures required in the contract (see condition 6 above) and any other authorizing document;
    c. empowered to order correction of acts that violate the environmental conditions of the Order, and any other authorizing document;
    d. responsible for documenting compliance with the environmental conditions of the Order, as well as any environmental conditions/permit requirements imposed by other federal, state, or local agencies; and
    e. responsible for maintaining status reports.

8. Beginning with the filing of its Implementation Plan, Columbia shall file updated status reports with the Secretary on a **biweekly** basis until all construction, abandonment, and restoration activities are complete. On request, these status reports would also be provided to other federal and state agencies with permitting responsibilities. Status reports shall include:

    a. an update on Columbia' efforts to obtain the necessary federal authorizations;
    b. project schedule, including current construction status of the project and work planned for the following reporting period;
    c. a listing of all problems encountered and each instance of noncompliance observed by the EI during the reporting period (both for the conditions imposed

by the Commission and any environmental conditions/permit requirements imposed by other federal, state, or local agencies);

d. a description of the corrective actions implemented in response to all instances of noncompliance;
e. the effectiveness of all corrective actions implemented;
f. a description of any landowner/resident complaints which may relate to compliance with the requirements of the Order, and the measures taken to satisfy their concerns; and
g. copies of any correspondence received by Columbia from other federal, state, or local permitting agencies concerning instances of noncompliance, and Columbia' response.

9. Columbia must receive written authorization from the Director of OEP, or the Director's designee, **before commencing construction or abandonment of any Project facilities**. To obtain such authorization, Columbia must file with the Secretary documentation that it has received all applicable authorizations required under federal law (or evidence of waiver thereof).

10. **Within 5 days of receipt of a water quality certification issued by Kentucky Energy and Environmental Cabinet – Department of Environmental Protection,** Columbia shall file the complete certification, including all conditions. All conditions attached to the water quality certification constitute mandatory conditions of the Certificate Order. **Prior to construction**, Columbia shall file, for review and written approval of the Director of OEP, or the Director's designee, any revisions to its project design necessary to comply with the water quality certification conditions.

11. **Within 5 days of the final determination of the use of the Nationwide Permit 12 issued by the United States Army Corps of Engineers,** Columbia shall file the complete water quality certification issued categorically by the Ohio Environmental Protection Agency, including all conditions. All conditions attached to the water quality certification constitute mandatory conditions of the Certificate Order. **Prior to construction**, Columbia shall file, for review and written approval by the Director of OEP, or the Director's designee, any revisions to its project design necessary to comply with the water quality certification conditions.

12. Columbia shall **not begin** construction activities **until**:

a. FERC staff receives comments from the FWS regarding the proposed action;
b. FERC staff completes ESA consultation with the FWS; and
c. Columbia has received written notification from the Director of OEP, or the Director's designee, that construction or use of mitigation may begin.

13. Columbia shall file a noise survey with the Secretary **no later than 60 days** after placing the Cold Spring Meter Station and Pendleton Meter Station into service. If a full flow rate noise survey at the stations' maximum design capacity is not possible, Columbia shall provide an interim survey at the maximum possible flow rate and provide the full flow rate survey **within 6 months**. If the noise attributable to the operation of the Cold

Spring Meter Station and Pendleton Meter Station under interim or full flow rate conditions exceeds an L_{dn} of 55 dBA at any nearby NSAs, Columbia shall file a report on what changes are needed and shall install additional noise controls to meet the level **within 1 year** of the in-service date. Columbia shall confirm compliance with the above requirement by filing a second noise survey with the Secretary **no later than 60 days** after it installs the additional noise controls.

APPENDIX A
PROPOSED SITE-SPECIFIC DEVIATIONS FROM THE FERC PLAN AND PROCEDURES

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
Line AM36						
Construction Corridor	0.00 – 24.62	N/A	Plan Section IV.A.2	Construction Corridor of 125 feet	Additional space needed to provide for safe and efficient construction of the pipeline through hilly terrain and steep slope conditions.	N/A
ATWS-AM36-001	0.06	SP10004	Procedures Section V.B.2	ATWS within 50 feet of waterbody	Additional temporary workspace (ATWS) location is necessary for additional spoil storage associated with stream crossing and limited usable workspace due to challenging terrain in adjacent areas.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-005	0.45	SP10007	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage at the Mary Ingles Hwy crossing and is located on the south side due to workspace restrictions on the north from a steep drop-off.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-016	2.45	SP10016	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage due to reduced usable workspace where two ravines intersect within the temporary workspace.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-018	2.69	SP10020	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage due to steep sidehill constraints and to support workspace crossover.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
ATWS-AM36-020	2.78	SP10029; SP10030; SP10031	Procedures Section V.B.2	ATWS within 50 feet of waterbody; ATWS within a waterbody	ATWS is required for additional spoil storage associated with the stream crossing to the north, and where an ephemeral stream limits usable temporary workspace.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from SP10029.
ATWS-AM36-021	2.80	SP10029	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS is necessary for additional spoil storage associated with the stream crossing to the south. Limited usable workspace due to challenging terrain in adjacent areas.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-023	3.12	SP10058	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS required for spoil storage for stream and road crossing. Confined due to wetland, confined by adjacent streams. ATWS is required for additional spoil storage associated with the stream crossing to the south, with temporary workspace constrained by a private drive and a rock ledge.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody
ATWS-AM36-025	3.15	SP10033	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS is required for additional spoil storage associated with the Hwy 154 road bore, as well as equipment access and grading due to the steep rise toward the roadway. Loss of use of temporary workspace where rock construction entrance will be installed.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody
ATWS-AM36-027	3.21	SP10034	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for personnel parking and travel access to the right-of-way at the top of the hill to the west and will connect to the proposed access road off Hwy 154. Limited usable space due to challenging terrain in adjacent areas.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
ATWS-AM36-035	4.34	SP10041; OWP10004; WP10005_PSS	Procedures Section V.B.2 and VI.B.1	ATWS within 50 feet of waterbodies; ATWS within 50 feet of wetland	ATWS location is necessary for additional spoil storage associated with the stream crossing to the east and to support pipeline and workspace crossover.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into adjacent waterbodies and the wetland. Spoil will not be stored within 10 feet from waterbodies and wetlands.
ATWS-AM36-036	4.34	SP10041; OWP10004; WP10005_PSS	Procedures Section V.B.2 and VI.B.1	ATWS within 50 feet of waterbodies; ATWS within 50 feet of wetland	ATWS location is necessary for additional spoil storage associated with the stream crossing to the east and to support pipeline and workspace crossover.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into adjacent waterbodies and the wetland. Spoil will not be stored within 10 feet from waterbodies and wetlands. ATWS is set back greater than 10 feet from waterbodies and wetland.
ATWS-AM36-038	4.70	SP10043	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS needed for spoil storage for stream crossing. Workspace is limited due to height of spoil storage due to power lines.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-043	5.00	SP10045; SP10046	Procedures Section V.B.2	ATWS within 50 feet of waterbody; ATWS within waterbody	ATWS location is necessary for additional spoil storage associated with the stream crossing to the east and needed at that location due to the location of the pipeline and workspace.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from SP10045.

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
ATWS-AM36-052	5.60	SP10050; SP10051	Procedures Section V.B.2	ATWS within 50 feet of waterbodies	ATWS location is necessary for spoil storage for stream crossing. Multiple streams within temporary workspace; steep terrain, pipeline bend to cross stream at perpendicular.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from waterbodies.
ATWS-AM36-071	6.89	SP7020	Procedures Section V.B.2	ATWS within waterbody	ATWS location is necessary for additional spoil storage associated with the stream crossing to the north and to support workspace crossover which minimizes stream impact.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody.
ATWS-AM36-072	6.95	SP7022	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage due to sidehill alignment at a pipeline point of inflection (PI), requiring a relief area for accumulated spoils.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-075	7.36	SP7002	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location required for spoil storage for two stream crossings and road and powerline crossings. Workspace limitations due to location of low powerlines.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-091	8.73	SP7009	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage to manage material runoff due to steep slopes.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
ATWS-AM36-113	11.28	SP7027; SP7028	Procedures Section V.B.2	ATWS within 50 feet of waterbodies	ATWS location is necessary for additional spoil storage associated with the PI, stream crossing to the north and the Stevens Branch Road crossing to the south. Constrained by location of stream and road.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-122	12.35	SP6007	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with the PI and stream crossing to the south.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-132	12.88	SP6010	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with the stream and road bore as well as PIs. Limited usable space in difficult terrain.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-134	13.04	SP6012	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with the stream crossing to the south and PI to the north. Limited usable workspace in challenging terrain.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-138	13.82	SP6145	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for spoil storage associated with the stream crossing to the south and PI to the north. Also needed for spoil storage due to powerline crossing.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
ATWS-AM36-142	14.12	SP6042	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage due to limited temporary workspace, constrained by a dwelling and swimming pool to the south and landowner request to limit clearing.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-148	14.59	SP6021; SP6091; SP6019; SP6020	Procedures Section V.B.2	ATWS within waterbodies; ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with the stream crossing and the Upper Lick Branch Rd crossing to the north and to support workspace crossover needed to avoid residential impacts to the north of Upper Lick Branch Road.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from SP6021 and SP6020.
ATWS-AM36-149	14.60	SP6020	Procedures Section V.B.2	ATWS within waterbody	ATWS location is necessary for additional spoil storage associated with the stream crossing and the Upper Lick Branch Rd crossing to the and to support workspace crossover needed to avoid residential impacts to the north of Upper Lick Branch Road.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. Spoil will not be stored within 10 feet from waterbody.
ATWS-AM36-152	15.00	SP6026	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage to support workspace crossover and cut and cap abandonment of Line AM-09. Also constrained by residence.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-158	15.79	SP10167	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with the stream crossing to the north.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
ATWS-AM36-160	16.02	SP10162	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with the stream crossing to the south and PI. Limited usable workspace available.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody.
ATWS-AM36-163	16.28	SP6035; SP6041; SP6043	Procedures Section V.B.2	ATWS within 50 feet of waterbodies	ATWS location is necessary for additional spoil storage associated with multiple stream crossings, the Four Mile Rd crossing, workspace crossover, and multiple stream crossings.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from waterbodies.
ATWS-AM36-164	16.31	SP6044	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with multiple stream crossings, the Four Mile Rd crossing, workspace crossover, and multiple stream crossings.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-171	17.14	SP6070	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage due to the stream crossing to the west and constraints due to adjacent steep terrain and PI.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-172	17.15	SP6070; SP6071	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is required for additional spoil storage associated with the stream crossings to the east and west. ATWS is within previously disturbed hayfield.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from SP6070.

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
ATWS-AM36-174	17.21	SP6071	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage for the stream crossing to the east. ATWS is within previously disturbed hayfield.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-184	18.55	SP6062	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with proposed bore of E Alexandria Pike Rd and adjacent stream and adjacent open cut of private drive and stream crossings.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody.
ATWS-AM36-190	19.20	SP6076	Procedures Section V.B.2	ATWS within waterbody	ATWS location is required for additional spoil storage associated with the stream crossing and E Low Gap Rd crossing to the east and to support workspace crossover, as well as a 3 PIs to minimize residential impacts and due to steep terrain (drop-off) along existing right of way.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody.
ATWS-AM36-191	19.25	SP6075	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage for stream crossing due to steep terrain, two PIs to the north and south.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody
ATWS-AM36-199	21.32	SP6092	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with the stream crossing to the east and hot line (Line AM-04) to the south.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-200	21.36	SP6093	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with the stream crossing and workspace crossover to minimize residential impacts.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
ATWS-AM36-204	21.74	SP3008	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with stream crossing to the north and to reduce residential impacts to the south.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-205	21.74	SP3008	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with stream crossing to the south.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-208	21.91	SP6099	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with the stream crossing to the north. Confined by steep terrain/rock outcroppings.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-213	22.11	SP6104	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage due to ridgetop alignment and a steep grade to the east, with placement at the base of the slope for spoil collection.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-215	22.28	SP6105	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for spoil storage associated with the Industrial Rd bore crossing to support safe construction at the road bore.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-217	22.33	SP6108	Procedures Section V.B.2	ATWS within waterbody	ATWS location is necessary for additional spoil storage due to steep terrain and ravines restricting usable workspace and multiple PIs.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies.

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
ATWS-AM36-220	23.06	SP6114; SP6115; SP6116	Procedures Section V.B.2	ATWS within 50 feet of waterbodies	ATWS location is necessary for additional spoil storage associated with the proposed bore of Winters Ln and streams, workspace crossover, and multiple PIs.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies.
ATWS-AM36-221	23.09	SP10105	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage associated with the proposed bore of Winters Ln and streams, workspace crossover, and multiple PIs.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-222	23.11	SP6116	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage and parking associated with the proposed bore of Winters Ln and streams, workspace crossover, and multiple PIs.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody. ATWS is set back greater than 10 feet from waterbody.
ATWS-AM36-225	23.83	SP3004; SP3005; SP3006; SP3007	Procedures Section V.B.2	ATWS within waterbodies	ATWS location is necessary for construction and grading of permanent access road PAR-AM36-23.82 to new meter station and associated spoil storage.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies.
ATWS-AM36-226	24.05	SP3007	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for construction and grading of permanent access road PAR-AM36-23.82 to new meter station and associated spoil storage.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbody.
ATWS-AM36-227	23.86	SP3003; SP3001	Procedures Section V.B.2	ATWS within waterbody; ATWS within 50 feet of waterbody	ATWS location is necessary for construction and grading of permanent access road PAR-AM36-23.82 to new meter station and associated spoil storage	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies.

Table A-1 Proposed Site-Specific Deviations to the FERC Plan and Procedures						
Workspace Type / ID	**Milepost**	**Waterbody / Wetland ID**	**Section of Plan and Procedures**	**Deviations to FERC Plan and Procedures**	**Justification**	**Equal Compliance Measures**
ATWS-AM36-233	24.56	SP6135	Procedures Section V.B.2	ATWS within a waterbody	ATWS location is necessary to accommodate HDD pullback pipe due to limited available workspace.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies.
Line EM-7						
ATWS-EM7-024	4.96	SP10096	Procedures Section V.B.2	ATWS within 50 feet of waterbody	ATWS location is necessary for additional spoil storage due to proposed bore of railroad and Scrubgrass Creek Rd. Constrained by terrain by railroad right of way to north.	Temporary erosion and sediment controls to be implemented as needed to prevent the flow of spoil or heavily silt-laden water into waterbodies.
Cold Spring Lateral						
Construction Corridor	0.00 – 0.36	N/A	Plan Section IV.A.2	Construction Corridor of 125 feet	Additional space needed to provide for safe and efficient construction of the pipeline through hilly terrain and steep slope conditions.	N/A
Source: Columbia's May 12, 2026, response to FERC's May 5, 2026, Environmental Information Request. FERC eLibrary Accession No. 20260512-5178.						

APPENDIX B
WATERBODIES CROSSED OR OTHERWISE AFFECTED BY THE PROJECT

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
New Pipeline Facilities									
Line AM36									
0.06	SP10004	Unnamed Tributary of Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2	2	Open-Cut/Timber Mats
0.18	SP10005	Unnamed Tributary of Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	3	Open-Cut/Timber Mats
0.35 / TAR-AM36-0.35 (SA-AM36-001)	SP10006 (1st impact)	Unnamed Tributary of Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Existing Culvert/Timber Mats
0.37 / SA-AM36-016	SP10006 (2nd impact)	Unnamed Tributary of Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Existing Culvert/Timber Mats
0.38	SP10006 (3rd impact)	Unnamed Tributary of Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2	2	Open-Cut/Timber Mats
0.39	SP10008	Unnamed Tributary of Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1	1	Open-Cut/Timber Mats
1.35 / TAR-AM36-1.35	SP10014 (1st impact)	Unnamed Tributary of Jimmie Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10015 (1st impact)	Unnamed Tributary of Jimmie Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
1.35	SP10015 (2nd impact)	Unnamed Tributary of Jimmie Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1	1	Open-Cut/Timber Mats
1.37	SP10014 (2nd impact)	Unnamed Tributary of Jimmie Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1	1	Open-Cut/Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
1.39	SP10013	Jimmie Creek	PCR, SCR, DWS	Warmwater	Perennial	Intermediate	22	18	Open-Cut/Timber Mats
1.57	SP10012	Unnamed Tributary of Jimmie Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	11	10	Open-Cut/Timber Mats
1.64	SP3017	Unnamed Tributary of Jimmie Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	3	Open-Cut/Timber Mats
1.87	SP10010	Unnamed Tributary of Jimmie Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	20	10	Open-Cut/ Timber Mats
1.87	SP10011	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	5	Timber Mats
2.44	SP10016	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	6	6	Open-Cut/Timber Mats
2.44	SP10018	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
2.61	SP10019	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	5	Open-Cut Timber Mats
2.71 / TAR-AM36-2.71	SP10020 (1st Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
2.71 / TAR-AM36-2.71	SP10020 (2nd Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
2.71	SP10020 (3rd Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1	1	Open-Cut/Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
2.71 / TAR-AM36-2.70 (CY-AM36-002)	SP10022	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	5	Timber Mats
2.77	SP10030 (1st Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
2.78	SP10021	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
2.78	SP10030 (2nd Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
2.78	SP10031	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
2.79	SP10029 (1st Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	5	4	Open-Cut/Timber Mats
3.11	SP10032	Stepstone Creek	PCR, SCR, DWS	Warmwater	Perennial	Minor	29	10	Open-Cut/Timber Mats
3.19	SP10033 (1st Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	4	Open-Cut/Timber Mats
3.19 / TAR-AM36-3.19	SP10033 (2nd Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
3.20	SP10034 (1st Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
3.21 / TAR-AM36-3.21	SP10034 (2nd Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
3.58 / TAR-AM36-3.58	SP10035 (1st Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
3.73	SP10035 (2nd Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1	8	Open-Cut/Timber Mats
3.76	SP10037 (1st Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	2	1	Open-Cut/Timber Mats
3.78 / TAR-AM36-3.78	SP10037 (2nd Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	0 [a]	2	Timber Mats
4.31	SP10041 (1st Impact)	Unnamed Tributary of Stepstone Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	9	5	Open-Cut/Timber Mats
4.60	SP10042	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	2	Open-Cut/Timber Mats
4.67	SP10043	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	6	5	Open-Cut/Timber Mats
4.76	SP10044	Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	4	Open-Cut/Timber Mats
4.98	SP10045 (1st Impact)	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	6	6	Open-Cut/Timber Mats
5.00	SP10046 (1st Impact)	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
5.00	SP10046 (2nd Impact)	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
5.45	SP10048 / SP10048_DT	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	7	5	Open-Cut/Timber Mats
5.60	SP10050	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	6	4	Open-Cut/Timber Mats
5.60	SP10051	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
5.79	SP10052	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	14	10	Open-Cut/Timber Mats
6.03	SP10053	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	12	10	Open-Cut/Timber Mats
6.03	SP10054	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	5	Timber Mats
6.71	SP7019 (1st Impact)	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	4	Open-Cut/Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
6.89	SP7020	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
6.91	SP7022 (1st Impact)	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	3	Open-Cut/Timber Mats
6.97	SP7022 (2nd Impact)	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0	3	Timber Mats
7.33	SP7001	Twelvemile Creek	PCR, SCR, DWS	Warmwater	Perennial	Intermediate	17	25	Open-Cut/Timber Mats
7.32 / TAR-AM36-7.32	SP10168	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	10	Existing Culvert/Timber Mats
7.35	SP7002	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	5	Open-Cut/Timber Mats
7.59	SP7004	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	9	6	Open-Cut/Timber Mats
8.11	SP7005	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	4	Open-Cut/Timber Mats
8.12 / CP Easement	SP3024	Unnamed Tributary to Fourmile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0	3	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
8.17	OWP7002	Manmade Pond	PCR, SCR, DWS	Warmwater	N/A	Intermediate	0 [a]	44	Timber Mats
8.34	SP7007	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	5	Open-Cut/Timber Mats
8.55	SP7008 (1st Impact)	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2	2	Open-Cut/Timber Mats
8.61	SP7008 (2nd Impact)	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2	2	Open-Cut/Timber Mats
8.62	SP7009	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
9.00	SP7029	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	3	3	Open-Cut/Timber Mats
10.17	SP7012	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	8	8	Open-Cut/Timber Mats
10.39	SP7013	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	18	5	Open-Cut/Timber Mats
10.45	SP7014	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
10.80	SP7015	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	16	8	Open-Cut/Timber Mats
11.30	SP7027	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
11.31	SP7028	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Perennial	Intermediate	24	15	Open-Cut/Timber Mats
11.59	SP6001	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	10	9	Open-Cut/Timber Mats
11.99	SP6003	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	10	8	Open-Cut/Timber Mats
12.13	SP9004_DT	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	3	3	Open-Cut/Timber Mats
12.17	SP9003_DT	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	3	3	Open-Cut/Timber Mats
12.23	SP6004	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	3	3	Open-Cut/Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
12.34	SP6007	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	6	5	Open-Cut/Timber Mats
12.64	SP9002_DT	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2	2	Open-Cut/Timber Mats
12.74	SP6008 / SP6008_DT	Unnamed Tributary of Brush Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	3	3	Open-Cut/Timber Mats
12.75	SP9001_DT (1st Impact)	Unnamed Tributary of Brush Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
12.77	SP9001_DT (2nd Impact)	Unnamed Tributary of Brush Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
12.93	SP6010	Brush Creek	PCR, SCR, DWS	Warmwater	Perennial	Intermediate	27	15	Open-Cut/Timber Mats
13.02	SP6012	Unnamed Tributary of Brush Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	19	9	Open-Cut/Timber Mats
13.27	SP6013	Unnamed Tributary of Brush Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	3	Open-Cut/Timber Mats
13.82	SP6145	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Intermittent	Intermediate	20	12	Open-Cut/Timber Mats
13.94	SP6144	Unnamed Tributary of Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2	1	Open-Cut/Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
14.38	SP6139 (1st Impact)	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
14.38	SP6140	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
14.40	SP6139 (2nd Impact)	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
14.58	SP6091	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
14.61	SP6019	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
14.62	SP6020 (1st Impact)	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Intermittent	Minor	7	6	Open-Cut/Timber Mats
14.62	SP6020 (2nd Impact)	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Intermittent	Minor	0 [a]	6	Timber Mats
14.64	SP3011	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
14.64	SP6023 (1st Impact)	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
14.65	SP6023 (2nd Impact)	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
14.95	SP6025	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	5	Open-Cut/Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
14.95	SP6026	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
15.13	SP6027	Unnamed Tributary of Lick Branch	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
15.46	SP10172 (1st Impact)	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	10	7	Open-Cut/Timber Mats
15.47 / TAR-AM36-15.47	SP10172 (2nd Impact)	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	7	Timber Mats
	SP10173	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10171	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
	SP10170	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
	SP10169	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
15.57	SP6031	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [c]	2	Bore/Timber Mats
15.61	SP6032	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	8	8	Bore/Timber Mats
15.81	SP10167	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	15	5	Open-Cut/Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
15.90	SP10165	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
15.91	SP10164	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
16.01	SP10162	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	13	10	Open-Cut/Timber Mats
16.20 / CP Easement	SP6036 (1st Impact)	Unnamed Tributary to Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	6	Timber Mats
16.21	SP6035	Four Mile Creek	PCR, SCR, DWS	Warmwater	Perennial	Intermediate	31	12	Open-Cut/Timber Mats
16.21	SP6036 (2nd Impact)	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	6	Timber Mats
16.24	SP6037	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
16.24	SP6038	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	5	Open-Cut/Timber Mats
16.26	SP6040	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	4	Open-Cut/Timber Mats
16.28	SP6041	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	6	5	Open-Cut/Timber Mats
16.28	SP6042	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
16.29	SP6043	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	5	Open-Cut/Timber Mats
16.33	SP6044 (1st Impact) / SP6045	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	4	Open-Cut/Timber Mats
16.35	SP6044 (2nd Impact)	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
16.47	SP6047	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	4	Open-Cut/Timber Mats
16.47	SP6048	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
16.87	SP6053	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	10	8	Open-Cut/Timber Mats
16.88	SP6054	Unnamed Tributary of Four Mile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
17.14	SP6070	Tug Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	19	10	Open-Cut/Timber Mats
17.20	SP6071	Tug Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	8	8	Open-Cut/Timber Mats
17.56	SP6067	Unnamed Tributary to Tug Creek	PCR, SCR, DWS	Warmwater	Perennial	Minor	10	10	Open-Cut/Timber Mats
17.82	SP6056	Unnamed Tributary to Tug Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
18.16	SP6057	Unnamed Tributary to Tug Creek	PCR, SCR, DWS	Warmwater	Intermittent	Intermediate	23	12	Open-Cut/Timber Mats
18.16	SP6058	Unnamed Tributary to Tug Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
18.19	SP6059	Unnamed Tributary to Tug Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
18.46	SP6060	Unnamed Tributary to Old Upper Tug Fork	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
18.52	SP6061	Unnamed Tributary to Old Upper Tug Fork	PCR, SCR, DWS	Warmwater	Intermittent	Minor	0 [a]	6	Timber Mats
18.52	SP6062	Unnamed Tributary to Old Upper Tug Fork	PCR, SCR, DWS	Warmwater	Perennial	Intermediate	16	12	Open-Cut/Timber Mats
18.57 / TAR-AM36-18.57	SP10147	Unnamed Tributary to Old Upper Tug Fork	PCR, SCR, DWS	Warmwater	Intermittent	Intermediate	0 [a]	15	Existing Culvert/Timber Mats
18.57	SP6063	Unnamed Tributary to Old Upper Tug Fork	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	4	Open-Cut/Timber Mats
18.84	SP6064 (1st Impact)	Unnamed Tributary to Old Upper Tug Fork	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2	2	Open-Cut/Timber Mats
18.95	SP6065 (1st Impact)	Unnamed Tributary to Old Upper Tug Fork	PCR, SCR, DWS	Warmwater	Intermittent	Minor	8	8	Open-Cut/Timber Mats
19.04	SP6072	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
19.23	SP6073	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
19.24	SP6076	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	14	9	Open-Cut/Timber mats
19.26 / TAR-AM36-19.26	SP10146	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Intermittent	Intermediate	0 [a]	15	Existing Culvert/Timber Mats
	SP6075	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	5	Existing Culvert/Timber Mats
	SP6178	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	0 [a]	9	Existing Culvert/Timber Mats
20.10	SP6079	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
20.21	SP6080	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Intermittent	Intermediate	19	15	Open-Cut/Timber Mats
20.42	SP6084	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	3	3	Open-Cut/Timber Mats
20.43	SP6083	Unnamed Tributary to Old Upper Tug Fork	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	7	3	Open-Cut/Timber Mats
20.50	SP6085	Owl Creek	PCR, SCR, DWS	Warmwater	Perennial	Minor	21	10	Open-Cut/Timber Mats
20.53	SP6086	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2	2	Open-Cut/Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
20.86	SP6087	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
20.86	SP6088	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
20.91	SP6089	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
21.20	SP6183	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	4	Timber Mats
21.32	SP6092 (1st Impact)	Unnamed Tributary to Owl Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	12	10	Open-Cut/Timber Mats
21.73	SP3008	Unnamed Tributary to Duck Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	5	Open-Cut/Timber Mats
21.93	SP6099	Unnamed Tributary to Duck Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	8	8	Open-Cut/Timber Mats
22.04	SP6101	Unnamed Tributary to Duck Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	6	6	Open-Cut/Timber Mats
22.04	SP6103	Unnamed Tributary to Duck Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
22.31	SP6106	Duck Creek	PCR, SCR, DWS	Warmwater	Perennial	Intermediate	30	22	Open-Cut/Timber Mats
22.46	SP6108	Unnamed Tributary to Duck Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	19	10	Open-Cut/Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
22.50 / TAR-AM36-22.50	SP6110	Unnamed Tributary to Duck Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Intermediate	0 [a]	12	Existing Culvert/Timber Mats
22.68	SP6111	Unnamed Tributary to Duck Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	5	Open-Cut/Timber Mats
22.94	SP6112	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2	2	Open-Cut/Timber Mats
23.02	SP6114 (1st Impact)	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	5	5	Open-Cut/Timber Mats
23.05	SP6114 (2nd Impact)	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	5	Timber Mats
23.08	SP6117	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2	2	Open-Cut
23.08	SP6116	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Intermittent	Minor	12	10	Open-Cut/Timber Mats
23.40 / TAR-AM36-23.40	SP10101	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10102	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10103	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10104	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
23.37	SP6120	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	4	Open-Cut/Timber Mats
23.51	SP6123	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
23.96	SP3004	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
24.01	SP3005	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
24.04	SP10098 (1st Impact)	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
24.04	SP3003	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
24.04	SP3007	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	5	Timber Mats
24.21	SP6181	Ohio River	ALH, WS, R	Warmwater	Perennial	Major	1,532 [c]	1,532	HDD
24.62	SP6135 (1st Impact)	Threemile Creek	ALH, WS, R	Warmwater	Intermittent	Minor	0 [a]	5	Timber Mats
24.62 / SA-AM36-015	SP6135 (2nd Impact)	Threemile Creek	ALH, WS, R	Warmwater	Intermittent	Minor	0 [a]	5	Timber Mats
Line EM-7									
0.70 / TAR-EM7-0.70	SP10063	Unnamed Tributary to Somerset Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
0.75	SP10064	Unnamed Tributary to Somerset Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1	1	Open-Cut/Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
1.09	SP10070	Unnamed Tributary to North Fork Somerset Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
1.13	SP10072	Unnamed Tributary to North Fork Somerset Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	6	Timber Mats
1.46	SP10074	Unnamed Tributary to North Fork Somerset Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
1.75	SP10075	Unnamed Tributary to North Fork Somerset Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4	1	Open-Cut/Timber Mats
2.41 / TAR-EM7-2.41	SP9007_DT	Unnamed Tributary to Somerset Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 a	4	Existing Culvert / Timber Mats
2.67	SP10076	Unnamed Tributary to North Fork Somerset Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
3.93	SP10083	Unnamed Tributary to North Fork Somerset Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
3.94	SP10084	Unnamed Tributary to North Fork Somerset Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
4.30	SP10086	Unnamed Tributary to Big Brushy Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
4.51	SP10093	Unnamed Tributary to Big Brushy Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1	1	Open-Cut/Timber Mats
4.53	SP10091	Unnamed Tributary to Big Brushy Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
4.8	SP10094	Unnamed Tributary to Big Brushy Creek	PCR, SCR, DWS	Warmwater	Intermittent	Intermediate	5	12	Open-Cut/Timber Mats
4.96	SP10096	Unnamed Tributary to Big Brushy Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	16	10	Open-Cut/Timber Mats
5.02	SP10097 (1st Impact)	Unnamed Tributary to Big Brushy Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
Black River Lateral									
There are no waterbodies impacted along the Black River Lateral.									
Cold Spring Lateral									
0.23	SP10129 (1st Impact)	Duck Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Intermediate	12	12	Open-Cut/Timber Mats
0.23	SP10129 (2nd Impact)	Duck Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Intermediate	0 [a]	12	Timber Mats
0.25	SP10131	Unnamed Tributary to Duck Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
Kellogg Station Lateral									
There are no waterbodies impacted along the Kellogg Station Lateral.									
Existing Pipeline Facilities									
Line AM04 Abandonment									

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
2.28 / ABD-WLK-001	SP6162	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1 [b]	1	Foot Traffic Only
2.30 / ABD-WLK-001	SP6166	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	4 [b]	4	Foot Traffic Only
2.31 / ABD-WLK-001	SP6168	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1 [b]	1	Foot Traffic Only
2.37 / ABD-WLK-001	SP6169	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2 [b]	1	Foot Traffic Only
2.67 / TAR-ABD-001	SP10020 (4th Impact)	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1 [b]	1	Timber Mats
2.74 / TAR-ABD-001	SP10029 (2nd Impact)	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	4 [b]	4	Timber Mats
2.97	SP6161	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	3 [b]	3	Timber Mats
3.04	SP6159	Stepstone Creek	PCR, SCR, DWS	Warmwater	Intermittent	Intermediate	22	15	Open-Cut/Timber Mats
3.15	SP6158	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	3 [b]	3	Timber Mat
3.99	SP10041 (2nd Impact)	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	6 [b]	5	Timber Mats
4.65 / TAR-ABD-005	SP10045 (2nd Impact)	Unnamed Tributary of Flag Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	6 [b]	6	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
6.37 / TAR-ABD-008	SP7019 (2nd Impact)	Unnamed Tributary to Flagg Spring Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	6 [b]	4	Timber Mats
8.82	SP7011	Unnamed Tributary to Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	6 [b]	4	Open-Cut /Timber Mats
10.7	SP6180	Unnamed Tributary to Twelvemile Creek	PCR, SCR, DWS	Warmwater	Intermittent	Intermediate	21	15	Open Cut/Timber Mats
12.20	SP6010	Brush Creek	PCR, SCR, DWS	Warmwater	Perennial	Intermediate	27	15	Open-Cut/Timber Mats
12.62	SP3012	Unnamed Tributary to Brush Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [b]	2	Timber Mats
14.33 / TAR-ABD-013	SP10110	Fourmile Creek	PCR, SCR, DWS	Warmwater	Intermittent	Intermediate	0 [a]	20	Timber Mats
	SP10111	Unnamed Tributary to Fourmile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
15.28 / TAR-ABD-014	SP6154	Unnamed Tributary to Fourmile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	3	Timber Mats
16.20 / ABD-WLK-007	SP3010	Unnamed Tributary to Fourmile Creek	PCR, SCR, DWS	Warmwater	Perennial	Minor	0 [b]	2	Foot Traffic Only
17.10	SP6064 (2nd Impact)	Unnamed Tributary to Old Upper Tug Fork	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [b]	8	Timber Mats
17.13 / TAR-ABD-017	SP6065 (2nd Impact)	Unnamed Tributary to Four Mile Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	0 [a]	6	Existing Culvert

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
Line AM09 Abandonment									
0.50	SP6171 (1st Impact)	Unnamed Tributary to Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [b]	2	Timber Mats
4.25	SP3015	Unnamed Tributary to Goose Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	2 [b]	2	Timber Mats
4.42	SP6156	Unnamed Tributary to Goose Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	3 [b]	2	Timber Mats
4.46	SP6155	Unnamed Tributary to Goose Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	1 [b]	1	Timber Mats
12.65 / TAR-ABD-012	SP6151	Unnamed Tributary to Twelvemile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Intermediate	0 [a]	15	Existing Culvert/Timber Mats
19.47	SP6092 (2nd Impact)	Unnamed Tributary to Fourmile Creek	PCR, SCR, DWS	Warmwater	Intermittent	Minor	6	10	Open-Cut /Timber Mats
20.11	SP10125	Unnamed Tributary to Fourmile Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
Line EKY Abandonment									
There are no waterbodies impacted along the Line EKY Abandonment.									
E-Loop Abandonment									
There are no waterbodies impacted along the E-Loop Abandonment.									

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
New Aboveground Facilities									
Line AM36									
TAR-AM36-0.00 (Foster RS)	SP10001	Unnamed Tributary to Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10002	Unnamed Tributary to Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10003	Unnamed Tributary to Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Existing Culvert/Timber Mats
TAR-AM36-0.61 (Bracken MS)	SP6171	Unnamed Tributary to Lick Run	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Existing Culvert/Timber Mats
PAR-AM36-23.82 (UL47 MS)	SP10098 (2nd Impact)	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	New Culvert
	SP3001	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	6	Timber Mats
TAR-AM36-24.00 (UL47 MS)	SP10098 (3rd Impact)	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Existing Culvert/Timber Mats
PAR-AM36-24.62 (Kellogg Station L/R)	SP6135 (2nd Impact)	Unnamed Tributary to Ohio River	PCR, SCR, DWS	Warmwater	Intermittent	Minor	0 [a]	5	Existing Culvert
TAR-BRV-0.08 (Black River MS)	SP10023 (1st Impact)	Unnamed Tributary to Jimmie Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
	SP10023 (2nd Impact)	Unnamed Tributary to Jimmie Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats

Table B-1 Waterbodies Crossed or Otherwise Affected by the Project									
Approximate Milepost / Project Component	**Feature ID**	**Waterbody Name**	**State Water Quality Classification**	**Fishery Certification**	**Flow Regime**	**FERC Classification**	**Pipeline Crossing Length (feet)**	**Approx Waterbody Width (feet)**	**Proposed Crossing Method**
	SP10023 (3rd Impact)	Unnamed Tributary to Jimmie Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
	SP10024 (1st Impact)	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10024 (2nd Impact)	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10025	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10026	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10027 (1st Impact)	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10027 (2nd Impact)	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
	SP10028	Unnamed Tributary to Stepstone Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	1	Timber Mats
Line EM7									
Carlisle MS	SP10097 (2nd Impact)	Unnamed Tributary to Big Brushy Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
TAR-EM7-5.03 (Carlisle MS)	SP10097 (3rd Impact)	Unnamed Tributary to Big Brushy Creek	PCR, SCR, DWS	Warmwater	Ephemeral	Minor	0 [a]	2	Timber Mats
Existing Aboveground Facilities									
There are no waterbodies impacted by existing aboveground facilities.									

Table B-1
Waterbodies Crossed or Otherwise Affected by the Project

Approximate Milepost / Project Component	Feature ID	Waterbody Name	State Water Quality Classification	Fishery Certification	Flow Regime	FERC Classification	Pipeline Crossing Length (feet)	Approx Waterbody Width (feet)	Proposed Crossing Method

Notes: Waterbody data is based on field delineations and desktop analysis.
Source: Columbia's November 10, 2025, Application, Resource Report 2, Appendix 2B. FERC eLibrary Accession No. 20251110-5151.

CP – cathodic protection
CY – contractor yard
DWS – Drinking Water Supply
HDD – horizontal directional drill
L/R – launcher/receiver
MS – meter station
N/A – Not Applicable
PAR – permanent access road
PCR – Primary Contract Recreation
RS – regulator station
SA – staging area
SCR – Secondary Contact Recreation
TAR – temporary access road
[a] Waterbody would not be crossed by the pipeline centerline but is located within the Project footprint.
[b] Waterbody is crossed by the existing pipeline centerline; however, the existing pipeline at this location would be abandoned in place.
[c] Waterbody is located along a segment of pipe that would be installed via HDD/bore.
[d] Waterbody would be crossed via an existing culvert and, therefore, would not be impacted.

APPENDIX C
SUMMARY OF VEGETATION/HABITAT EFFECTS FOR THE PROJECT

Table C-1 Summary of Habitat Effects for the Project (acres) [1]														
Facility	**Agricultural**		**Forest**		**Wetlands**		**Developed**		**Open Land**		**Open Water**		**Project Total**	
	Temp. [a]	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**
New Pipeline Facilities														
Line AM36														
Right-of-Way	10.70	3.22	218.40	68.62	0.78	0.46	11.65	4.24	117.62	31.83	1.78	1.78	**360.93**	**110.15**
Additional Temporary Workspace	0.54	0.00	18.88	0.00	0.00	0.00	3.30	0.00	15.17	0.00	0.00	0.00	**37.89**	**0.00**
Contractor/Staging/Pipe Yards	0.14	0.00	0.05	0.00	0.00	0.00	7.31	0.00	20.68	0.00	0.00	0.00	**28.18**	**0.00**
Access Roads	0.00	0.00	2.52	0.02	0.02	0.00	1.48	0.02	4.16	0.03	0.00	0.00	**8.18**	**0.07**
Cathodic Protection	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.49	0.00	0.00	0.00	**0.51**	**0.00**
Line EM-7														
Right-of-Way	15.84	1.12	4.92	0.08	0.17	0.02	0.22	0.01	40.41	0.51	0.00	0.00	**61.56**	**1.74**
Additional Temporary Workspace	1.23	0.00	0.23	0.00	0.02	0.00	0.11	0.00	1.27	0.00	0.00	0.00	**2.86**	**0.00**
Contractor/Staging/Pipe Yards	6.98	0.00	0.00	0.00	0.00	0.00	0.63	0.00	3.65	0.00	0.00	0.00	**11.26**	**0.00**
Access Roads	0.72	0.00	0.00	0.00	0.00	0.00	2.44	0.00	1.01	0.00	0.00	0.00	**4.17**	**0.00**
Black River Lateral														
Right-of-Way	0.00	0.00	0.81	0.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.81**	**0.53**
Cold Spring Lateral														
Right-of-Way	0.00	0.00	2.28	0.84	0.06	0.02	0.28	0.00	1.83	0.17	0.00	0.00	**4.45**	**1.03**
Additional Temporary Workspace	0.00	0.00	0.20	0.00	0.00	0.00	0.00	0.00	0.64	0.00	0.00	0.00	**0.84**	**0.00**
Kellogg Station Lateral														
Right-of-Way	0.00	0.00	0.07	0.07	0.00	0.00	0.00	0.00	0.29	0.21	0.00	0.00	**0.36**	**0.28**
New Pipeline Facilities Subtotal	***36.15***	***4.34***	***248.38***	***70.16***	***1.05***	***0.50***	***27.42***	***4.27***	***207.22***	***32.75***	***1.78***	***1.78***	***522.00***	***113.80***
Existing Pipeline Facilities														
Line AM04 Abandonment														
Right-of-Way	0.04	0.00	0.45	0.00	0.00	0.00	1.15	0.00	4.95	0.00	0.00	0.00	**6.59**	**0.00**
Access Roads	0.30	0.00	0.31	0.00	0.00	0.00	0.53	0.00	2.19	0.00	0.00	0.00	**3.33**	**0.00**
Line AM09 Abandonment														
Right-of-Way	0.09	0.00	0.17	0.00	0.05	0.00	0.63	0.00	5.59	0.00	0.00	0.00	**6.53**	**0.00**
Access Roads	0.00	0.00	0.00	0.00	0.00	0.00	1.66	0.00	0.20	0.00	0.00	0.00	**1.86**	**0.00**

Table C-1 Summary of Habitat Effects for the Project (acres) [1]														
Facility	**Agricultural**		**Forest**		**Wetlands**		**Developed**		**Open Land**		**Open Water**		**Project Total**	
	Temp. [a]	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**
Line EKY Abandonment														
Right-of-Way	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	**0.01**	**0.00**
E-Loop Abandonment [c]														
Right-of-Way	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**	**0.00**
Existing Pipeline Facilities Subtotal	***0.43***	***0.00***	***0.93***	***0.00***	***0.05***	***0.00***	***3.97***	***0.00***	***12.94***	***0.00***	***0.00***	***0.00***	***18.32***	***0.00***
Pipeline Facilities Subtotal	***36.58***	***4.34***	***249.31***	***70.16***	***1.10***	***0.50***	***31.39***	***4.27***	***220.16***	***32.75***	***1.78***	***1.78***	***540.32***	***113.80***
New Aboveground Facilities														
Line AM36														
Bracken Meter Station / Tap	0.00	0.00	0.05	0.05	0.00	0.00	0.54	0.11	0.82	0.53	0.00	0.00	**1.41**	**0.69**
Black River Tap	0.00	0.00	<0.01	<0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**	**0.00**
MLV-1 / Gubsur Mill Tap	0.00	0.00	0.00	0.00	0.00	0.00	0.33	0.33	0.11	0.11	0.00	0.00	**0.44**	**0.44**
MLV-2	0.00	0.00	0.00	0.00	0.00	0.00	0.11	0.11	0.40	0.40	0.00	0.00	**0.51**	**0.51**
Parkview Meter Station / Tap	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.56	0.49	0.00	0.00	**0.59**	**0.49**
MLV-3 / Cold Spring Launcher & Receiver	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.37	0.37	0.00	0.00	**0.37**	**0.37**
UL47 Meter Station	0.00	0.00	0.64	0.64	0.00	0.00	0.08	0.05	0.40	0.24	0.00	0.00	**1.12**	**0.93**
Kellogg Station / Launcher & Receiver	0.00	0.00	0.27	0.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.27**	**0.27**
Access Roads	0.00	0.00	0.94	0.90	0.00	0.00	1.72	1.52	0.69	0.66	0.00	0.00	**3.35**	**3.08**
New Aboveground Facilities Subtotal	***0.00***	***0.00***	***1.90***	***1.86***	***0.00***	***0.00***	***2.81***	***2.12***	***3.35***	***2.80***	***0.00***	***0.00***	***8.06***	***6.78***
Existing Aboveground Facilities														
Line AM36														
Foster Regulator Station	0.00	0.00	1.16	0.17	0.00	0.00	2.55	0.52	0.00	0.00	0.00	0.00	**3.71**	**0.69**
Black River Meter Station	0.00	0.00	0.35	0.04	0.00	0.00	0.00	0.00	0.26	0.00	0.00	0.00	**0.61**	**0.04**
Pendleton Meter Station	0.00	0.00	0.34	0.34	0.00	0.00	0.08	0.01	0.25	0.08	0.00	0.00	**0.67**	**0.43**
Alexandria Meter Station	0.00	0.00	0.00	0.00	0.00	0.00	2.17	0.07	0.51	0.51	0.00	0.00	**2.68**	**0.58**
Cold Spring Meter Station	0.00	0.00	0.12	0.00	0.00	0.00	1.26	0.63	1.21	0.01	0.00	0.00	**2.59**	**0.64**
Access Roads	0.00	0.00	0.24	0.00	0.00	0.00	5.82	2.66	0.08	0.05	0.00	0.00	**6.14**	**2.71**
Line EM-7														

Table C-1
Summary of Habitat Effects for the Project (acres) [1]

Facility	Agricultural		Forest		Wetlands		Developed		Open Land		Open Water		Project Total	
	Temp. [a]	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**
Moorefield Site	0.00	0.00	0.00	0.00	0.00	0.00	0.29	0.00	<0.01	0.00	0.00	0.00	**0.29**	**0.00**
Carlisle Site	0.00	0.00	0.00	0.00	0.00	0.00	0.42	0.00	0.00	0.00	0.00	0.00	**0.42**	**0.00**
Access Roads	0.00	0.00	0.00	0.00	0.00	0.00	0.16	0.00	0.03	0.00	0.00	0.00	**0.19**	**0.00**
Existing Aboveground Facilities Subtotal	***0.00***	***0.00***	***2.21***	***0.55***	***0.00***	***0.00***	***12.75***	***3.89***	***2.34***	***0.65***	***0.00***	***0.00***	***17.30***	***5.09***
Aboveground Facilities Subtotal	***0.00***	***0.00***	***4.11***	***2.41***	***0.00***	***0.00***	***15.56***	***6.01***	***5.69***	***3.45***	***0.00***	***0.00***	***25.36***	***11.87***
Project Total	**36.58**	**4.34**	**253.42**	**72.57**	**1.10**	**0.50**	**46.95**	**10.28**	**225.85**	**36.20**	**1.78**	**1.78**	**565.68**	**125.67**

Source: Columbia's November 10, 2025, Application, Resource Report 3, table 3.3-1. FERC eLibrary Accession No. 20251110-5151.
[a] Habitat affected during construction is inclusive of operation impacts (permanent).
[b] Habitat affected during operation consists only of operational areas associated with new permanent easement and permanent access roads, as well as new permanent impacts at the expanded aboveground facilities.
[c] Habitat impacts associated with this workspace are captured in the Line EKY Abandonment Right-of-Way.

APPENDIX D
SUMMARY OF LAND USE EFFECTS FOR THE PROJECT

Table D-1 Summary of Land Use Effects for the Project (acres)																
Facility	**Agricultural**		**Forested Land**		**Wetlands**		**Industrial**		**Residential**		**Open Land**		**Open Water**		**Project Total**	
	Temp. [a]	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**
New Pipeline Facilities																
Line AM36																
Right-of-Way	10.70	3.22	218.40	68.62	0.78	0.46	4.33	1.98	7.32	2.26	117.62	31.83	1.78	1.78	**360.93**	**110.15**
Additional Temporary Workspace	0.54	0.00	18.88	0.00	0.00	0.00	2.15	0.00	1.15	0.00	15.17	0.00	0.00	0.00	**37.89**	**0.00**
Contractor/Staging/ Pipe Yards	0.14	0.00	0.05	0.00	0.00	0.00	6.88	0.00	0.43	0.00	20.68	0.00	0.00	0.00	**28.18**	**0.00**
Access Roads	0.00	0.00	2.52	0.02	0.02	0.00	0.89	0.00	0.59	0.02	4.16	0.03	0.00	0.00	**8.18**	**0.07**
Cathodic Protection	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.49	0.00	0.00	0.00	**0.51**	**0.00**
Line EM-7																
Right-of-Way	15.84	1.12	4.92	0.08	0.17	0.02	0.16	0.01	0.06	0.00	40.41	0.51	0.00	0.00	**61.56**	**1.74**
Additional Temporary Workspace	1.23	0.00	0.23	0.00	0.02	0.00	0.01	0.00	0.10	0.00	1.27	0.00	0.00	0.00	**2.86**	**0.00**
Contractor/Staging/ Pipe Yards	6.98	0.00	0.00	0.00	0.00	0.00	<0.01	0.00	0.63	0.00	3.65	0.00	0.00	0.00	**11.26**	**0.00**
Access Roads	0.72	0.00	0.00	0.00	0.00	0.00	0.03	0.00	2.41	0.00	1.01	0.00	0.00	0.00	**4.17**	**0.00**
Black River Lateral																
Right-of-Way	0.00	0.00	0.81	0.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.81**	**0.53**
Cold Spring Lateral																
Right-of-Way	0.00	0.00	2.28	0.84	0.06	0.02	0.02	0.00	0.26	0.00	1.83	0.17	0.00	0.00	**4.45**	**1.03**
Additional Temporary Workspace	0.00	0.00	0.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.64	0.00	0.00	0.00	**0.84**	**0.00**

Table D-1 Summary of Land Use Effects for the Project (acres)																
Facility	**Agricultural**		**Forested Land**		**Wetlands**		**Industrial**		**Residential**		**Open Land**		**Open Water**		**Project Total**	
	Temp. [a]	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**
Kellogg Station Lateral																
Right-of-Way	0.00	0.00	0.07	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.29	0.21	0.00	0.00	**0.36**	**0.28**
New Pipeline Facilities Subtotal	***36.15***	***4.34***	***248.38***	***70.16***	***1.05***	***0.50***	***14.47***	***1.99***	***12.95***	***2.28***	***207.22***	***32.75***	***1.78***	***1.78***	***522.00***	***113.80***
Existing Pipeline Facilities																
Line AM04 Abandonment																
Right-of-Way	0.04	0.00	0.45	0.00	0.00	0.00	0.09	0.00	1.06	0.00	4.95	0.00	0.00	0.00	**6.59**	**0.00**
Access Roads	0.30	0.00	0.31	0.00	0.00	0.00	0.03	0.00	0.50	0.00	2.19	0.00	0.00	0.00	**3.33**	**0.00**
Line AM09 Abandonment																
Right-of-Way	0.09	0.00	0.17	0.00	0.05	0.00	0.03	0.00	0.60	0.00	5.59	0.00	0.00	0.00	**6.53**	**0.00**
Access Roads	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.66	0.00	0.20	0.00	0.00	0.00	**1.86**	**0.00**
Line EKY Abandonment																
Right-of-Way	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	**0.01**	**0.00**
E-Loop Abandonment [c]																
Right-of-Way	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**	**0.00**
Existing Pipeline Facilities Subtotal	***0.43***	***0.00***	***0.93***	***0.00***	***0.05***	***0.00***	***0.15***	***0.00***	***3.82***	***0.00***	***12.94***	***0.00***	***0.00***	***0.00***	***18.32***	***0.00***
Pipeline Facilities Subtotal	***36.58***	***4.34***	***249.31***	***70.16***	***1.10***	***0.50***	***14.62***	***1.99***	***16.77***	***2.28***	***220.16***	***32.75***	***1.78***	***1.78***	***540.32***	***113.80***
New Aboveground Facilities																
Line AM36																
Bracken Meter Station	0.00	0.00	0.05	0.05	0.00	0.00	0.54	0.11	0.00	0.00	0.82	0.53	0.00	0.00	**1.41**	**0.69**
Black River Tap	0.00	0.00	<0.01	<0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**	**0.00**
MLV-1 / Gubsur Mill Meter Station	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.33	0.33	0.11	0.11	0.00	0.00	**0.44**	**0.44**
MLV-2	0.00	0.00	0.00	0.00	0.00	0.00	0.11	0.11	0.00	0.00	0.40	0.40	0.00	0.00	**0.51**	**0.51**

Table D-1 Summary of Land Use Effects for the Project (acres)																
Facility	**Agricultural**		**Forested Land**		**Wetlands**		**Industrial**		**Residential**		**Open Land**		**Open Water**		**Project Total**	
	Temp. [a]	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**	**Temp. [a]**	**Op. [b]**
Parkview Meter Station	0.00	0.00	0.00	0.00	0.00	0.00	0.03	<0.01	0.00	0.00	0.56	0.49	0.00	0.00	**0.59**	**0.49**
MLV-3 / Cold Spring Lateral Launcher/ Receiver	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.37	0.37	0.00	0.00	**0.37**	**0.37**
UL47 Meter Station	0.00	0.00	0.64	0.64	0.00	0.00	0.08	0.05	0.00	0.00	0.40	0.24	0.00	0.00	**1.12**	**0.93**
Kellogg Station Launcher/ Receiver	0.00	0.00	0.27	0.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.27**	**0.27**
Access Roads	0.00	0.00	0.94	0.90	0.00	0.00	1.37	1.17	0.35	0.35	0.69	0.66	0.00	0.00	**3.35**	**3.08**
New Aboveground Facilities Subtotal	***0.00***	***0.00***	***1.90***	***1.86***	***0.00***	***0.00***	***2.13***	***1.44***	***0.68***	***0.68***	***3.35***	***2.80***	***0.00***	***0.00***	***8.06***	***6.78***
Existing Aboveground Facilities																
Line AM36																
Foster Regulator Station	0.00	0.00	1.16	0.17	0.00	0.00	2.55	0.52	0.00	0.00	0.00	0.00	0.00	0.00	**3.71**	**0.69**
Black River Meter Station	0.00	0.00	0.35	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.26	0.00	0.00	0.00	**0.61**	**0.04**
Pendleton Meter Station	0.00	0.00	0.34	0.34	0.00	0.00	0.08	0.01	0.00	0.00	0.25	0.08	0.00	0.00	**0.67**	**0.43**
Alexandria Meter Station	0.00	0.00	0.00	0.00	0.00	0.00	2.17	0.07	0.00	0.00	0.51	0.51	0.00	0.00	**2.68**	**0.58**
Cold Spring Meter Station	0.00	0.00	0.12	0.00	0.00	0.00	1.24	0.63	0.02	<0.01	1.21	0.01	0.00	0.00	**2.59**	**0.64**
Access Roads	0.00	0.00	0.24	0.00	0.00	0.00	3.22	0.06	2.60	2.60	0.08	0.05	0.00	0.00	**6.14**	**2.71**
Line EM-7																
Moorefield Site	0.00	0.00	0.00	0.00	0.00	0.00	0.25	0.00	0.04	0.00	<0.01	0.00	0.00	0.00	**0.29**	**0.00**
Carlisle Meter Station	0.00	0.00	0.00	0.00	0.00	0.00	0.42	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.42**	**0.00**
Access Roads	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.12	0.00	0.03	0.00	0.00	0.00	**0.19**	**0.00**

Table D-1
Summary of Land Use Effects for the Project (acres)

Facility	Agricultural		Forested Land		Wetlands		Industrial		Residential		Open Land		Open Water		Project Total	
	Temp. [a]	Op. [b]	Temp. [a]	Op. [b]	Temp. [a]	Op. [b]	Temp. [a]	Op. [b]	Temp. [a]	Op. [b]	Temp. [a]	Op. [b]	Temp. [a]	Op. [b]	Temp. [a]	Op. [b]
Existing Aboveground Facilities Subtotal	***0.00***	***0.00***	***2.21***	***0.55***	***0.00***	***0.00***	***9.97***	***1.29***	***2.78***	***2.60***	***2.34***	***0.65***	***0.00***	***0.00***	***17.30***	***5.09***
Aboveground Facilities Subtotal	***0.00***	***0.00***	***4.11***	***2.41***	***0.00***	***0.00***	***12.10***	***2.73***	***3.46***	***3.28***	***5.69***	***3.45***	***0.00***	***0.00***	***25.36***	***11.87***
Project Total	**36.58**	**4.34**	**253.42**	**72.57**	**1.10**	**0.50**	**26.72**	**4.72**	**20.23**	**5.56**	**225.85**	**36.20**	**1.78**	**1.78**	**565.68**	**125.67**

Source: Columbia's November 10, 2025, Application, Resource Report 8, table 8.2-1. FERC eLibrary Accession No. 20251110-5151.
[a] Land affected during construction is inclusive of operation impacts (permanent).
[b] Land affected during operation consists only of operational areas associated with new permanent easement and permanent access roads, as well as new permanent impacts at new and expanded aboveground facilities.
[c] Land impacts associated with this workspace are captured in the Line EKY Abandonment Right-of-Way.

APPENDIX E
SITE-SPECIFIC RESIDENTIAL CONSTRUCTION PLANS

TRACT NUMBER: KY-CA-158.000

THIS DRAWING DOCUMENTS MITIGATION MEASURES THAT WILL BE IMPLEMENTED FOR ALL RESIDENCES WITHIN 25 FEET OF THE PROPOSED CONSTRUCTION WORK AREA. CONTRACTOR SHALL COMPLY WITH THE FOLLOWING CONSTRUCTION MITIGATION REQUIREMENTS IN ADDITION TO THOSE LISTED IN THE CONSTRUCTION SPECIFICATIONS.

CONSTRUCTION REQUIREMENTS

1. CONTRACTOR SHALL ERECT AND MAINTAIN A SAFETY FENCE BETWEEN THE CONSTRUCTION ZONE AND THE ADJACENT RESIDENCES EXTENDING 100 FEET ON EITHER SIDE OF THE RESIDENCE TO ENSURE THAT EQUIPMENT, MATERIALS AND SPOIL REMAIN WITHIN THE CONSTRUCTION WORK AREA.
2. FOR RESIDENCES WITHIN 25 FEET OF THE CENTERLINE, THE TRENCH WILL NOT BE EXCAVATED UNTIL THE PIPELINE IS READY FOR REMOVAL OR INSTALLATION AND WILL BE BACKFILLED IN THE SAME DAY AS PIPE REMOVAL / INSTALLATION WHENEVER PRACTICABLE. WHERE SAME DAY BACKFILLING IS NOT COMPLETED, OPEN DITCHES WILL BE BARRICADED / FENCED OFF OR PLATED DURING OVERNIGHT PERIODS.
3. MATURE TREES AND LANDSCAPING WILL NOT BE REMOVED FROM WITHIN THE EDGE OF THE CONSTRUCTION WORK AREA UNLESS NECESSARY FOR SAFE OPERATION OF CONSTRUCTION EQUIPMENT OR AS SPECIFIED IN LANDOWNER AGREEMENTS.
4. OTHER EXISTING PHYSICAL FEATURES THAT NEED TO BE PROTECTED, SUCH AS SEPTIC SYSTEMS AND WELLS, WILL BE ENCLOSED IN SAFETY FENCE TO AVOID DISTURBANCE DURING CONSTRUCTION.
5. IMMEDIATELY AFTER BACKFILLING THE TRENCH, ALL LAWN AND LANDSCAPING WILL BE RESTORED TO FINAL RESTORATION, OR TEMPORARY RESTORATION PENDING WEATHER AND SOIL CONDITIONS. IF SEASONAL OR WEATHER CONDITIONS PREVENT COMPLIANCE WITH THESE TIME FRAMES, TEMPORARY EROSION CONTROLS MUST BE MONITORED AND MAINTAINED UNTIL CONDITIONS ALLOW COMPLETION OF RESTORATION.
6. CONTRACTOR SHALL UTILIZE WATER TRUCKS AS NECESSARY TO MINIMIZE FUGITIVE DUST FROM CONSTRUCTION ACTIVITIES NEAR RESIDENCES/BUSINESSES. WATERING WILL OCCUR AT A FREQUENCY TO MAINTAIN ADEQUATE EXPOSED SOIL MOISTURE LEVELS NECESSARY TO SUPPRESS DUST EMISSION. DISTURBED OR EXPOSED SOIL WILL BE MONITORED CONTINUOUSLY THROUGHOUT THE CONSTRUCTION PROCESS.
7. ACCESS TO RESIDENCES BY CAR WILL BE MAINTAINED AT ALL TIMES, OR OTHER ACCOMMODATIONS WILL BE MADE WITH EACH RESPECTIVE LANDOWNER. STEEL PLATES WILL BE KEPT ON SITE AT EACH DRIVEWAY CROSSING SO THAT A TEMPORARY PLATFORM CAN BE MADE ACROSS THE TRENCH AS REQUIRED (E.G., EMERGENCY VEHICLES). IN THE EVENT OF AN EMERGENCY, STEEL PLATES CAN TYPICALLY BE INSTALLED ACROSS THE OPEN TRENCH WITHIN FIVE MINUTES.
8. CONTRACTOR SHALL MAINTAIN AGREED UPON ACCESS TO THE IMPACTED AREA DURING CONSTRUCTION.
9. CONTRACTOR SHALL LIMIT WORK IN THIS AREA TO DAYTIME HOURS (BETWEEN 7:00 AM & 7:00 PM), UNLESS OTHERWISE AGREED UPON WITH LANDOWNER/OCCUPANT AND IN ACCORDANCE WITH APPLICABLE REGULATIONS.
10. LANDOWNER/OCCUPANT SHALL BE NOTIFIED BY CERTIFIED LETTER OF PROPOSED CONSTRUCTION ACTIVITIES 45 DAYS PRIOR TO CONSTRUCTION WORK.
11. NO REFUELING OR FUEL STORAGE IS AUTHORIZED WITHIN 200' OF A WELL.
12. SAFETY FENCE MAY BE OMITTED IN AREAS WHERE FENCING IS PRE-EXISTING.
13. SPOIL PILES WILL BE STABILIZED, WATERED, OR COVERED AS NEEDED TO MINIMIZE THE EFFECTS OF WIND EROSION AND REDUCE FUGITIVE DUST.
14. VEHICLE SPEED WILL BE LIMITED TO 15 MPH TO REDUCE THE POTENTIAL OF DUST.
15. CONSTRUCTION EQUIPMENT NOT ACTIVELY IN USE MAY NOT IDLE FOR LONGER THAN 5 MINUTES WHEN LOCATED WITHIN 100 FEET OF A RESIDENCE.

LEGEND

- EXISTING PIPELINE
- FOREIGN PIPELINE
- PROPOSED PIPELINE
- LIMIT OF DISTURBANCE
- COUNTY BOUNDARY
- CL ROAD
- EDGE OF ROAD
- ROADWAY/RAILWAY R.O.W.
- CL RAILROAD
- FENCE
- POWER LINE
- BURIED POWER LINE
- ELECTRIC CABLE
- WATER LINE
- CABLE LINE
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- STORM SEWER
- SANITARY SEWER
- TELEPHONE LINE
- BURIED TELEPHONE LINE
- CULVERT
- TOP OF BANK
- TOE OF BANK
- WATERWAY
- BAR DITCH
- GAURD RAIL
- STONE/ROCK FENCE
- STRUCTURES
- PERMANENT EASEMENT
- EXISTING EASEMENT
- PERMANENT CONTOURS
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- STAGING AREA
- CP PERMANENT EASE.
- FACILITY
- CONTRACTOR / PIPE YARD
- ABANDONMENT WORKSPACE
- ST-CO-000.000 PROPERTY TRACT NUMBER
- ENVIRONMENTAL SURVEY BOUNDARY
- WETLAND
- STREAMS
- OPEN WATER
- WETLAND ID
- STREAM ID
- OPEN WATER ID
- MILEPOST
- ACCESS ROAD
- AC-0001 ACCESS ROAD NUMBER
- POWER POLE
- WATER WELL
- MANHOLE
- GUY WIRE / ANCHOR
- SANITARY SEWER MANHOLE
- STORM SEWER MANHOLE
- GRAVE MARKER
- ELECTRICAL PEDESTAL
- TELEPHONE PEDESTAL
- TELEVISION PEDESTAL
- CATCH BASIN
- 2 CONTOURS
- SAFETY FENCE

0 50' 100' 200'
GRAPHIC SCALE

N W E S

NOTES:
1. PRELIMINARY - NOT FOR CONSTRUCTION.
2. PHOTOGRAPHY IS BY KUCERA INTERNATIONAL INC., FLOWN DECEMBER, 08, 2024.
3. COORDINATE SYSTEM: KENTUCKY STATE PLANE NORTH, NAD83, U.S. SURVEY FEET.
4. TC ENERGY - COLUMBIA GAS TRANSMISSION , LLC.

REFERENCE DRAWINGS	
DRAWING No	TITLE
85731-03-ML-15-101	ACCESS ROAD LAYOUT

REVISION			APPROVAL						
REV No	DATE	DESCRIPTION	PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
B	2025/10/15	ISSUED FOR FERC	M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO. | DATE | PERMIT NUMBER:

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661 TRC PROJECT NO.: 574139

AM36 – SEGMENT 3 (AM36 CAMPBELL CO KY TO MLV-1)

FIA # 85731 | CHAINAGE: | DISCIPLINE # 03

RESIDENTIAL DETAIL KY-CA-158.000

SCALE AS NOTED | DRAWING No 85731-03-ML-15-200 | REV B

ID: TC_UD_BR_24x36.DWG 24X36

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

PLOTTED SIZE: ARCH D (24x36)

TIME: DESIGNFILE:

TRACT NUMBER: AM09–022.00

THIS DRAWING DOCUMENTS MITIGATION MEASURES THAT WILL BE IMPLEMENTED FOR ALL RESIDENCES WITHIN 25 FEET OF THE PROPOSED CONSTRUCTION WORK AREA. CONTRACTOR SHALL COMPLY WITH THE FOLLOWING CONSTRUCTION MITIGATION REQUIREMENTS IN ADDITION TO THOSE LISTED IN THE CONSTRUCTION SPECIFICATIONS.

CONSTRUCTION REQUIREMENTS

1. CONTRACTOR SHALL ERECT AND MAINTAIN A SAFETY FENCE BETWEEN THE CONSTRUCTION ZONE AND THE ADJACENT RESIDENCES EXTENDING 100 FEET ON EITHER SIDE OF THE RESIDENCE TO ENSURE THAT EQUIPMENT, MATERIALS AND SPOIL REMAIN WITHIN THE CONSTRUCTION WORK AREA.
2. FOR RESIDENCES WITHIN 25 FEET OF THE CENTERLINE, THE TRENCH WILL NOT BE EXCAVATED UNTIL THE PIPELINE IS READY FOR REMOVAL OR INSTALLATION AND WILL BE BACKFILLED IN THE SAME DAY AS PIPE REMOVAL / INSTALLATION WHENEVER PRACTICABLE. WHERE SAME DAY BACKFILLING IS NOT COMPLETED, OPEN DITCHES WILL BE BARRICADED / FENCED OFF OR PLATED DURING OVERNIGHT PERIODS.
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- STREAMS
- OPEN WATER
- WETLAND ID
- STREAM ID
- OPEN WATER ID
- MILEPOST
- ACCESS ROAD
- AR–0001 ACCESS ROAD NUMBER
- POWER POLE
- WATER WELL
- MANHOLE
- GUY WIRE / ANCHOR
- SANITARY SEWER MANHOLE
- STORM SEWER MANHOLE
- GRAVE MARKER
- ELECTRICAL PEDESTAL
- TELEPHONE PEDESTAL
- TELEVISION PEDESTAL
- CATCH BASIN
- 2 CONTOURS
- SAFETY FENCE

0 50' 100' 200'
GRAPHIC SCALE

N W E S

NOTES:
1. PRELIMINARY - NOT FOR CONSTRUCTION.
2. PHOTOGRAPHY IS BY KUCERA INTERNATIONAL INC., FLOWN DECEMBER, 08, 2024.
3. COORDINATE SYSTEM: KENTUCKY STATE PLANE NORTH, NAD83, U.S. SURVEY FEET.
4. TC ENERGY - COLUMBIA GAS TRANSMISSION , LLC.

REFERENCE DRAWINGS

DRAWING No	TITLE
85737–03–ML–15–100	ACCESS ROAD LAYOUT

REVISION

REV No	DATE	DESCRIPTION
B	2025/10/15	ISSUED FOR FERC

APPROVAL

PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO. | DATE | PERMIT NUMBER:

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661
TRC PROJECT NO.: 574139

AM36 – SEGMENT 5 (AM36 MLV–2 TO MLV–3)

FIA #	CHAINAGE:	DISCIPLINE #
85737		03

RESIDENTIAL DETAIL AM09–022.00

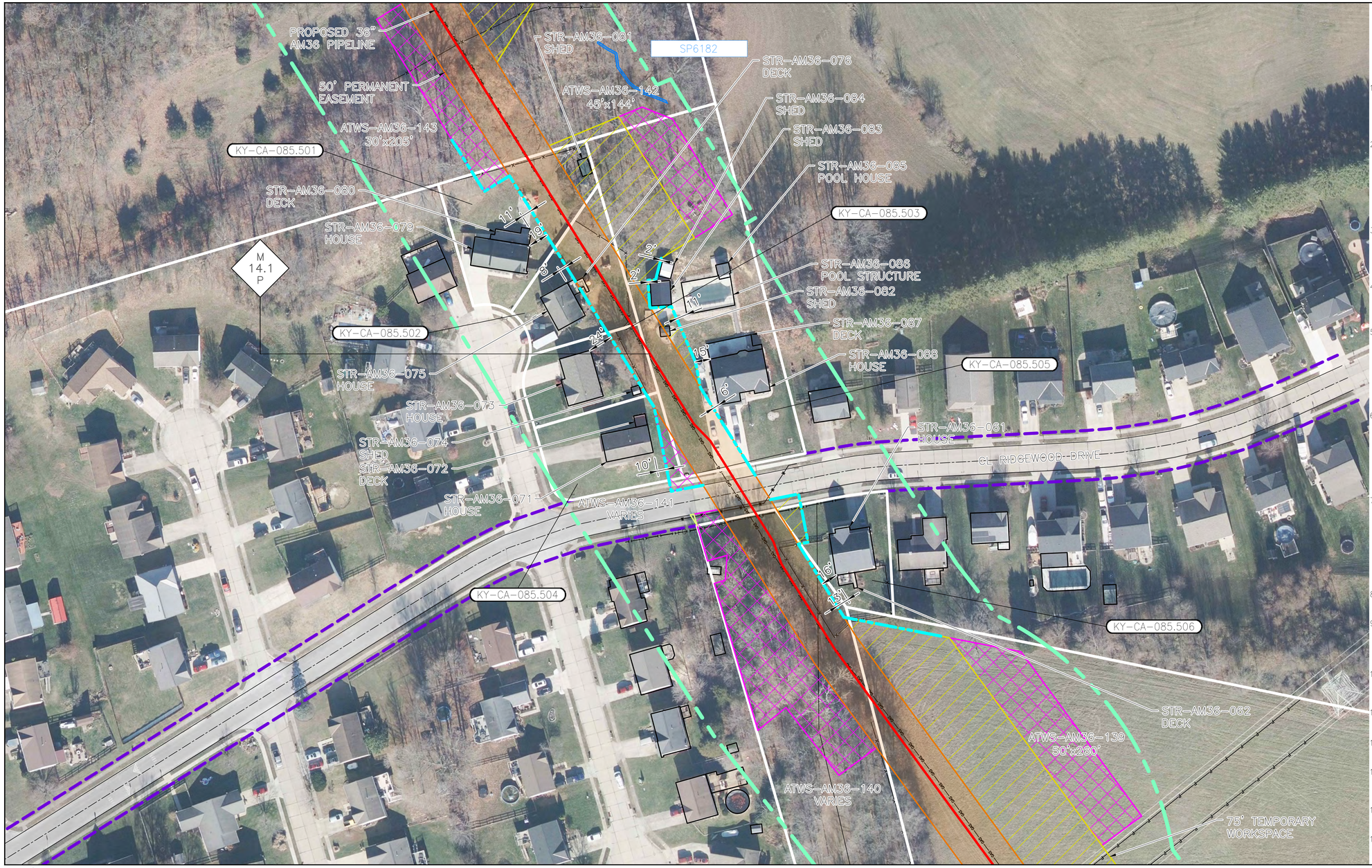
SCALE	DRAWING No	REV
AS NOTED	85737–03–ML–15–200	B

PLOTTED SIZE: ARCH D (24x36)

TC_UD_BR_24x36.DWG 24X36

ID TIME DESIGNFILE

TRACT NUMBER: KY-CA-085.501, KY-CA-085.502, KY-CA-085.503, KY-CA-085.504, KY-CA-085.505 & KY-CA-085.506



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0 50' 100' 200' GRAPHIC SCALE

N W E S

NOTES:
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2. PHOTOGRAPHY IS BY KUCERA INTERNATIONAL INC., FLOWN DECEMBER, 08, 2024.
3. COORDINATE SYSTEM: KENTUCKY STATE PLANE NORTH, NAD83, U.S. SURVEY FEET.
4. TC ENERGY - COLUMBIA GAS TRANSMISSION , LLC.

REFERENCE DRAWINGS

DRAWING No	TITLE
85737-03-ML-02-005	ALIGNMENT SHEET

REVISION

REV No	DATE	DESCRIPTION
B	2025/10/15	ISSUED FOR FERC

APPROVAL

PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO. | DATE | PERMIT NUMBER:

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661 TRC PROJECT NO.: 574139

AM36 – SEGMENT 5 (AM36 MLV-2 TO MLV-3)

FIA # 85737 | CHAINAGE: | DISCIPLINE # 03

RESIDENTIAL DETAIL MP 14.1

SCALE AS NOTED | DRAWING No 85737-03-ML-15-201 | REV B

PLOTTED SIZE: ARCH D (24x36)

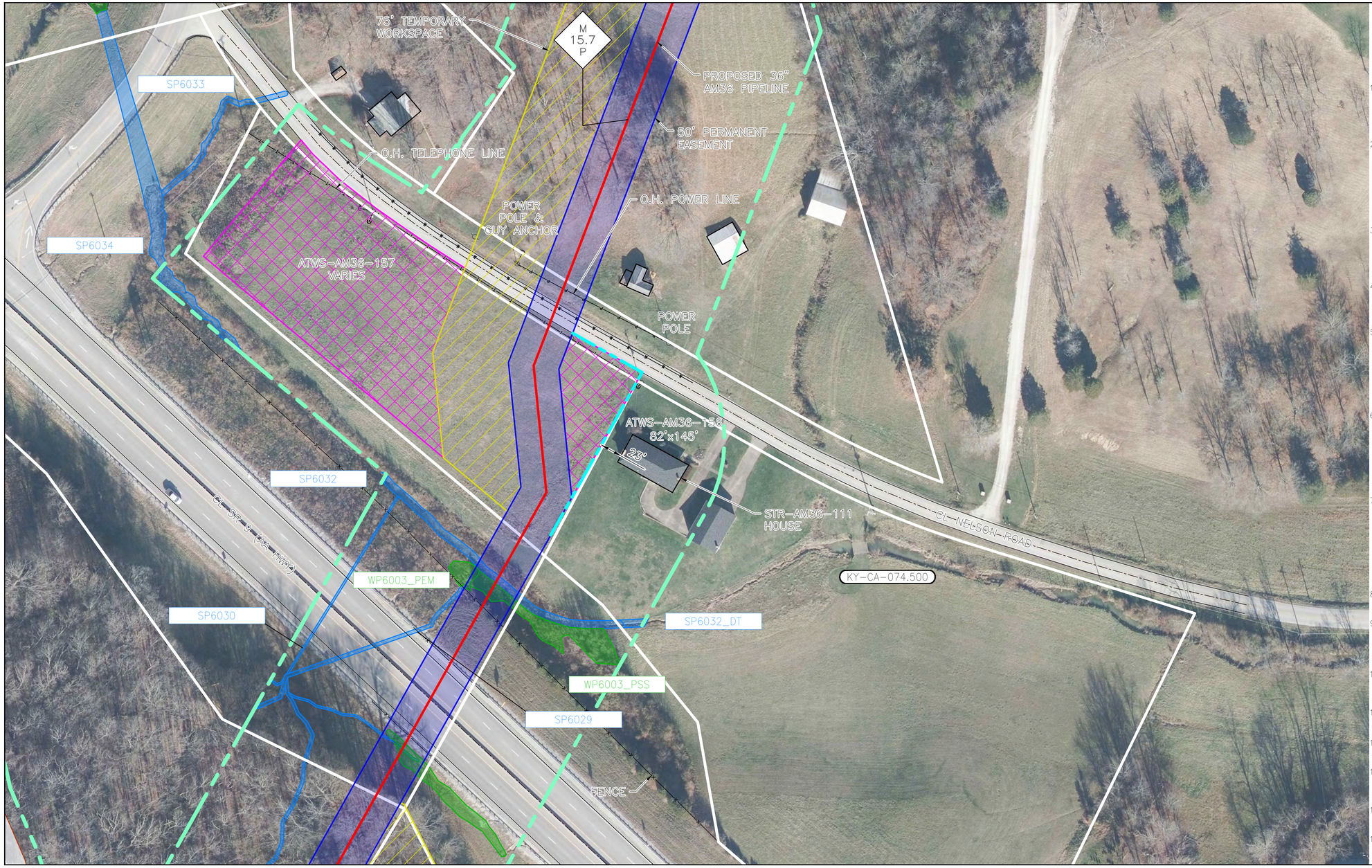
TC_UD_BR_24x36.DWG 24X36

ID

TIME

DESIGNFILE

TRACT NUMBER: KY-CA-074.500



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N W E S

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3. COORDINATE SYSTEM: KENTUCKY STATE PLANE NORTH, NAD83, U.S. SURVEY FEET.
4. TC ENERGY - COLUMBIA GAS TRANSMISSION , LLC.

REFERENCE DRAWINGS

DRAWING No	TITLE
85737-03-ML-02-008	ALIGNMENT SHEET

REVISION

REV No	DATE	DESCRIPTION
B	2025/10/15	ISSUED FOR FERC

APPROVAL

PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO. | DATE | PERMIT NUMBER:

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661 TRC PROJECT NO.: 574139

AM36 – SEGMENT 5 (AM36 MLV-2 TO MLV-3)

FIA #	CHAINAGE:	DISCIPLINE #
85737		03

RESIDENTIAL DETAIL
MP 15.7

SCALE	DRAWING No	REV
AS NOTED	85737-03-ML-15-202	B

PLOTTED SIZE: ARCH D (24x36)

ID: TC_UD_BR_24x36.DWG 24X36

TIME: DESIGNFILE:

TRACT NUMBER: LAT-7-KY-CA-002.000, LAT-7-KY-CA-002.002 & LAT-7-KY-CA-006.000

THIS DRAWING DOCUMENTS MITIGATION MEASURES THAT WILL BE IMPLEMENTED FOR ALL RESIDENCES WITHIN 25 FEET OF THE PROPOSED CONSTRUCTION WORK AREA. CONTRACTOR SHALL COMPLY WITH THE FOLLOWING CONSTRUCTION MITIGATION REQUIREMENTS IN ADDITION TO THOSE LISTED IN THE CONSTRUCTION SPECIFICATIONS.

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0 50' 100' 200' GRAPHIC SCALE

N W E S

NOTES:
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3. COORDINATE SYSTEM: KENTUCKY STATE PLANE NORTH, NAD83, U.S. SURVEY FEET.
4. TC ENERGY - COLUMBIA GAS TRANSMISSION , LLC.

REFERENCE DRAWINGS

DRAWING No	TITLE
85737-03-ML-15-103	ACCESS ROAD LAYOUT

REVISION

REV No	DATE	DESCRIPTION
B	2025/10/15	ISSUED FOR FERC

APPROVAL

PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO. | DATE | PERMIT NUMBER:

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661 TRC PROJECT NO.: 574139

AM36 – SEGMENT 5 (AM36 MLV-2 TO MLV-3)

FIA # 85737 | CHAINAGE: | DISCIPLINE # 03

RESIDENTIAL DETAIL
LAT 7-KY-CA-002.000, LAT 7-KY-CA-002.002 & LAT 7-KY-CA-006.000

SCALE AS NOTED | DRAWING No 85737-03-ML-15-203 | REV B

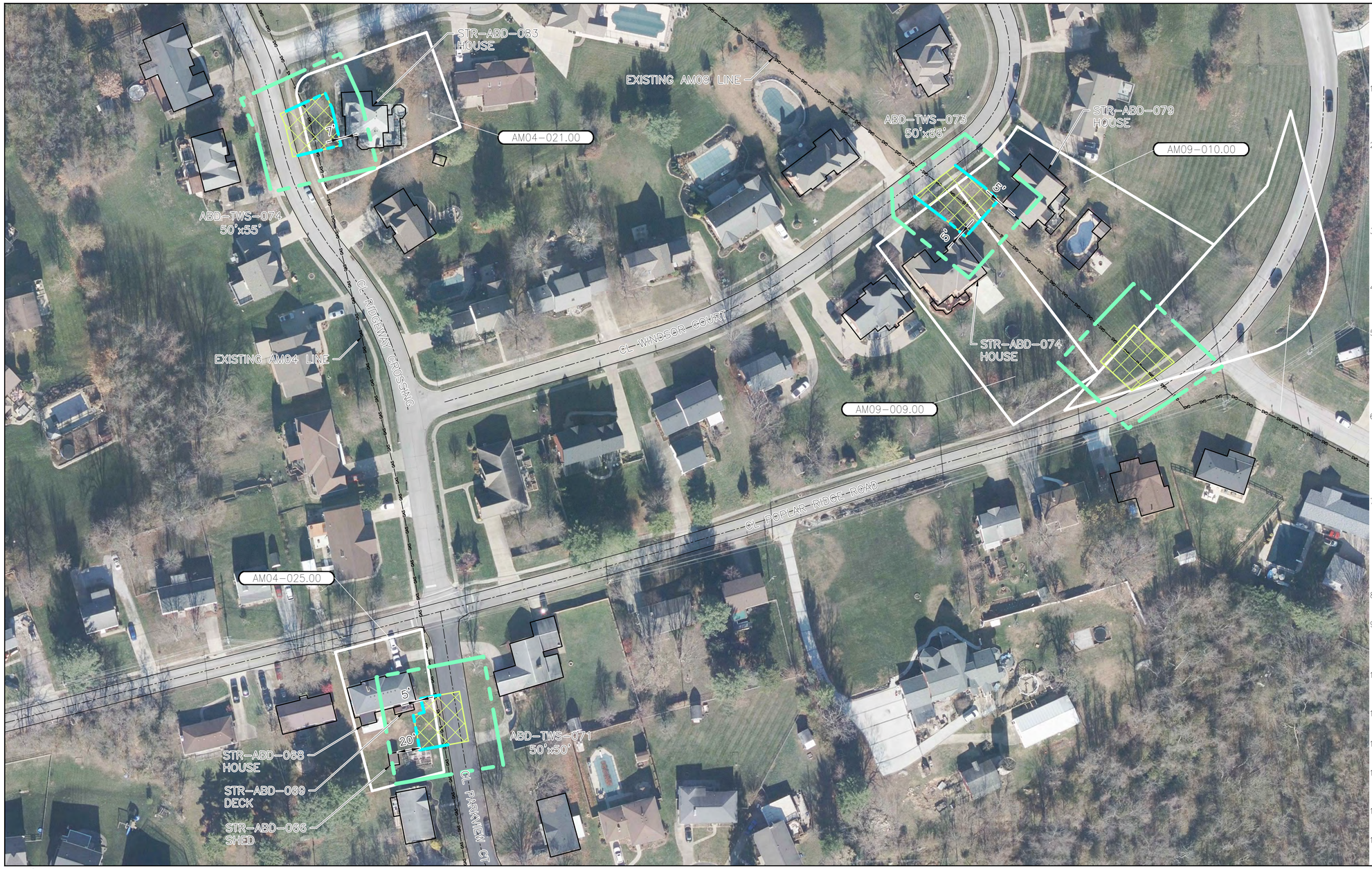
ID TC_UD_BR_24x36.DWG 24X36

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

PLOTTED SIZE: ARCH D (24x36)

TIME DESIGNFILE

TRACT NUMBER: AM04–021.00, AM04–025.00, AM09–009.00 & AM09–010.00



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- ST–CO–000.000 PROPERTY TRACT NUMBER
- ENVIRONMENTAL SURVEY BOUNDARY
- WETLAND
- STREAMS
- OPEN WATER
- WETLAND ID
- STREAM ID
- OPEN WATER ID
- MILEPOST
- ACCESS ROAD
- AR–0001 ACCESS ROAD NUMBER
- POWER POLE
- WATER WELL
- MANHOLE
- GUY WIRE / ANCHOR
- SANITARY SEWER MANHOLE
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- GRAVE MARKER
- ELECTRICAL PEDESTAL
- TELEPHONE PEDESTAL
- TELEVISION PEDESTAL
- CATCH BASIN
- 2 CONTOURS
- SAFETY FENCE

0 50' 100' 200' GRAPHIC SCALE

N W E S

NOTES:
1. PRELIMINARY - NOT FOR CONSTRUCTION.
2. PHOTOGRAPHY IS BY KUCERA INTERNATIONAL INC., FLOWN DECEMBER, 08, 2024.
3. COORDINATE SYSTEM: KENTUCKY STATE PLANE NORTH, NAD83, U.S. SURVEY FEET.
4. TC ENERGY - COLUMBIA GAS TRANSMISSION , LLC.

REFERENCE DRAWINGS

DRAWING No	TITLE
85737–03–ML–15–104	WORKSPACE DETAIL

REVISION

REV No	DATE	DESCRIPTION
B	2025/10/15	ISSUED FOR FERC

APPROVAL

PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO. | DATE | PERMIT NUMBER:

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661 TRC PROJECT NO.: 574139

AM36 – SEGMENT 5 (AM36 MLV–2 TO MLV–3)

FIA # 85737 | CHAINAGE: | DISCIPLINE # 03

RESIDENTIAL DETAIL AM04–021.00, AM04–025.00, AM09–009.00 & AM09–010.00

SCALE AS NOTED | DRAWING No 85737–03–ML–15–204 | REV B

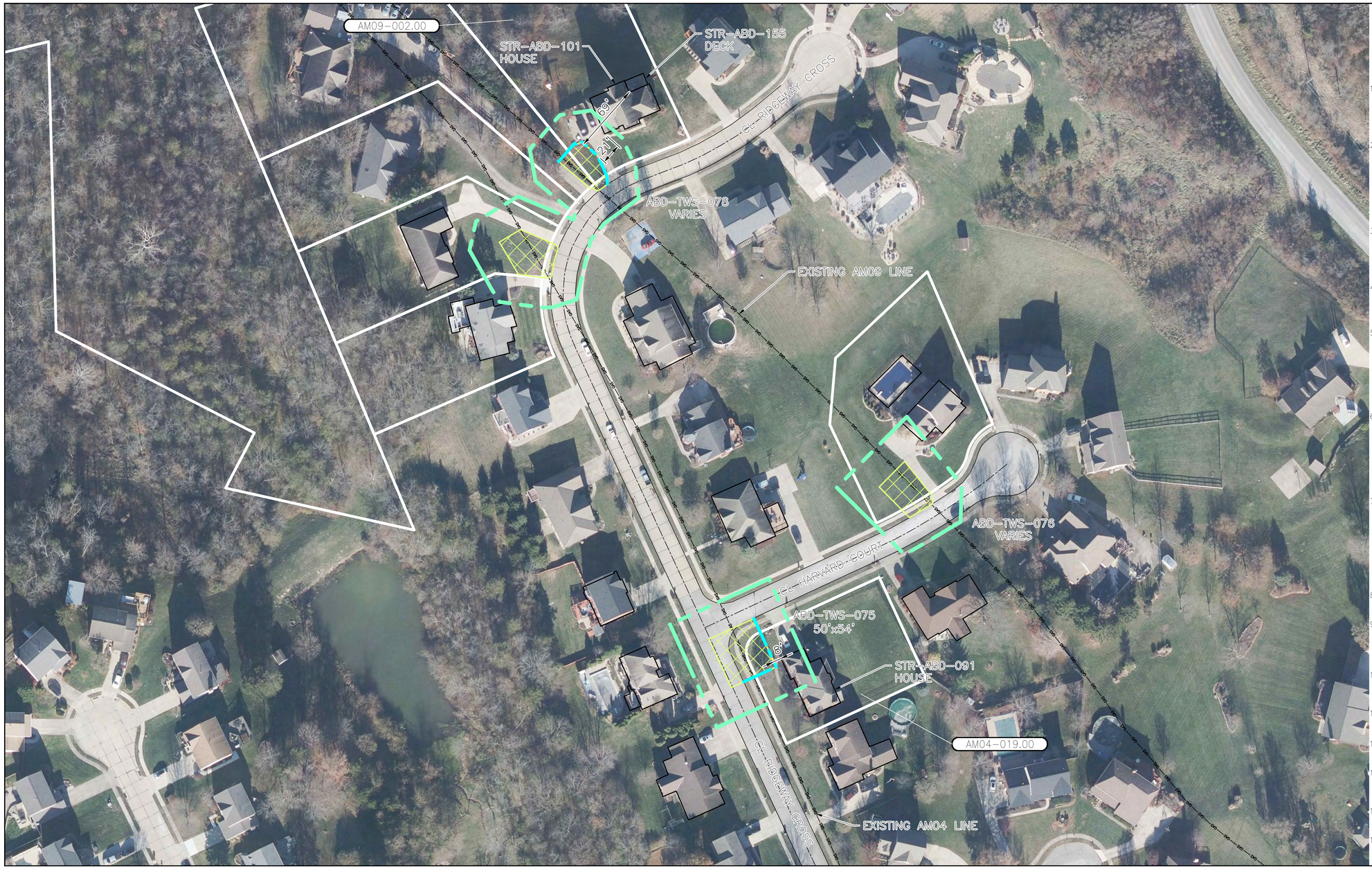
ID TC_UD_BR_24x36.DWG 24X36

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

PLOTTED SIZE: ARCH D (24x36)

TIME DESIGNFILE

TRACT NUMBER: AM09–002.00 & AM04–019.00



THIS DRAWING DOCUMENTS MITIGATION MEASURES THAT WILL BE IMPLEMENTED FOR ALL RESIDENCES WITHIN 25 FEET OF THE PROPOSED CONSTRUCTION WORK AREA. CONTRACTOR SHALL COMPLY WITH THE FOLLOWING CONSTRUCTION MITIGATION REQUIREMENTS IN ADDITION TO THOSE LISTED IN THE CONSTRUCTION SPECIFICATIONS.

CONSTRUCTION REQUIREMENTS

1. CONTRACTOR SHALL ERECT AND MAINTAIN A SAFETY FENCE BETWEEN THE CONSTRUCTION ZONE AND THE ADJACENT RESIDENCES EXTENDING 100 FEET ON EITHER SIDE OF THE RESIDENCE TO ENSURE THAT EQUIPMENT, MATERIALS AND SPOIL REMAIN WITHIN THE CONSTRUCTION WORK AREA.
2. FOR RESIDENCES WITHIN 25 FEET OF THE CENTERLINE, THE TRENCH WILL NOT BE EXCAVATED UNTIL THE PIPELINE IS READY FOR REMOVAL OR INSTALLATION AND WILL BE BACKFILLED IN THE SAME DAY AS PIPE REMOVAL / INSTALLATION WHENEVER PRACTICABLE. WHERE SAME DAY BACKFILLING IS NOT COMPLETED, OPEN DITCHES WILL BE BARRICADED / FENCED OFF OR PLATED DURING OVERNIGHT PERIODS.
3. MATURE TREES AND LANDSCAPING WILL NOT BE REMOVED FROM WITHIN THE EDGE OF THE CONSTRUCTION WORK AREA UNLESS NECESSARY FOR SAFE OPERATION OF CONSTRUCTION EQUIPMENT OR AS SPECIFIED IN LANDOWNER AGREEMENTS.
4. OTHER EXISTING PHYSICAL FEATURES THAT NEED TO BE PROTECTED, SUCH AS SEPTIC SYSTEMS AND WELLS, WILL BE ENCLOSED IN SAFETY FENCE TO AVOID DISTURBANCE DURING CONSTRUCTION.
5. IMMEDIATELY AFTER BACKFILLING THE TRENCH, ALL LAWN AND LANDSCAPING WILL BE RESTORED TO FINAL RESTORATION, OR TEMPORARY RESTORATION PENDING WEATHER AND SOIL CONDITIONS. IF SEASONAL OR WEATHER CONDITIONS PREVENT COMPLIANCE WITH THESE TIME FRAMES, TEMPORARY EROSION CONTROLS MUST BE MONITORED AND MAINTAINED UNTIL CONDITIONS ALLOW COMPLETION OF RESTORATION.
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LEGEND

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- FOREIGN PIPELINE
- PROPOSED PIPELINE
- LIMIT OF DISTURBANCE
- COUNTY BOUNDARY
- CL ROAD
- EDGE OF ROAD
- ROADWAY/RAILWAY R.O.W.
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- BURIED POWER LINE
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0 50' 100' 200' GRAPHIC SCALE

N W E S

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3. COORDINATE SYSTEM: KENTUCKY STATE PLANE NORTH, NAD83, U.S. SURVEY FEET.
4. TC ENERGY - COLUMBIA GAS TRANSMISSION , LLC.

REFERENCE DRAWINGS

DRAWING No	TITLE
85737–03–ML–15–104	WORKSPACE DETAIL

REVISION

REV No	DATE	DESCRIPTION
B	2025/10/15	ISSUED FOR FERC

APPROVAL

PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO. | DATE | PERMIT NUMBER:

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661 TRC PROJECT NO.: 574139

AM36 – SEGMENT 5 (AM36 MLV–2 TO MLV–3)

FIA # 85737 | CHAINAGE: | DISCIPLINE # 03

RESIDENTIAL DETAIL
AM09–002.00 & AM04–019.00

SCALE AS NOTED | DRAWING No 85737–03–ML–15–205 | REV B

ID

TC_UD_BR_24x36.DWG 24X36

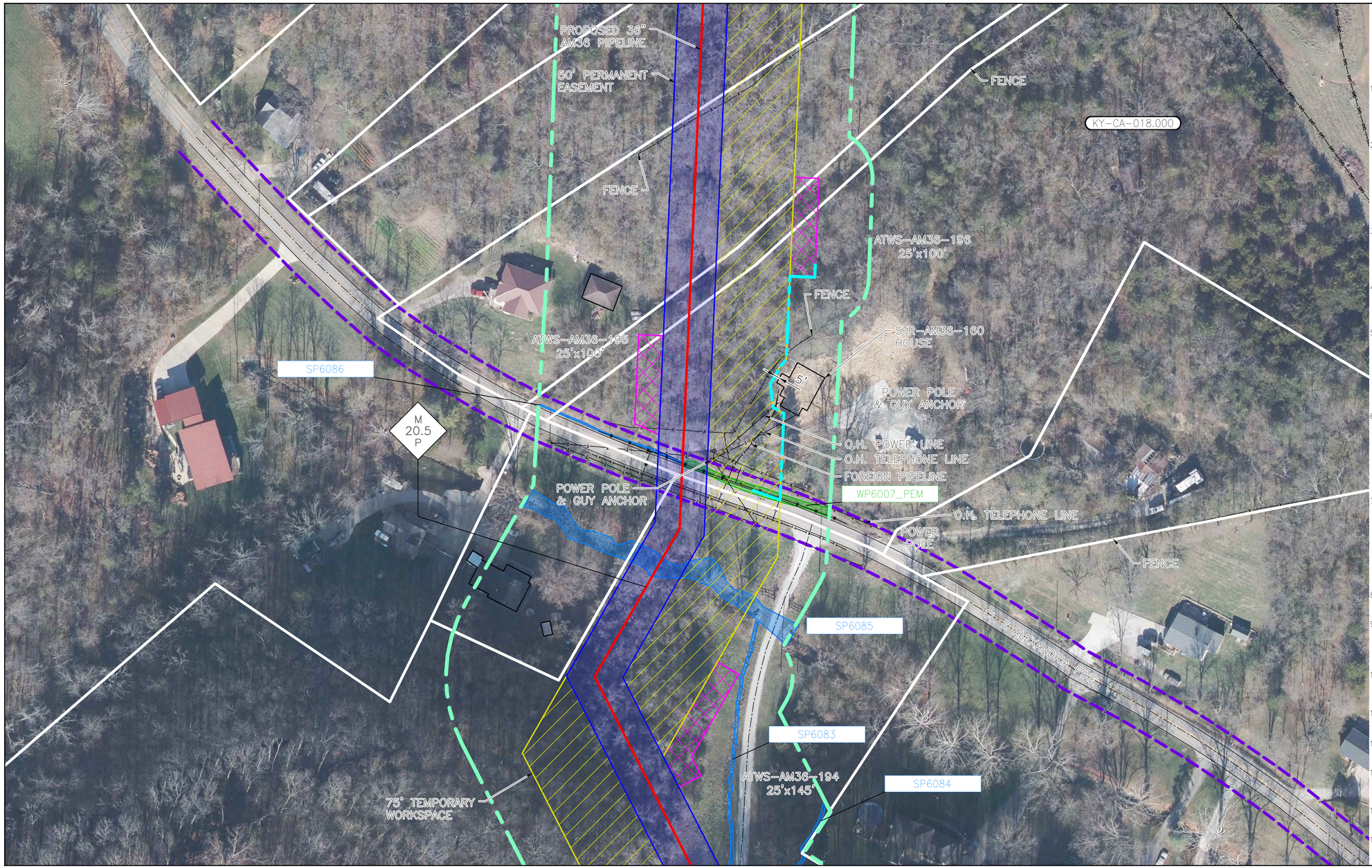
CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

PLOTTED SIZE: ARCH D (24x36)

TIME

DESIGNFILE

TRACT NUMBER: KY-CA-018.000



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0 50' 100' 200' GRAPHIC SCALE

N W E S

NOTES:
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4. TC ENERGY - COLUMBIA GAS TRANSMISSION , LLC.

REFERENCE DRAWINGS

DRAWING No	TITLE
85737-03-ML-02-020	ALIGNMENT SHEET

REVISION

REV No	DATE	DESCRIPTION
B	2025/10/15	ISSUED FOR FERC

APPROVAL

PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO. | DATE | PERMIT NUMBER:

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661 TRC PROJECT NO.: 574139

AM36 – SEGMENT 5 (AM36 MLV-2 TO MLV-3)

FIA #	CHAINAGE:	DISCIPLINE #
85737		03

RESIDENTIAL DETAIL
MP 20.5

SCALE	DRAWING No	REV
AS NOTED	85737-03-ML-15-206	B

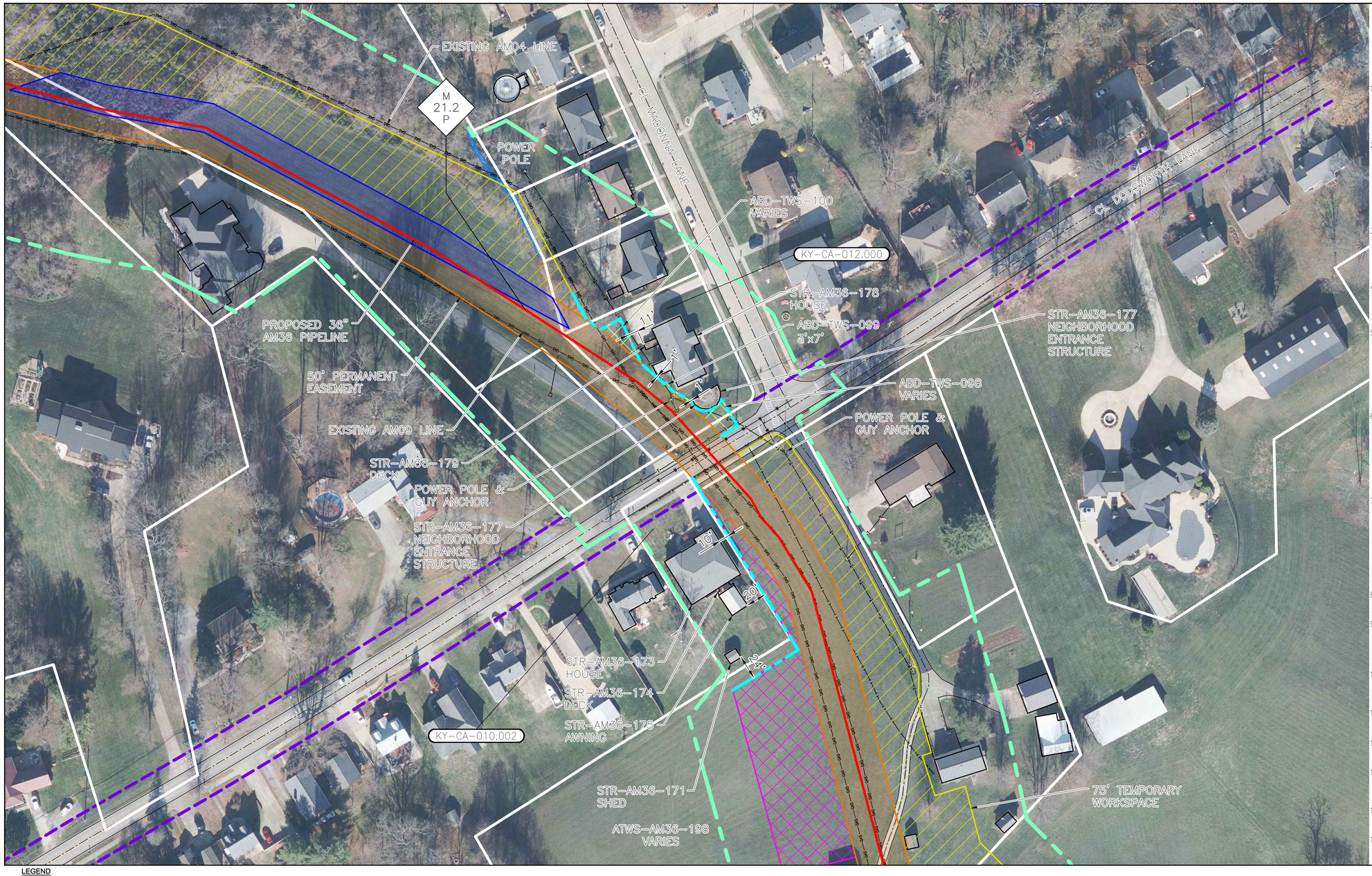
TIME: DESIGNFILE:

TC_UD_BR_24x36.DWG 24X36

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

PLOTTED SIZE: ARCH D (24x36)

TRACT NUMBER: KY-CA-010.020 & KY-CA-012.000



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N W E S

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4. TC ENERGY - COLUMBIA GAS TRANSMISSION , LLC.

REFERENCE DRAWINGS	
DRAWING No	TITLE
85737-03-ML-02-021	ALIGNMENT SHEET

REVISION			APPROVAL						
REV No	DATE	DESCRIPTION	PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
B	2025/10/15	ISSUED FOR FERC	M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT	PERMIT/ ENG. APPROVAL
	DATE

REV. NO.	DATE	PERMIT NUMBER:

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661 TRC PROJECT NO.: 574139

AM36 – SEGMENT 5 (AM36 MLV-2 TO MLV-3)

FIA #	CHAINAGE:	DISCIPLINE #
85737		03

RESIDENTIAL DETAIL
MP 21.2

SCALE	DRAWING No	REV
AS NOTED	85737-03-ML-15-207	B

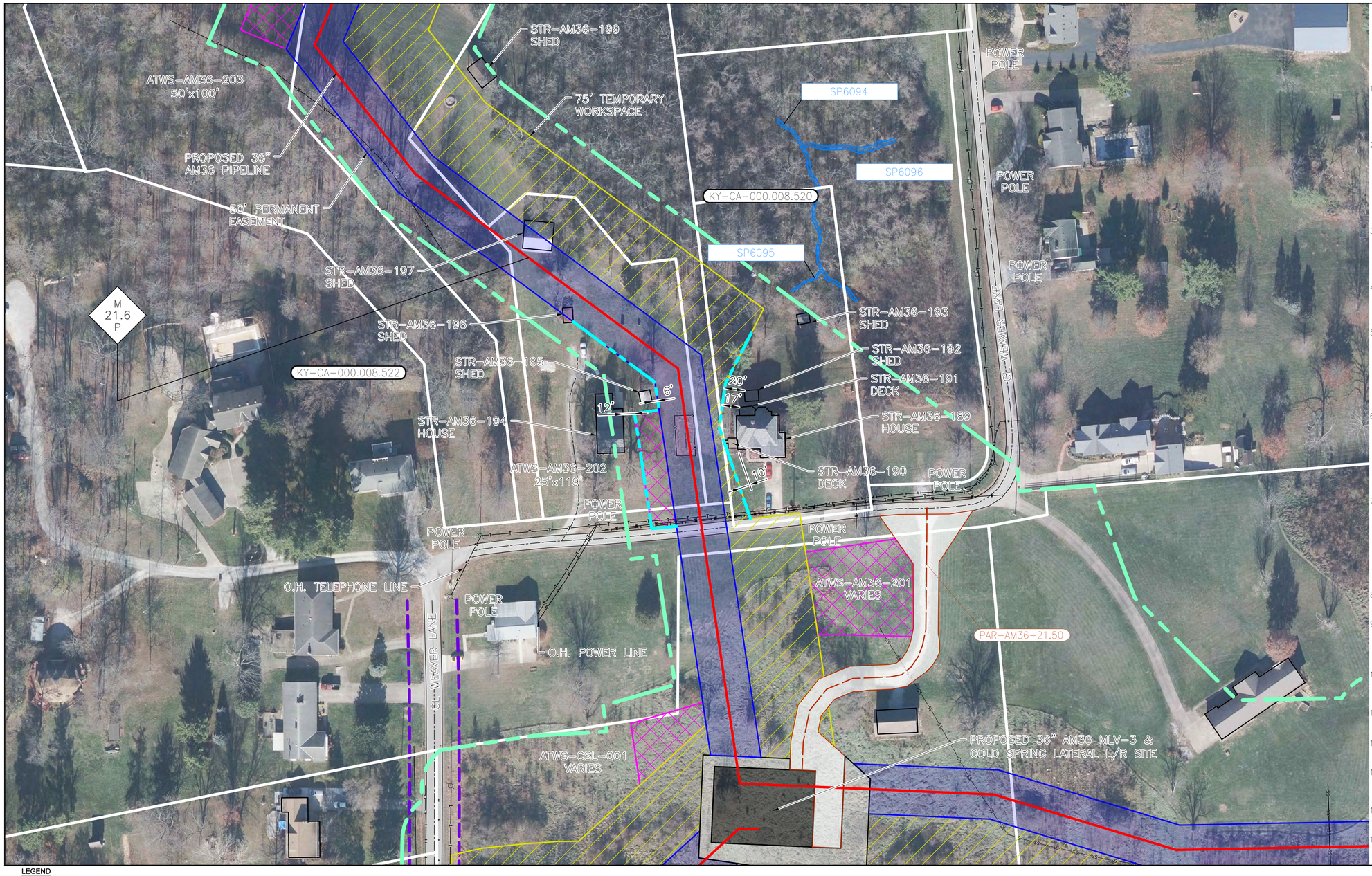
TC_UD_BR_24x36.DWG 24X36

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

PLOTTED SIZE: ARCH D (24x36)

ID TIME DESIGNFILE

TRACT NUMBER: KY-CA-000.008.520 & KY-CA-000.008.522



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0 50' 100' 200' GRAPHIC SCALE

N W E S

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4. TC ENERGY - COLUMBIA GAS TRANSMISSION , LLC.

REFERENCE DRAWINGS

DRAWING No	TITLE
85739-03-ML-02-001	ALIGNMENT SHEET

REVISION

REV No	DATE	DESCRIPTION
B	2025/10/15	ISSUED FOR FERC

APPROVAL

PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO. | DATE | PERMIT NUMBER:

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661 TRC PROJECT NO.: 574139

AM36 – SEGMENT 6 (AM36 MLV-3 TO HAMILTON CO. LINE)

FIA # 85739 | CHAINAGE: | DISCIPLINE # 03

RESIDENTIAL DETAIL
MP 21.6

SCALE AS NOTED | DRAWING No 85739-03-ML-15-200 | REV B

TC_UD_BR_24x36.DWG 24X36

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

PLOTTED SIZE: ARCH D (24x36)

ID

TIME

DESIGNFILE

TRACT NUMBER: AM04–004.00 & AM04–004.50

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GRAPHIC SCALE

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REFERENCE DRAWINGS

DRAWING No	TITLE
85739–03–ML–15–101	WORKSPACE DETAIL

REVISION

REV No	DATE	DESCRIPTION
B	2025/10/15	ISSUED FOR FERC

APPROVAL

PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO.	DATE	PERMIT NUMBER:

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661
TRC PROJECT NO.: 574139

AM36 – SEGMENT 6 (AM36 MLV–3 TO HAMILTON CO. LINE)

FIA #	CHAINAGE:	DISCIPLINE #
85739		03

RESIDENTIAL DETAIL
AM04–004.00 & AM04–004.50

SCALE	DRAWING No	REV
AS NOTED	85739–03–ML–15–201	B

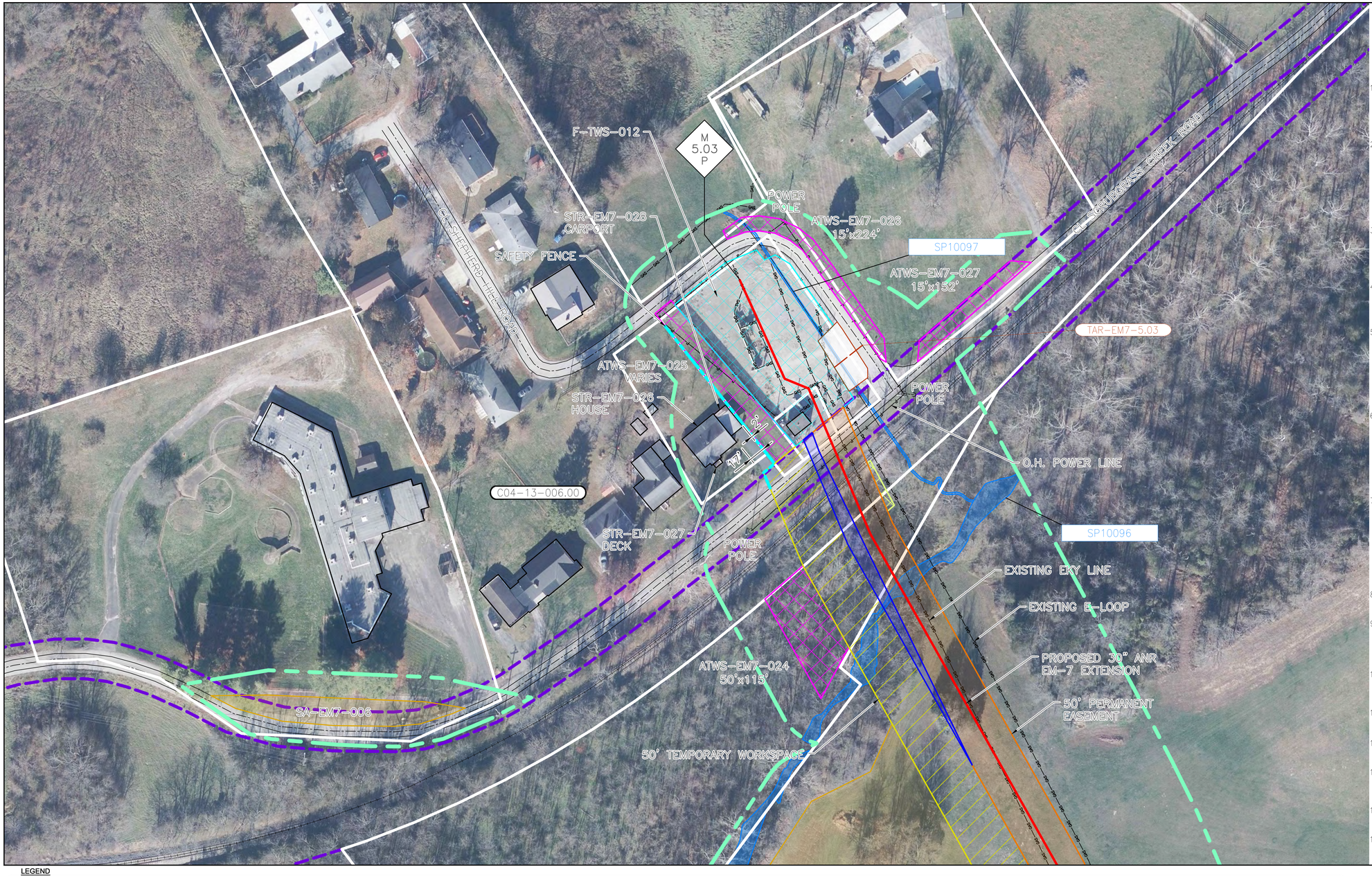
TIME----- -----DESIGNFILE

ID

TC_UD_BR_24x36.DWG 24X36

PLOTTED SIZE: ARCH D (24x36)

TRACT NUMBER: C04-13-006.00



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REFERENCE DRAWINGS

DRAWING No	TITLE
23689-03-ML-02-010	ALIGNMENT SHEET

REVISION

REV No	DATE	DESCRIPTION
B	2025/10/15	ISSUED FOR FERC

APPROVAL

PROJECT CODE	DRAFTER	DRAFTING CHECKER	DESIGNER	DESIGN CHECKER	PROJECT MANAGER	COMPANY
M.002371	HWK	KDF	SJO	JTD	SLS	TRC

PROFESSIONAL ENGINEER/RPT | PERMIT/ ENG. APPROVAL | DATE

REV. NO. | DATE | PERMIT NUMBER:

CADD DRAWING: DO NOT MAKE MANUAL REVISIONS

TC Energy

TRC FIRM REGISTRATION NO.: KY 1661 TRC PROJECT NO.: 574139

PL-EM7-FROM E-LOOP-215 TO EM7-225-SOMERSET CRK TO SHARPSBURG

FIA # 26389 | CHAINAGE: | DISCIPLINE # 03

RESIDENTIAL DETAIL
MP 5.03

SCALE AS NOTED | DRAWING No 26389-03-ML-15-200 | REV B

PLOTTED SIZE: ARCH D (24x36)

TC_UD_BR_24x36.DWG 24X36

ID

TIME

DESIGNFILE

APPENDIX F
SUMMARY OF GREENHOUSE GAS EMISSIONS FOR THE PROJECT

Table F-1: Total Construction Greenhouse Gas Emissions for the Project					
Construction Activity	**CO_2 (tons)**	**CH_4 (tons)**	**N_2O (tons)**	**CO_2e (tons)**	**CO_2e (metric tons)**
Non-Road Equipment	2,523.28	0.03	0.01	2,525.98	2,291.53
On-Road Equipment	4,459.14	0.15	0.03	4,928.93	4,471.46
Purging	4.20	729.42	--	17,837.06	16,181.53
Blowdown	0.53	92.51	--	2,590.91	2,350.43
Open Burn	--	5.42	--	151.67	137.59
PROJECT TOTAL	**6,986.61**	**735.02**	**0.04**	**28,034.55**	**25,432.55**

Source: Columbia's November 10, 2025, Application, Resource Report 9, Appendix 9A - Construction Emission Calculations. FERC eLibrary Accession No. 20251110-5151.

Note: Construction emission impacts conservatively assume all construction is completed in one year.

CO_2 = carbon dioxide; CH_4 = methane; N_2O = nitrogen dioxide; CO_2e = carbon dioxide equivalent.

Table F-2: Total Direct Operating Greenhouse Gas Emissions for the Project					
Pipeline Facility / Emission Source	**N_2O (tpy)**	**CH_4 (tpy)**	**CO_2 (tpy)**	**CO_2e (tpy)**	**CO_2e (metric tons/year)**
Line AM36					
Foster Regulator Station					
Component Fugitives	--	2.34	0.00	65.66	59.57
Venting	--	1.19	0.01	33.33	30.24
Bracken Meter Station					
Component Fugitives	--	5.26	0.01	147.17	133.51
Heater	1.53E-03	0.02	809.53	810.36	735.15
Black River Meter Station					
Component Fugitives	--	0.51	0.00	14.30	12.97
Pendleton Meter Station					
Component Fugitives	--	3.45	0.01	96.74	87.76
Heater	5.14E-04	0.01	272.58	272.86	247.53
MLV-1 / Gubsur Mill Meter Station					
Component Fugitives	--	1.79	0.00	50.21	45.55
MLV-2 and Alexandria Meter Station					
Component Fugitives	--	6.73	0.01	188.59	171.08
Heater	8.78E-03	0.09	4,659.61	4,664.39	4,231.47
Parkview Meter Station					
Component Fugitives	--	1.53	0.00	42.83	38.86
Heaters	1.00E-04	0.00	53.27	53.37	48.39
MLV-3 / Cold Spring Lateral Launcher/Receiver					

Table F-2: Total Direct Operating Greenhouse Gas Emissions for the Project					
Pipeline Facility / Emission Source	**N_2O (tpy)**	**CH_4 (tpy)**	**CO_2 (tpy)**	**CO_2e (tpy)**	**CO_2e (metric tons/year)**
Component Fugitives	--	2.98	0.01	83.48	75.73
Venting	--	0.61	0.00	17.01	15.43
Cold Spring Meter Station					
Component Fugitives	--	11.23	0.02	314.33	285.15
Venting	--	0.61	0.00	17.01	15.43
Heaters	1.23E-02	0.12	6,543.55	6,550.27	5,942.32
UL47 Meter Station					
Component Fugitives	--	4.80	0.01	134.47	121.99
Heater	9.07E-04	0.01	481.11	481.60	436.90
Kellogg Station Launcher/Receiver					
Component Fugitives	--	1.27	0.00	35.44	32.15
Venting	--	1.19	0.01	33.33	30.24
Pipeline Fugitives	--	0.09	-	2.56	2.32
Line EM-7					
Moorefield Site					
Component Fugitives	--	2.40	0.01	67.08	60.86
Venting	--	0.22	0.00	6.12	5.55
Carlisle Meter Station					
Component Fugitives	--	2.53	0.01	70.87	64.29
Venting	--	0.22	0.00	6.12	5.55
Pipeline Fugitives	--	0.11	-	3.13	2.84
Black River Lateral					
Pipeline Fugitives	--	0.00	-	0.01	0.01
Cold Spring Lateral					
Pipeline Fugitives	--	0.00	-	0.04	0.03
Kellogg Station Lateral					
Pipeline Fugitives	--	0.00	-	0.01	0.01
Project Total	**2.42E-02**	**51.30**	**12,819.78**	**14,262.65**	**12,938.88**

Source: Columbia's November 10, 2025, Application, Resource Report 9, Appendix 9A - Construction Emission Calculations. FERC eLibrary Accession No. 20251110-5151.
CS = compressor station; CO_2 = carbon dioxide; CH_4 = methane; N_2O = nitrogen dioxide; CO_2e = carbon dioxide equivalent; tpy = tons per year.

APPENDIX G
NOISE EFFECTS AT NSAS FOR THE PROJECT METER STATIONS

Table G-1 Noise Modeling Results at NSAs for the Alexandra Meter Station							
NSAs	**Distance and Direction from Proposed Weather Canopy**	**Measured L_d (dBA)**	**Measured L_n (dBA)**	**Measured Ambient L_{dn} (dBA)**	**Modeled Sound Level of Modified Meter Station L_{dn} (dBA)**	**Sound Level of Ambient plus Modeled Noise L_{dn} (dBA)**	**Potential Increase Above Ambient (dB)**
A-01	840 ft. W	50.80	38.40	50.10	40.60	50.60	0.50
A-02	1,280 ft. NW	50.80	38.40	50.10	37.20	50.30	0.20
A-03	1,645 ft. NW	50.80	38.40	50.10	32.50	50.20	0.10
A-04	1,760 ft. NE	50.80	38.40	50.10	35.60	50.30	0.20
A-05	1,545 ft. E	50.80	38.40	50.10	30.60	50.10	0.00
A-06	1,255 ft. SE	50.80	38.40	50.10	41.10	50.60	0.50
A-07	2,000 ft. S	50.80	38.40	50.10	35.90	50.30	0.20
A-08	2,665 ft. SW	50.80	38.40	50.10	19.50	50.10	0.00
A-09	1,700 ft. W	50.80	38.40	50.10	29.90	50.10	0.00
A-10	1,355 ft. W	50.80	38.40	50.10	36.40	50.30	0.20
Source: Columbia's November 10, 2025, Application, Resource Report 9, Appendix 9C, Alexandria Meter Station Noise Assessment. FERC eLibrary Accession No. 20251110-5151.							

Table G-2: Noise Modeling Results at NSAs for the Cold Spring Meter Station							
NSAs	**Distance and Direction from Proposed Weather Canopy**	**Measured L_d (dBA)**	**Measured L_n (dBA)**	**Measured Ambient L_{dn} (dBA)**	**Modeled Sound Level of Modified Meter Station L_{dn} (dBA)**	**Sound Level of Ambient plus Modeled Noise L_{dn} (dBA)**	**Potential Increase Above Ambient (dB)**
C-01	250 ft. E	47.50	42.40	60.00	53.90	61.00	1.00
C-02	190 ft. SE	47.50	42.40	60.00	50.50	60.50	0.50
C-03	180 ft. S	47.50	42.40	60.00	48.30	60.30	0.30
C-04	200 ft. S	47.50	42.40	60.00	45.80	60.20	0.20
C-05	255 ft. S	47.50	42.40	60.00	43.80	60.10	0.10
C-06	310 ft. SW	59.50	50.80	60.00	43.40	60.10	0.10
C-07	525 ft. NW	59.50	50.80	50.00	45.40	51.30	1.30
C-08	565 ft. NW	59.50	50.80	50.00	45.00	51.20	1.20
C-09	890 ft. N	59.50	50.80	50.00	40.50	50.50	0.50
C-10	495 ft. NE	59.50	50.80	50.00	46.50	51.60	1.60
Source: Columbia's November 10, 2025, Application, Resource Report 9, Appendix 9D Cold Spring Meter Station Noise Assessment. FERC eLibrary Accession No. 20251110-5151.							

Table G-3: Noise Modeling Results at NSAs for the Gubsur Mill Meter Station							
NSAs	**Distance and Direction from Proposed Weather Canopy**	**Measured L_d (dBA)**	**Measured L_n (dBA)**	**Measured Ambient L_{dn} (dBA)**	**Modeled Sound Level of New Meter Station L_{dn} (dBA)**	**Sound Level of Ambient plus Modeled Noise L_{dn} (dBA)**	**Potential Noise Increase Above Ambient (dB)**
G-01	305 ft. N	60.90	54.50	62.60	28.30	62.60	0.00
G-02	710 ft. NE	60.90	54.50	62.60	20.30	62.60	0.00
G-03	1,280 ft. SE	62.20	55.00	63.50	12.10	63.50	0.00
G-04	635 ft. SW	62.20	55.00	63.50	21.50	63.50	0.00
Source: Columbia's November 10, 2025, Application, Resource Report 9, Appendix 9E Gubsur Mill Meter Station Noise Assessment. FERC eLibrary Accession No. 20251110-5151.							

Table G-4: Noise Modeling Results at NSAs for the Parkview Meter Station							
NSAs	**Distance and Direction from Proposed Weather Canopy**	**Measured L_d (dBA)**	**Measured L_n (dBA)**	**Measured Ambient L_{dn} (dBA)**	**Modeled Sound Level of New Meter Station L_{dn} (dBA)**	**Sound Level of Ambient plus Modeled Noise L_{dn} (dBA)**	**Potential Noise Increase Above Ambient (dB)**
PAR-01	1,900 ft. E	62.00	52.00	62.00	13.40	62.00	0.00
PAR-02	2,080 ft. E	62.00	52.00	62.00	7.70	62.00	0.00
PAR-03	1,980 ft. E	62.00	52.00	62.00	1.20	62.00	0.00
PAR-04	1,360 ft. SE	62.00	52.00	62.00	24.30	62.00	0.00
PAR-05	480 ft. S	62.00	52.00	62.00	33.80	62.00	0.00
PAR-06	985 ft. SW	62.00	52.00	62.00	23.40	62.00	0.00
PAR-07	920 ft. N	62.00	52.00	62.00	9.70	62.00	0.00
PAR-08	755 ft. N	62.00	52.00	62.00	16.60	62.00	0.00
PAR-09	490 ft. N	62.00	52.00	62.00	21.70	62.00	0.00
Source: Columbia's November 10, 2025, Application, Resource Report 9, Appendix 9F Parkview Meter Station Noise Assessment. FERC eLibrary Accession No. 20251110-5151.							

Table G-5: Noise Modeling Results at NSAs for the Pendleton Meter Station							
NSAs	**Distance and Direction from Proposed Weather Canopy**	**Measured L_d (dBA)**	**Measured L_n (dBA)**	**Measured Ambient L_{dn} (dBA)**	**Modeled Sound Level of Modified Meter Station L_{dn} (dBA)**	**Sound Level of Ambient plus Modeled Noise L_{dn} (dBA)**	**Potential Increase Above Ambient (dB)**
PEN-01	280 ft. E	66.10	58.00	66.90	50.9	67.00	0.10

Table G-5: Noise Modeling Results at NSAs for the Pendleton Meter Station							
NSAs	**Distance and Direction from Proposed Weather Canopy**	**Measured L_d (dBA)**	**Measured L_n (dBA)**	**Measured Ambient L_{dn} (dBA)**	**Modeled Sound Level of Modified Meter Station L_{dn} (dBA)**	**Sound Level of Ambient plus Modeled Noise L_{dn} (dBA)**	**Potential Increase Above Ambient (dB)**
PEN-02	870 ft. SE	66.10	58.00	66.90	39.60	66.90	0.00
PEN-03	945 ft. S	66.10	58.00	66.90	38.50	49.60	0.30
PEN-04	855 ft. SW	66.10	58.00	66.90	36.00	49.50	0.20
PEN-05	985 ft. SW	66.10	58.00	66.90	36.20	49.50	0.20
PEN-06	1,135 ft. W	48.30	40.50	49.30	32.50	49.40	0.10
PEN-07	665 ft. W	48.30	40.50	49.30	31.00	49.40	0.10
PEN-08	855 ft. W	48.30	40.50	49.30	26.70	49.30	0.00
PEN-09	400 ft. NW	48.30	40.50	49.30	42.80	50.20	0.90
PEN-10	1,115 ft. NE	48.30	40.50	49.30	18.10	66.90	0.00
Source: Columbia's November 10, 2025, Application, Resource Report 9, Appendix 9G Pendleton Meter Station Noise Assessment. FERC eLibrary Accession No. 20251110-5151							

Table G-6: Noise Modeling Results at NSAs for the UL47 Meter Station							
NSAs	**Distance and Direction to from Proposed Weather Canopy**	**Measured L_d (dBA)**	**Measured L_n (dBA)**	**Measured Ambient L_{dn} (dBA)**	**Modeled Sound Level of New Meter Station L_{dn} (dBA)**	**Sound Level of Ambient plus Modeled Noise L_{dn} (dBA)**	**Potential Increase Above Ambient (dB)**
UL-01	1,270 ft. W	60.40	56.50	63.70	34.70	63.70	0.00
UL-02	1,175 ft. W	60.40	56.50	63.70	35.60	63.70	0.00
UL-03	1,225 ft. W	60.40	56.50	63.70	35.40	63.70	0.00
UL-04	1,160 ft. W	60.40	56.50	63.70	35.70	63.70	0.00
UL-05	1,170 ft. W	60.40	56.50	63.70	35.60	63.70	0.00
UL-06	1,170 ft. NW	60.40	56.50	63.70	35.10	63.70	0.00
UL-07	1,375 ft. N	60.40	56.50	63.70	33.50	63.70	0.00
UL-08	1,085 ft. N	60.40	56.50	63.70	27.50	63.70	0.00
UL-09	1,135 ft. N	60.40	56.50	63.70	25.00	63.70	0.00
UL-10	1,195 ft. N	60.40	56.50	63.70	23.80	63.70	0.00
UL-11	1,460 ft. N	60.40	56.50	63.70	19.30	63.70	0.00
UL-12	1,480 ft. NE	60.40	56.50	63.70	16.80	63.70	0.00
UL-13	1,450 ft. NE	60.40	56.50	63.70	18.90	63.70	0.00
UL-014	1,355 ft. NE	60.40	56.50	63.70	21.50	63.70	0.00

Table G-6: Noise Modeling Results at NSAs for the UL47 Meter Station							
NSAs	**Distance and Direction to from Proposed Weather Canopy**	**Measured L_d (dBA)**	**Measured L_n (dBA)**	**Measured Ambient L_{dn} (dBA)**	**Modeled Sound Level of New Meter Station L_{dn} (dBA)**	**Sound Level of Ambient plus Modeled Noise L_{dn} (dBA)**	**Potential Increase Above Ambient (dB)**
UL-15	1,145 ft. NE	60.40	56.50	63.70	26.40	63.70	0.00
UL-16	745 ft. E	60.40	56.50	63.70	25.10	63.70	0.00
UL-17	705 ft. SE	60.40	56.50	63.70	29.00	63.70	0.00
UL-18	885 ft. SE	60.40	56.50	63.70	30.80	63.70	0.00
UL-19	1,360 ft. SE	60.40	56.50	63.70	33.70	63.70	0.00
UL-20	1,640 ft. SW	60.40	56.50	63.70	30.90	63.70	0.00
UL-21	1,550 ft. SW	60.40	56.50	63.70	32.80	63.70	0.00
Source: Columbia's November 10, 2025, Application, Resource Report 9, Appendix 9H UL47 Meter Station Noise Assessment. FERC eLibrary Accession No. 20251110-5151							

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